



**Experimental version for
testing purpose only!**

My private, unofficial Version of:

SUSE Linux Enterprise Server 15 SP7

System Analysis and Tuning Guide

System Analysis and Tuning Guide

SUSE Linux Enterprise Server 15 SP7

This guide supports administrators in problem detection, resolution and optimization.

File generated at 2025-11-15 14:27

This is my own, **experimental version** of a Document from SUSE company. The only purpose of this document is the test of an alternative publishing mechanism. **Errors in the publishing mechanism may lead to wrong content.** You can find the original version of this document at documentation.suse.com.

The books and articles exist as XML sources, conformant to the DocBook standard. SUSE publishes them with the DocBook XSLT 1.0 Stylesheets, which generate XSL-FO, and Apache FOP, which in turn generates PDF.

This version is based on the same DocBook sources, but published with the new [xslTNG Stylesheets](#), which produce XHTML+CSS, and an rendering engine like *Antenna House* or *Weasyprint* to generate PDF. The only purpose of this version is a "*real life test*" of the new publishing mechanism, together with an "*DocBook TNG Framework*" that i wrote. It helps me to use and customize the xslTNG Stylesheets.
— Frank Steimke, Bremen, Germany

Contents

1 Preface 9

1 Available documentation 9

2 Improving the documentation 9

3 Documentation conventions 10

4 Support 12

Support statement for SUSE Linux Enterprise Server 12 • Technology previews 13

I Basics 1

1 General notes on system tuning 2

1.1 Be sure what problem to solve 2

1.2 Rule out common problems 3

1.3 Finding the bottleneck 3

1.4 Step-by-step tuning 3

II System monitoring 4

2 System monitoring utilities 5

2.1 Multi-purpose tools 5

vmstat 5 • **dstat** 9 • System activity information: **sar** 10

2.2 System information 15

Device load information: **iostat** 15 • Processor activity monitoring: **mpstat** 16 • Processor frequency monitoring: **turbostat** 17 • Task monitoring: **pidstat** 17 • Kernel ring buffer: **dmesg** 17 • List of open files: **lsof** 18 • Kernel and udev event sequence viewer: **udevadm monitor** 18

2.3 Processes 19

Inter-process communication: **ipcs** 19 • Process list: **ps** 19 • Process tree: **pstree** 21 • Table of processes: **top** 21 • IBM Z hypervisor monitor: **hyptop** 22 • A top-like I/O monitor: **iotop** 24 • Modify a process's niceness: **nice** and **renice** 24

2.4 Memory 25

Memory usage: **free** 25 • Detailed memory usage: `/proc/meminfo` 25 • Process memory usage: `smaps` 30 • `numaTOP` 30

2.5 Networking 31

Basic network diagnostics: **ip** 31 • Show the network usage of processes: **nethogs** 31 • Ethernet cards in detail: `ethtool` 32 • Show the network status: **ss** 33

2.6 The `/proc` file system 34

procinfo 36 • System control parameters: `/proc/sys/` 37

2.7 Hardware information 38

PCI resources: **lspci** 38 • USB devices: **lsusb** 39 • Monitoring and tuning the thermal subsystem: **tmon** 40 • MCELog: machine check exceptions (MCE) 40 • AMD64/Intel 64: `dmidecode`: DMI table decoder 41 • **POWER**: list hardware 42

2.8 Files and file systems 42

Determine the file type: **file** 42 • File systems and their usage: **mount**, **df** and **du** 43 • Additional information about ELF binaries 43 • File properties: **stat** 44

2.9 User information 44

User accessing files: **fuser** 44 • Who is doing what: **w** 45

2.10 Time and date 45

Time measurement with **time** 45

2.11 Graph your data: RRDtool 46

How RRDtool works 46 • A practical example 47 • More information 50

3 System log files 52

3.1 System log files in `/var/log/` 52

3.2 Viewing and parsing log files 54

3.3 Managing log files with **logrotate** 54

3.4 Monitoring log files with **logwatch** 55

3.5 Configuring mail forwarding for `root` 56

3.6 Forwarding log messages to a central syslog server 58

Set up the central syslog server 58 • Set up the client machines 59 • More information 60

3.7 Using **logger** to make system log entries 60

III Kernel monitoring 61

4 SystemTap—filtering and analyzing system data 62

4.1 Conceptual overview 62

SystemTap scripts 62 • Tapsets 63 • Commands and privileges 63 • Important files and directories 64

4.2 Installation and setup 64

4.3 Script syntax 66

Probe format 66 • SystemTap events (probe points) 67 • SystemTap handlers (probe body) 68

4.4 Example script 72

4.5 User space probing 73

4.6 More information 74

5 Kernel probes 75

5.1 Supported architectures 75

5.2 Types of kernel probes 75

Kprobes 76 • Jprobes 76 • Return probe 76

5.3 Kprobes API 76

5.4 debugfs Interface 77

Listing registered kernel probes 77 • Globally enabling/disabling kernel probes 78

5.5 More information 78

6 Hardware-based performance monitoring with Perf 79

6.1 Hardware-based monitoring 79

6.2 Sampling and counting 79

6.3 Installing Perf 79

6.4 Perf subcommands 80

6.5 Counting particular types of event 81

6.6 Recording events specific to particular commands 81

6.7 More information 82

7 OProfile—system-wide profiler 83

- 7 . 1 Conceptual overview 83
- 7 . 2 Installation and requirements 83
- 7 . 3 Available OProfile utilities 83
- 7 . 4 Using OProfile 84
 - Creating a report 84 • Getting event configurations 85
- 7 . 5 Generating reports 86
- 7 . 6 More information 86

8 Dynamic debug—kernel debugging messages 88

- 8 . 1 Benefits of dynamic debugging 88
- 8 . 2 Checking the status of dynamic debug 88
- 8 . 3 Using dynamic debug 88
- 8 . 4 Viewing the dynamic debug messages 89

IV Resource management 90

9 General system resource management 91

- 9 . 1 Planning the installation 91
 - Partitioning 91 • Installation scope 91 • Default target 92
- 9 . 2 Disabling unnecessary services 92
- 9 . 3 File systems and disk access 93
 - File systems 93 • Time stamp update policy 93 • Prioritizing disk access with **ionice** 94

10 Kernel control groups 95

- 10 . 1 Overview 95
 - Hybrid cgroup hierarchy 95
- 10 . 2 Resource accounting 95
- 10 . 3 Setting resource limits 96
- 10 . 4 Preventing fork bombs with **TasksMax** 96
 - Finding the current default **TasksMax** values 96 • Overriding the **DefaultTasksMax** value 97 • Default **TasksMax** limit on users 98

10 . 5 I/O control with cgroups 99

Prerequisites 99 • Configuring control quantities 100 • I/O control behavior and setting expectations 100 • Resource control in user sessions 101

10 . 6 More information 101

11 Automatic Non-Uniform Memory Access (NUMA) balancing 102

11 . 1 Implementation 102

11 . 2 Configuration 102

11 . 3 Monitoring 103

11 . 4 Impact 104

12 Power management 106

12 . 1 Power management at CPU Level 106

C-states (processor operating states) 106 • P-states (processor performance states) 107 • Turbo features 107

12 . 2 In-kernel governors 107

12 . 3 The `cpupower` tools 108

Viewing current settings with `cpupower`

109 • Viewing kernel idle statistics with `cpupower` 109 • Monitoring kernel and hardware statistics with `cpupower` 110 • Modifying current settings with `cpupower` 111

12 . 4 Special tuning options 112

Tuning options for P-states 112

12 . 5 Troubleshooting 112

12 . 6 More information 113

12 . 7 Monitoring power consumption with `powerTOP` 113

13 Adaptive and dynamic tuning using TuneD 116

13 . 1 Introduction to TuneD 116

13 . 2 Components of TuneD 116

13 . 3 Managing TuneD 117

Installing TuneD 117 • Enabling `tuned` 118 • Starting `tuned` 118 • Disabling `tuned` 119

13 . 4 TuneD profiles 120

Supported TuneD profiles 120 • Managing TuneD profiles 122

- 13 . 5 TuneD configuration 124
 - Global TuneD configuration 125 • Supported profile configuration 126 • Custom profile configuration 127
- 13 . 6 TuneD plug-ins 129
 - Supported TuneD plug-ins 129
- 13 . 7 More information 144

V Kernel tuning 146

14 Tuning I/O performance 147

- 14 . 1 Switching I/O scheduling 147
- 14 . 2 Available I/O elevators with blk-mq I/O path 147
 - MQ-DEADLINE 148 • NONE 149 • BFQ (Budget Fair Queueing) 149 • KYBER 150
- 14 . 3 I/O barrier tuning 151

15 Tuning the task scheduler 152

- 15 . 1 Introduction 152
 - Preemption 152 • Timeslice 152 • Process priority 153
- 15 . 2 Process classification 153
- 15 . 3 Completely Fair Scheduler 153
 - How CFS works 154 • Grouping processes 154 • Kernel configuration options 155 • Terminology 155 • Changing real-time attributes of processes with **chrt** 156 • Runtime tuning with **sysctl** 156 • Debugging interface and scheduler statistics 160
- 15 . 4 More information 162

16 Tuning the memory management subsystem 163

- 16 . 1 Memory usage 163
 - Anonymous memory 163 • Pagecache 164 • Buffercache 164 • Buffer heads 164 • Writeback 164 • Readahead 164 • VFS caches 165
- 16 . 2 Reducing memory usage 165
 - Reducing malloc (anonymous) usage 165 • Reducing kernel memory overheads 165 • Memory controller (memory cgroups) 165

- 16 . 3 Virtual memory manager (VM) tunable parameters 166
 - Reclaim ratios 166 • Writeback parameters 167 • Timing differences of I/O writes between SUSE Linux Enterprise 12 and SUSE Linux Enterprise 11 168 • Readahead parameters 169 • Transparent HugePage parameters 170 • khugepaged parameters 171 • Further VM parameters 172

- 16 . 4 Monitoring VM behavior 172

17 Tuning the network 174

- 17 . 1 Configurable kernel socket buffers 174
- 17 . 2 Detecting network bottlenecks and analyzing network traffic 175
- 17 . 3 Netfilter 176
- 17 . 4 Improving the network performance with receive packet steering (RPS) 176

18 Tuning SUSE Linux Enterprise for SAP 178

- 18 . 1 Tuning SLE Systems with `sapconf` 5 178
 - Verifying `sapconf` setup 179 • Enabling and disabling `sapconf` and viewing its status 179 • Configuring `sapconf` 180 • Removing `sapconf` 181 • For more information 181 • Using **tuned** together with `sapconf` 182

VI Handling system dumps 183

19 Tracing tools 184

- 19 . 1 Tracing system calls with `strace` 184
- 19 . 2 Tracing library calls with `ltrace` 187
- 19 . 3 Debugging and profiling with Valgrind 188
 - Installation 188 • Supported architectures 188 • General information 188 • Default options 189 • How Valgrind works 190 • Messages 190 • Error messages 192
- 19 . 4 More information 192

20 Kexec and Kdump 193

- 20 . 1 Introduction 193
- 20 . 2 Required packages 193
- 20 . 3 Kexec internals 194
- 20 . 4 Calculating `crashkernel` allocation size 195

20 . 5	Basic Kexec usage	198
20 . 6	How to configure Kexec for routine reboots	199
20 . 7	Basic Kdump configuration	199
	Manual Kdump configuration	200 • YaST configuration 201 • Kdump over SSH 202
20 . 8	Analyzing the crash dump	203
	Kernel binary formats	204
20 . 9	Advanced Kdump configuration	207
20 . 10	More information	208
21	Using systemd-coredump to debug application crashes	210
21 . 1	Use and configuration	210
VII	Synchronized clocks with Precision Time Protocol	213
22	Precision Time Protocol	214
22 . 1	Introduction to PTP	214
	PTP Linux implementation	214
22 . 2	Using PTP	214
	Network driver and hardware support	214 • Using ptp4l 215 • ptp4l configuration file 216 • Delay measurement 217 • PTP management client: pmc 217
22 . 3	Synchronizing the clocks with phc2sys	218
	Verifying time synchronization	219
22 . 4	Examples of configurations	220
22 . 5	PTP and NTP	221
	NTP to PTP synchronization	221 • Configuring PTP-NTP bridge 221
A	GNU licenses	223
A . 1	GNU Free Documentation License	223

Preface

Available documentation

Online documentation

Our documentation is available online at <https://documentation.suse.com>. Browse or download the documentation in various formats.



Latest updates

The latest updates are usually available in the English-language version of this documentation.

SUSE Knowledgebase

If you run into an issue, check out the Technical Information Documents (TIDs) that are available online at <https://www.suse.com/support/kb/>. Search the SUSE Knowledgebase for known solutions driven by customer need.

Release notes

For release notes, see <https://www.suse.com/releasenotes/>.

In your system

For offline use, the release notes are also available under `/usr/share/doc/release-notes` on your system. The documentation for individual packages is available at `/usr/share/doc/packages`.

Many commands are also described in their *manual pages*. To view them, run **man**, followed by a specific command name. If the **man** command is not installed on your system, install it with **sudo zypper install man**.

Improving the documentation

Your feedback and contributions to this documentation are welcome. The following channels for giving feedback are available:

Service requests and support

For services and support options available for your product, see <https://www.suse.com/support/>.

To open a service request, you need a SUSE subscription registered at SUSE Customer Center. Go to <https://scc.suse.com/support/requests>, log in, and click *Create New*.

Bug reports

Report issues with the documentation at <https://bugzilla.suse.com/>.

To simplify this process, click the *Report an issue* icon next to a headline in the HTML version of this document. This preselects the right product and category in Bugzilla and adds a link to the current section. You can start typing your bug report right away.

A Bugzilla account is required.

Contributions

To contribute to this documentation, click the *Edit source document* icon next to a headline in the HTML version of this document. This will take you to the source code on GitHub, where you can open a pull request.

A GitHub account is required.



Edit source document only available for English

The *Edit source document* icons are only available for the English version of each document. For all other languages, use the *Report an issue* icons instead.

For more information about the documentation environment used for this documentation, see the repository's README.

Mail

You can also report errors and send feedback concerning the documentation to `doc-team@suse.com`. Include the document title, the product version, and the publication date of the document. Additionally, include the relevant section number and title (or provide the URL) and provide a concise description of the problem.

Documentation conventions

The following notices and typographic conventions are used in this document:

- `/etc/passwd`: Directory names and file names
- *PLACEHOLDER*: Replace *PLACEHOLDER* with the actual value
- `PATH`: An environment variable

- **ls**, **--help**: Commands, options, and parameters
- **user**: The name of a user or group
- **package_name**: The name of a software package
- **Alt**, **Alt-F1**: A key to press or a key combination. Keys are shown in uppercase as on a keyboard.
- *File*, *File > Save As*: menu items, buttons
- **x86_64**► This paragraph is only relevant for the AMD64/Intel 64 architectures. The arrows mark the beginning and the end of the text block.◀
- **zseries;power**► This paragraph is only relevant for the architectures IBM Z and POWER. The arrows mark the beginning and the end of the text block.◀
- *Chapter 1, “Example chapter”*: A cross-reference to another chapter in this guide.
- Commands that must be run with root privileges. You can also prefix these commands with the **sudo** command to run them as a non-privileged user:

```
#command>sudo command
```

- Commands that can be run by non-privileged users:

```
>command
```

- Commands can be split into two or multiple lines by a backslash character (\) at the end of a line. The backslash informs the shell that the command invocation will continue after the end of the line:

```
>echo a b \
c d
```

- A code block that shows both the command (preceded by a prompt) and the respective output returned by the shell:

```
>command
output
```

- Notices



Warning notice

Vital information you must be aware of before proceeding. Warns you about security issues, potential loss of data, damage to hardware, or physical hazards.



Important notice

Important information you should be aware of before proceeding.



Note notice

Additional information, for example about differences in software versions.



Tip notice

Helpful information, like a guideline or a piece of practical advice.

• Compact Notices



Note

Additional information, for example about differences in software versions.



Tip

Helpful information, like a guideline or a piece of practical advice.

Support

Find the support statement for SUSE Linux Enterprise Server and general information about technology previews below. For details about the product lifecycle, see <https://www.suse.com/lifecycle>.

If you are entitled to support, find details on how to collect information for a support ticket at <https://documentation.suse.com/sles-15/html/SLES-all/cha-adm-support.html>.

Support statement for SUSE Linux Enterprise Server

To receive support, you need an appropriate subscription with SUSE. To view the specific support offers available to you, go to <https://www.suse.com/support/> and select your product.

The support levels are defined as follows:

L1

Problem determination, which means technical support designed to provide compatibility information, usage support, ongoing maintenance, information gathering and basic troubleshooting using available documentation.

L2

Problem isolation, which means technical support designed to analyze data, reproduce customer problems, isolate a problem area and provide a resolution for problems not resolved by Level 1 or prepare for Level 3.

L3

Problem resolution, which means technical support designed to resolve problems by engaging engineering to resolve product defects which have been identified by Level 2 Support.

For contracted customers and partners, SUSE Linux Enterprise Server is delivered with L3 support for all packages, except for the following:

- Technology previews.
- Sound, graphics, fonts, and artwork.
- Packages that require an additional customer contract.
- Some packages shipped as part of the module *Workstation Extension* are L2-supported only.
- Packages with names ending in `-devel` (containing header files and similar developer resources) will only be supported together with their main packages.

SUSE will only support the usage of original packages. That is, packages that are unchanged and not recompiled.

Technology previews

Technology previews are packages, stacks, or features delivered by SUSE to provide glimpses into upcoming innovations. Technology previews are included for your convenience to give you a chance to test new technologies within your environment. We would appreciate your feedback. If you test a technology preview, please contact your SUSE representative and let them know about your experience and use cases. Your input is helpful for future development.

Technology previews have the following limitations:

- Technology previews are still in development. Therefore, they may be functionally incomplete, unstable, or otherwise *not* suitable for production use.
- Technology previews are *not* supported.
- Technology previews may only be available for specific hardware architectures.
- Details and functionality of technology previews are subject to change. As a result, upgrading to subsequent releases of a technology preview may be impossible and require a fresh installation.
- SUSE may discover that a preview does not meet customer or market needs, or does not comply with enterprise standards. Technology previews can be removed from a product at any time. SUSE does not commit to providing a supported version of such technologies in the future.

For an overview of technology previews shipped with your product, see the release notes at <https://www.suse.com/releasenotes>.

Part I. Basics

1 General notes on system tuning 2

Chapter 1. General notes on system tuning

Procedure 1.1. General approach when tuning a system

1. Specify the problem that needs to be solved.
2. In case the degradation is new, identify any recent changes to the system.
3. Identify why the issue is considered a performance problem.
4. Specify a metric that can be used to analyze performance. This metric could for example be latency, throughput, the maximum number of users that are simultaneously logged in, or the maximum number of active users.
5. Measure current performance using the metric from the previous step.
6. Identify the subsystems where the application is spending the most time.
7.
 1. Monitor the system and/or the application.
 2. Analyze the data, categorize where time is being spent.
8. Tune the subsystem identified in the previous step.
9. Remeasure the current performance without monitoring using the same metric as before.
10. If performance is still not acceptable, start over with *step 3*.

1.1. Be sure what problem to solve

Before starting to tuning a system, try to describe the problem as exactly as possible. A statement like “The system is slow!” is not a helpful problem description. For example, it could make a difference whether the system speed needs to be generally improved, or only at peak times.

Furthermore, make sure you can apply a measurement to your problem, otherwise you cannot verify if the tuning was a success or not. You should always be able to compare “before” and “after”. Which metrics to use depends on the scenario or application you are looking into. For example, relevant Web server metrics could be expressed in terms of the following:

Latency

The time to deliver a page

Throughput

Number of pages served per second or megabytes transferred per second

Active users

The maximum number of users that can download pages while still receiving pages within an acceptable latency

1.2. Rule out common problems

A performance problem often is caused by network or hardware problems, bugs or configuration issues. Make sure to rule out problems such as the ones listed below before attempting to tune your system:

- Check the output of the systemd journal (see Chapter 21, **journalctl**: query the systemd journal in “[Administration Guide](#)”) for unusual entries.
- Check (using **top** or **ps**) whether a certain process misbehaves by eating up unusual amounts of CPU time or memory.
- Check for network problems by inspecting `/proc/net/dev`.
- In case of I/O problems with physical disks, make sure it is not caused by hardware problems (check the disk with the `smartmontools`) or by a full disk.
- Ensure that background jobs are scheduled to be carried out in times the server load is low. Those jobs should also run with low priority (set via **nice**).
- If the machine runs several services using the same resources, consider moving services to another server.
- Last, make sure your software is up-to-date.

1.3. Finding the bottleneck

Finding the bottleneck is the hardest part when tuning a system. SUSE Linux Enterprise Server offers many tools to help you with this task. See *Part II, “System monitoring”* for detailed information on general system monitoring applications and log file analysis. If the problem requires a long-time in-depth analysis, the Linux kernel offers means to perform such analysis. See *Part III, “Kernel monitoring”* for coverage.

Once you have collected the data, it needs to be analyzed. First, inspect if the server's hardware (memory, CPU, bus) and its I/O capacities (disk, network) are sufficient. If these basic conditions are met, the system can benefit from tuning.

1.4. Step-by-step tuning

Make sure to carefully plan the tuning itself. It is of vital importance to only do one step at a time. Only by doing so can you measure whether the change made an improvement or even had a negative impact. Each tuning activity should be measured over a sufficient time period to ensure you can do an analysis based on significant data. If you cannot measure a positive effect, do not make the change permanent. Chances are that it can have a negative effect in the future.

Part II. System monitoring

2 System monitoring utilities **5**

3 System log files **52**

Chapter 2. System monitoring utilities



Gathering and Analyzing System Information with **supportconfig**

Apart from the utilities presented in the following, SUSE Linux Enterprise Server also contains **supportconfig**, a tool to create reports about the system such as: current kernel version, hardware, installed packages, partition setup and much more. These reports are used to provide the SUSE support with needed information in case a support ticket is created. However, they can also be analyzed for known issues to help resolve problems faster. For this purpose, SUSE Linux Enterprise Server provides both an appliance and a command line tool for Supportconfig Analysis (SCA). See Chapter 48, Gathering system information for support in “[Administration Guide](#)” for details.

For each of the described commands, examples of the relevant outputs are presented. In the examples, the first line is the command itself (after the tux > or root #). Omissions are indicated with square brackets ([. . .]) and long lines are wrapped where necessary. Line breaks for long lines are indicated by a backslash (\).

```
>command -x -y
output line 1
output line 2
output line 3 is annoyingly long, so long that \
    we need to break it
output line 4
[... ]
output line 98
output line 99
```

The descriptions have been kept short so that we can include as many utilities as possible. Further information for all the commands can be found in the manual pages. Most of the commands also understand the parameter `--help`, which produces a brief list of possible parameters.

2.1. Multi-purpose tools

While most Linux system monitoring tools monitor only a single aspect of the system, there are a few tools with a broader scope. To get an overview and find out which part of the system to examine further, use these tools first.

2.1. vmstat

vmstat collects information about processes, memory, I/O, interrupts and CPUs:

```
vmstat [options] [delay [count]]
```

When called without values for delay and count, it displays average values since the last reboot. When called with a value for delay (in seconds), it displays values for the given period (two

seconds in the examples below). The value for count specifies the number of updates vmstat should perform. If not specified, it runs until manually stopped.

Example 2.1. vmstat output on a lightly used machine

```
>vmstat 2
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st
 1  0   44264  81520    424 935736    0    0    12    25   27   34   1   0  98   0
0
0  0   44264  81552    424 935736    0    0     0     0   38   25   0   0 100   0
0
0  0   44264  81520    424 935732    0    0     0     0   23   15   0   0 100   0
0
0  0   44264  81520    424 935732    0    0     0     0   36   24   0   0 100   0
0
0  0   44264  81552    424 935732    0    0     0     0   51   38   0   0 100   0
0
```

Example 2.2. vmstat output on a heavily used machine (CPU bound)

```
>vmstat 2
procs -----memory----- ---swap-- -----io----- -system-- -----
cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa
st
32  1   26236 459640 110240 6312648    0    0  9944     2 4552 6597 95   5   0   0
0
23  1   26236 396728 110336 6136224    0    0  9588     0 4468 6273 94   6   0   0
0
35  0   26236 554920 110508 6166508    0    0  7684 27992 4474 4700 95   5   0   0
0
28  0   26236 518184 110516 6039996    0    0 10830     4 4446 4670 94   6   0   0
0
21  5   26236 716468 110684 6074872    0    0  8734 20534 4512 4061 96   4   0   0
0
```



First line of output

The first line of the vmstat output always displays average values since the last reboot.

The columns show the following:

r

Shows the number of processes in a runnable state. These processes are either executing or waiting for a free CPU slot. If the number of processes in this column is constantly higher than the number of CPUs available, this may be an indication of insufficient CPU power.

b

Shows the number of processes waiting for a resource other than a CPU. A high number in this column may indicate an I/O problem (network or disk).

swpd

The amount of swap space (KB) currently used.

free

The amount of unused memory (KB).

inact

Recently unused memory that can be reclaimed. This column is only visible when calling **vmstat** with the parameter -a (recommended).

active

Recently used memory that normally does not get reclaimed. This column is only visible when calling **vmstat** with the parameter -a (recommended).

buff

File buffer cache (KB) in RAM that contains file system metadata. This column is not visible when calling **vmstat** with the parameter -a.

cache

Page cache (KB) in RAM with the actual contents of files. This column is not visible when calling **vmstat** with the parameter -a.

si / so

Amount of data (KB) that is moved from swap to RAM (**si**) or from RAM to swap (**so**) per second. High **so** values over a long period of time may indicate that an application is leaking memory and the leaked memory is being swapped out. High **si** values over a long period of time could mean that an application that was inactive for a long time is now active again. Combined high **si** and **so** values for prolonged periods of time are evidence of swap thrashing and may indicate that more RAM needs to be installed in the system because there is not enough memory to hold the working set size.

bi

Number of blocks per second received from a block device (for example, a disk read). Swapping also impacts the values shown here. The block size may vary between file systems but can be determined using the **stat** utility. If throughput data is required then **iostat** may be used.

bo

Number of blocks per second sent to a block device (for example, a disk write). Swapping also impacts the values shown here.

in

Interrupts per second. A high value may indicate a high I/O level (network and/or disk), but could also be triggered for other reasons such as inter-processor interrupts triggered by another activity. Make sure to also check `/proc/interrupts` to identify the source of interrupts.

cs

Number of context switches per second. This is the number of times that the kernel replaces executable code of one program in memory with that of another program.

us

Percentage of CPU usage executing application code.

sy

Percentage of CPU usage executing kernel code.

id

Percentage of CPU time spent idling. If this value is zero over a longer time, your CPUs are working to full capacity. This is not necessarily a bad sign—rather refer to the values in columns *r* and *b* to determine if your machine is equipped with sufficient CPU power.

wa

If "wa" time is non-zero, it indicates throughput lost because of waiting for I/O. This may be inevitable, for example, if a file is being read for the first time, background writeback cannot keep up, and so on. It can also be an indicator for a hardware bottleneck (network or hard disk). Lastly, it can indicate a potential for tuning the virtual memory manager (refer to *Chapter 16, Tuning the memory management subsystem*).

st

Percentage of CPU time stolen from a virtual machine.

See **`vmstat -h`** for more options.

2.2. dstat

dstat is a replacement for tools such as **vmstat**, **iostat**, **netstat**, or **ifstat**. **dstat** displays information about the system resources in real time. For example, you can compare disk usage in combination with interrupts from the IDE controller, or compare network bandwidth with the disk throughput (in the same interval).

By default, its output is presented in readable tables. Alternatively, CSV output can be produced which is suitable as a spreadsheet import format.

It is written in Python and can be enhanced with plug-ins.

This is the general syntax:

```
dstat [-afv] [OPTIONS...] [DELAY [COUNT]]
```

All options and parameters are optional. Without any parameter, **dstat** displays statistics about CPU (-c, --cpu), disk (-d, --disk), network (-n, --net), paging (-g, --page), and the interrupts and context switches of the system (-y, --sys); it refreshes the output every second ad infinitum:

```
#dstat
You did not select any stats, using -cdngy by default.
----total-cpu-usage---- -dsk/total- -net/total- ---paging-- ---system--
usr sys idl wai hiq siq| read  writ| recv  send|  in   out| int  csw
  0   0 100   0   0   0| 15k   44k|    0    0|    0   82B| 148  194
  0   0 100   0   0   0|    0    0| 5430B 170B|    0    0| 163  187
  0   0 100   0   0   0|    0    0| 6363B 842B|    0    0| 196  185
```

-a, --all

equal to -cdngy (default)

-f, --full

expand -C, -D, -I, -N and -S discovery lists

-v, --vmstat

equal to -pmgdsc, -D total

DELAY

delay in seconds between each update

COUNT

the number of updates to display before exiting

The default delay is 1 and the count is unspecified (unlimited).

For more information, see the man page of **dstat** and its Web page at <http://dag.wieers.com/home-made/dstat/>.

2.3. System activity information: **sar**

sar can generate extensive reports on almost all important system activities, among them CPU, memory, IRQ usage, I/O, and networking. It can also generate reports in real time. The **sar** command gathers data from the `/proc` file system.



sysstat package

The **sar** command is a part of the **sysstat** package. Install it with YaST, or with the **zypper in sysstat** command. **sysstat.service** does not start by default, and must be enabled and started with the following command:

```
>sudo systemctl enable --now sysstat
```

2.1. Generating reports with **sar**

To generate reports in real time, call **sar** with an interval (seconds) and a count. To generate reports from files specify a file name with the option `-f` instead of interval and count. If file name, interval and count are not specified, **sar** attempts to generate a report from `/var/log/sa/saDD`, where *DD* stands for the current day. This is the default location to where **sadc** (the system activity data collector) writes its data. Query multiple files with multiple `-f` options.

```
sar 2 10 # real time report, 10 times every 2 seconds
sar -f ~/reports/sar_2014_07_17 # queries file sar_2014_07_17
sar # queries file from today in /var/log/sa/
cd /var/log/sa && \
sar -f sa01 -f sa02 # queries files /var/log/sa/0[12]
```

Find examples for useful **sar** calls and their interpretation below. For detailed information on the meaning of each column, refer to the man (1) of **sar**.



sysstat reporting when the service stops

When the **sysstat** service is stopped (for example, during reboot or shutdown), the tool still collects last-minute statistics by automatically running the **/usr/lib64/sa/sa1 -S ALL 1 1** command. The collected binary data is stored in the system activity data file.

2.1. CPU usage report: **sar**

When called with no options, **sar** shows a basic report about CPU usage. On multi-processor machines, results for all CPUs are summarized. Use the option **-P ALL** to also see statistics for individual CPUs.

```
#sar 10 5
Linux 6.4.0-150700.38-default (jupiter)      03/11/2024
_x86_64_      (2 CPU)
```

	CPU	%user	%nice	%system	%iowait	%steal	%idle
17:51:29	all	57,93	0,00	9,58	1,01	0,00	31,47
17:51:39	all	32,71	0,00	3,79	0,05	0,00	63,45
17:51:49	all	47,23	0,00	3,66	0,00	0,00	49,11
17:51:59	all	53,33	0,00	4,88	0,05	0,00	41,74
17:52:09	all	56,98	0,00	5,65	0,10	0,00	37,27
17:52:19	all	49,62	0,00	5,51	0,24	0,00	44,62
Average:	all						

%iowait displays the percentage of time that the CPU was idle while waiting for an I/O request. If this value is much higher than zero over a long period of time, there is a bottleneck in the I/O system (network or hard disk). If the *%idle* value is zero over a long period of time, your CPU is working at capacity.

2.2. Memory usage report: **sar -r**

Generate an overall picture of the system memory (RAM) by using the option **-r**:

```
#sar -r 10 5
Linux 6.4.0-150700.38-default (jupiter)      03/11/2024
_x86_64_      (2 CPU)
```

	kbmemfree	kbmemused	%memused	kbbuffers	kbcached	kbcommit	%commit
	kbactive	kbinact	kbdirty				
17:55:27	104232	1834624	94.62	20	627340	2677656	66.24
17:55:37	802052	828024	1744				
17:55:47	98584	1840272	94.92	20	624536	2693936	66.65
17:55:57	808872	826932	2012				
17:56:07	87088	1851768	95.51	20	605288	2706392	66.95
17:56:17	827260	821304	1588				
17:56:27	86268	1852588	95.55	20	599240	2739224	67.77
17:56:37	829764	820888	3036				
17:56:47	104260	1834596	94.62	20	599864	2730688	67.56
17:56:57	811284	821584	3164				
Average:	96086	1842770	95.04	20	611254	2709579	67.03
17:57:07	815846	823746	2309				

The columns *kbcommit* and *%commit* show an approximation of the maximum amount of memory (RAM and swap) that the current workload could need. While *kbcommit* displays the absolute number in kilobytes, *%commit* displays a percentage.

2.3. Paging statistics report: **sar -B**

Use the option **-B** to display the kernel paging statistics.

```
#sar -B 10 5
Linux 6.4.0-150700.38-default (jupiter)          03/11/2024
_x86_64_          (2 CPU)

18:23:01 pgpgin/s pgpgout/s fault/s majflt/s pgfree/s pgscank/s pgscand/s
pgsteal/s %vmeff
18:23:11    366.80      11.60   542.50      1.10  4354.80      0.00      0.00
0.00    0.00
18:23:21      0.00     333.30  1522.40      0.00 18132.40      0.00      0.00
0.00    0.00
18:23:31    47.20     127.40  1048.30      0.10 11887.30      0.00      0.00
0.00    0.00
18:23:41    46.40       2.50   336.10      0.10  7945.00      0.00      0.00
0.00    0.00
18:23:51      0.00     583.70  2037.20      0.00 17731.90      0.00      0.00
0.00    0.00
Average:   92.08     211.70  1097.30      0.26 12010.28      0.00      0.00
0.00    0.00
```

The *majflt/s* (major faults per second) column shows how many pages are loaded from disk into memory. The source of the faults may be file accesses or faults. At times, many major faults are normal. For example, during application start-up time. If major faults are experienced for the entire lifetime of the application it may be an indication that there is insufficient main memory, particularly if combined with large amounts of direct scanning (*pgscand/s*).

The *%vmeff* column shows the number of pages scanned (*pgscand/s*) in relation to the ones being reused from the main memory cache or the swap cache (*pgsteal/s*). It is a measurement of the efficiency of page reclaim. Healthy values are either near 100 (every inactive page swapped out is being reused) or 0 (no pages have been scanned). The value should not drop below 30.



Understanding and addressing *%vmeff* values exceeding 100% in modern Linux kernels

A *%vmeff* value exceeding 100% means that the number of pages reclaimed (*pgsteal*) is reported as higher than the number of pages scanned (*pgscank*) during memory reclamation. This is unexpected because under normal circumstances, the kernel should not reclaim more pages than it scans.

In modern Linux kernels (5.x and higher), changes in memory management and page reclamation accounting make *%vmeff* less reliable as a measure of virtual memory efficiency. These changes include separating background and direct reclaim metrics, tracking memory operations with greater granularity, introducing deferred reclamation, and adding support for NUMA-aware and cgroup-based memory management. Such updates can cause tools like **sar** to misinterpret metrics like *pgsteal* and *pgscank*, resulting in *%vmeff* value exceeding 100% in certain situations.

If *%vmeff* exceeds 100%, avoid using it as a performance indicator. Instead, analyze memory performance by monitoring *pgpgin/s* and *pgpgout/s* for paging activity, *majorflt/s* and *fault/s* for page faults, and memory utilization details in */proc/meminfo* (for example, *Active*, *Inactive*, *Dirty* and *Writeback*). Additionally, correlate these metrics with application-level performance and I/O patterns to identify potential memory bottlenecks or inefficiencies.

2.4. Block device statistics report: **sar -d**

Use the option **-d** to display the block device (hard disk, optical drive, USB storage device, etc.). Make sure to use the additional option **-p** (pretty-print) to make the *DEV* column readable.

```
#sar -d -p 10 5
Linux 6.4.0-150700.38-default (jupiter)          03/11/2024
_x86_64_          (2 CPU)
```

	DEV	tps	rd_sec/s	wr_sec/s	avgrq-sz	avgqu-sz	await
18:46:09	svctm %util						
18:46:19	sda	1.70	33.60	0.00	19.76	0.00	0.47
0.47	0.08						
18:46:19	sr0	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00						
18:46:19	svctm %util						
18:46:29	sda	8.60	114.40	518.10	73.55	0.06	7.12
0.93	0.80						
18:46:29	sr0	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00						
18:46:29	svctm %util						
18:46:39	sda	40.50	3800.80	454.90	105.08	0.36	8.86
0.69	2.80						
18:46:39	sr0	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00						
18:46:39	svctm %util						
18:46:49	sda	1.40	0.00	204.90	146.36	0.00	0.29
0.29	0.04						
18:46:49	sr0	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00						
18:46:49	svctm %util						
18:46:59	sda	3.30	0.00	503.80	152.67	0.03	8.12
1.70	0.56						
18:46:59	sr0	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00						
Average:	DEV	tps	rd_sec/s	wr_sec/s	avgrq-sz	avgqu-sz	await
svctm	%util						
Average:	sda	11.10	789.76	336.34	101.45	0.09	8.07
0.77	0.86						
Average:	sr0	0.00	0.00	0.00	0.00	0.00	0.00
0.00	0.00						

Compare the *Average* values for *tps*, *rd_sec/s*, and *wr_sec/s* of all disks. Constantly high values in the *svctm* and *%util* columns could be an indication that I/O subsystem is a bottleneck.

If the machine uses multiple disks, then it is best if I/O is interleaved evenly between disks of equal speed and capacity. It is necessary to take into account whether the storage has multiple tiers. Furthermore, if there are multiple paths to storage, then consider the extent of link saturation when balancing how storage is used.

2.5. Network statistics reports: **sar -n** *KEYWORD*

The option **-n** lets you generate multiple network related reports. Specify one of the following keywords along with the **-n**:

- **DEV**: Generates a statistic report for all network devices

- *EDEV*: Generates an error statistics report for all network devices
- *NFS*: Generates a statistic report for an NFS client
- *NFSD*: Generates a statistic report for an NFS server
- *SOCK*: Generates a statistic report on sockets
- *ALL*: Generates all network statistic reports

2.2. Visualizing **sar** data

sar reports are not always easy to parse for humans. kSar, a Java application visualizing your **sar** data, creates easy-to-read graphs. It can even generate PDF reports. kSar takes data generated in real time, and past data from a file. kSar is licensed under the BSD license and is available from <https://sourceforge.net/projects/ksar/>.

2.2. System information

2.1. Device load information: **iostat**

To monitor the system device load, use **iostat**. It generates reports that can be useful for better balancing the load between physical disks attached to your system.

To be able to use **iostat**, install the package `sysstat`.

The first **iostat** report shows statistics collected since the system was booted. Subsequent reports cover the time since the previous report.

```
>iostat
Linux 6.4.0-150700.38-default (jupiter)      03/11/2024
_x86_64_      (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           17.68    4.49    4.24    0.29    0.00   73.31

Device:            tps    kB_read/s    kB_wrtn/s    kB_read  kB_wrtn
sdb                 2.02         36.74        45.73    3544894    4412392
sda                 1.05          5.12        13.47     493753    1300276
sdc                 0.02          0.14         0.00      13641         37
```

Invoking **iostat** in this way helps you find out whether throughput is different from your expectation, but not why. Such questions can be better answered by an extended report which can be generated by invoking **iostat -x**. Extended reports additionally include, for example, information on average queue sizes and average wait times. It may also be easier to evaluate the data if idle block devices are excluded using the `-z` switch. Find definitions for each of the displayed column titles in the man page of **iostat** (**man 1 iostat**).

You can also specify that a certain device should be monitored at specified intervals. For example, to generate five reports at three-second intervals for the device `sda`, use:

```
>iostat -p sda 3 5
```


To show statistics of network file systems (NFS), there are two similar utilities:

- **nfsiostat-sysstat** is included with the package `sysstat`.
- **nfsiostat** is included with the package `nfs-client`.



Using **iostat** in multipath setups

The **iostat** command might not show all controllers that are listed by **nvme list-subsys**. By default, **iostat** filters out all block devices with no I/O. To make **iostat** show *all* devices, use the following command:

```
> iostat -p ALL
```

2.2. Processor activity monitoring: **mpstat**

The utility **mpstat** examines activities of each available processor. If your system has one processor only, the global average statistics are reported.

The timing arguments work the same way as with the **iostat** command. Entering **mpstat 2 5** prints five reports for all processors in two-second intervals.

```
#mpstat 2 5
Linux 6.4.0-150700.38-default (jupiter)          03/11/2024
_x86_64_          (2 CPU)

13:51:10 CPU   %usr  %nice  %sys  %iowait  %irq  %soft  %steal  %guest
%gnice  %idle
13:51:12 all   8,27   0,00  0,50    0,00  0,00   0,00   0,00   0,00
0,00  91,23
13:51:14 all  46,62   0,00  3,01    0,00  0,00   0,25   0,00   0,00
0,00  50,13
13:51:16 all  54,71   0,00  3,82    0,00  0,00   0,51   0,00   0,00
0,00  40,97
13:51:18 all  78,77   0,00  5,12    0,00  0,00   0,77   0,00   0,00
0,00  15,35
13:51:20 all  51,65   0,00  4,30    0,00  0,00   0,51   0,00   0,00
0,00  43,54
Average: all  47,85   0,00  3,34    0,00  0,00   0,40   0,00   0,00
0,00  48,41
```

From the **mpstat** data, you can see:

- The ratio between the `%usr` and `%sys`. For example, a ratio of 10:1 indicates the workload is primarily running application code, and analysis should focus on the application. A ratio of 1:10 indicates the workload is primarily kernel-bound, and tuning the kernel is worth considering. Alternatively, determine why the application is kernel-bound and see if that can be alleviated.
- Whether there is a subset of CPUs that are nearly fully utilized even if the system is lightly loaded overall. Few hot CPUs can indicate that the workload is not parallelized and could benefit from executing on a machine with a smaller number of faster processors.

2.3. Processor frequency monitoring: **turbostat**

turbostat shows frequencies, load, temperature, and power of AMD64/Intel 64 processors. It can operate in two modes: If called with a command, the command process is forked and statistics are displayed upon command completion. When run without a command, it displays updated statistics every five seconds. **turbostat** requires the kernel module `msr` to be loaded.

```
>sudo turbostat find /etc -type d -exec true {} \;
0.546880 sec
  CPU Avg_MHz   Busy% Bzy_MHz TSC_MHz
    -      416   28.43   1465   3215
     0      631   37.29   1691   3215
     1      416   27.14   1534   3215
     2      270   24.30   1113   3215
     3      406   26.57   1530   3214
     4      505   32.46   1556   3214
     5      270   22.79   1184   3214
```

The output depends on the CPU type and may vary. To display more details such as temperature and power, use the `--debug` option. For more command line options and an explanation of the field descriptions, refer to **man 8 turbostat**.

2.4. Task monitoring: **pidstat**

If you need to see what load a particular task applies to your system, use **pidstat** command. It prints activity of every selected task or all tasks managed by Linux kernel if no task is specified. You can also set the number of reports to be displayed and the time interval between them.

For example, **pidstat -C firefox 2 3** prints the load statistic for tasks whose command name includes the string “firefox”. There are three reports printed at two second intervals.

```
#pidstat -C firefox 2 3
Linux 6.4.0-150700.38-default (jupiter)          03/11/2024
_x86_64_          (2 CPU)

14:09:11      UID      PID      %usr %system  %guest   %CPU   CPU   Command
14:09:13      1000      387      22,77    0,99    0,00   23,76    1   firefox

14:09:13      UID      PID      %usr %system  %guest   %CPU   CPU   Command
14:09:15      1000      387      46,50    3,00    0,00   49,50    1   firefox

14:09:15      UID      PID      %usr %system  %guest   %CPU   CPU   Command
14:09:17      1000      387      60,50    7,00    0,00   67,50    1   firefox

Average:      UID      PID      %usr %system  %guest   %CPU   CPU   Command
Average:      1000      387      43,19    3,65    0,00   46,84    -   firefox
```

Similarly, **pidstat -d** can be used to estimate how much I/O tasks are doing, whether they are sleeping on that I/O and how many clock ticks the task was stalled.

2.5. Kernel ring buffer: **dmesg**

The Linux kernel keeps certain messages in a ring buffer. To view these messages, enter the command **dmesg -T**.

Older events are logged in the systemd journal. See Chapter 21, **journalctl**: query the systemd journal in “[Administration Guide](#)” for more information on the journal.

2.6. List of open files: **lsuf**

To view a list of all the files open for the process with process ID *PID*, use `-p`. For example, to view all the files used by the current shell, enter:

```
#lsuf -p $$
COMMAND  PID USER  FD   TYPE DEVICE SIZE/OFF  NODE NAME
bash     8842 root   cwd   DIR   0,32    222   6772 /root
bash     8842 root   rtd   DIR   0,32    166    256 /
bash     8842 root   txt   REG   0,32  656584 31066 /bin/bash
bash     8842 root   mem   REG   0,32 1978832 22993 /lib64/libc-2.19.so
[...]
bash     8842 root    2u   CHR  136,2      0t0     5 /dev/pts/2
bash     8842 root   255u  CHR  136,2      0t0     5 /dev/pts/2
```

The special shell variable `$$`, whose value is the process ID of the shell, has been used.

When used with `-i`, **lsuf** lists currently open Internet files as well:

```
#lsuf -i
COMMAND  PID USER  FD   TYPE DEVICE SIZE/OFF  NODE NAME
wickedd-d  917 root    8u  IPv4  16627      0t0  UDP *:bootpc
wickedd-d  918 root    8u  IPv6  20752      0t0  UDP
[fe80::5054:ff:fe72:5ead]:dhcpv6-client
sshd     3152 root    3u  IPv4  18618      0t0  TCP *:ssh (LISTEN)
sshd     3152 root    4u  IPv6  18620      0t0  TCP *:ssh (LISTEN)
master   4746 root   13u  IPv4  20588      0t0  TCP localhost:smtp (LISTEN)
master   4746 root   14u  IPv6  20589      0t0  TCP localhost:smtp (LISTEN)
sshd     8837 root    5u  IPv4  293709     0t0  TCP jupiter.suse.de:ssh-
>venus.suse.de:33619 (ESTABLISHED)
sshd     8837 root    9u  IPv6  294830     0t0  TCP localhost:x11 (LISTEN)
sshd     8837 root   10u  IPv4  294831     0t0  TCP localhost:x11 (LISTEN)
```

2.7. Kernel and udev event sequence viewer: **udevadm monitor**

udevadm monitor listens to the kernel uevents and events sent out by a udev rule and prints the device path (DEVPATH) of the event to the console. This is a sequence of events while connecting a USB memory stick:



Monitoring udev events

Only root user is allowed to monitor udev events by running the **udevadm** command.

```

UEVENT[1138806687] add@/devices/pci0000:00/0000:00:1d.7/usb4/4-2/4-2.2
UEVENT[1138806687] add@/devices/pci0000:00/0000:00:1d.7/usb4/4-2/4-2.2/4-2.2
UEVENT[1138806687] add@/class/scsi_host/host4
UEVENT[1138806687] add@/class/usb_device/usbdev4.10
UDEV [1138806687] add@/devices/pci0000:00/0000:00:1d.7/usb4/4-2/4-2.2
UDEV [1138806687] add@/devices/pci0000:00/0000:00:1d.7/usb4/4-2/4-2.2/4-2.2
UDEV [1138806687] add@/class/scsi_host/host4
UDEV [1138806687] add@/class/usb_device/usbdev4.10
UEVENT[1138806692] add@/devices/pci0000:00/0000:00:1d.7/usb4/4-2/4-2.2/4-2.2
UEVENT[1138806692] add@/block/sdb
UEVENT[1138806692] add@/class/scsi_generic/sg1
UEVENT[1138806692] add@/class/scsi_device/4:0:0:0
UDEV [1138806693] add@/devices/pci0000:00/0000:00:1d.7/usb4/4-2/4-2.2/4-2.2
UDEV [1138806693] add@/class/scsi_generic/sg1
UDEV [1138806693] add@/class/scsi_device/4:0:0:0
UDEV [1138806693] add@/block/sdb
UEVENT[1138806694] add@/block/sdb/sdb1
UDEV [1138806694] add@/block/sdb/sdb1
UEVENT[1138806694] mount@/block/sdb/sdb1
UEVENT[1138806697] umount@/block/sdb/sdb1

```

2.3. Processes

2.1. Inter-process communication: **ipcs**

The command **ipcs** produces a list of the IPC resources currently in use:

```

#ipcs
----- Message Queues -----
key          msqid          owner          perms          used-bytes      messages

----- Shared Memory Segments -----
key          shmid          owner          perms          bytes           nattch          status
0x00000000   65536          tux            600            524288          2              dest
0x00000000   98305          tux            600            4194304         2              dest
0x00000000   884738         root           600            524288          2              dest
0x00000000   786435         tux            600            4194304         2              dest
0x00000000   12058628       tux            600            524288          2              dest
0x00000000   917509         root           600            524288          2              dest
0x00000000   12353542       tux            600            196608          2              dest
0x00000000   12451847       tux            600            524288          2              dest
0x00000000   11567114       root           600            262144          1              dest
0x00000000   10911763       tux            600            2097152         2              dest
0x00000000   11665429       root           600            2336768         2              dest
0x00000000   11698198       root           600            196608          2              dest
0x00000000   11730967       root           600            524288          2              dest

----- Semaphore Arrays -----
key          semid          owner          perms          nsems
0xa12e0919   32768          tux            666            2

```

2.2. Process list: **ps**

The command **ps** produces a list of processes. Most parameters must be written without a minus sign. Refer to **ps --help** for a brief help or to the man page for extensive help.

To list all processes with user and command line information, use **ps aux**:

```
>ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  34376  4608 ?        Ss   Jul24    0:02 /usr/lib/
systemd/systemd
root           2  0.0  0.0      0     0 ?        S    Jul24    0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    Jul24    0:00 [ksoftirqd/0]
root           5  0.0  0.0      0     0 ?        S<   Jul24    0:00 [kworker/0:0H]
root           6  0.0  0.0      0     0 ?        S    Jul24    0:00 [kworker/u2:0]
root           7  0.0  0.0      0     0 ?        S    Jul24    0:00 [migration/0]
[...]
tux          12583  0.0  0.1 185980  2720 ?        Sl   10:12    0:00 /usr/lib/gvfs/
gvfs-mtp-volume-monitor
tux          12587  0.0  0.1 198132  3044 ?        Sl   10:12    0:00 /usr/lib/gvfs/
gvfs-gphoto2-volume-monitor
tux          12591  0.0  0.1 181940  2700 ?        Sl   10:12    0:00 /usr/lib/gvfs/
gvfs-goa-volume-monitor
tux          12594  8.1 10.6 1418216 163564 ?      Sl   10:12    0:03 /usr/bin/gnome-
shell
tux          12600  0.0  0.3 393448  5972 ?        Sl   10:12    0:00 /usr/lib/gnome-
settings-daemon-3.0/gsd-printer
tux          12625  0.0  0.6 227776 10112 ?        Sl   10:12    0:00 /usr/lib/gnome-
control-center-search-provider
tux          12626  0.5  1.5 890972 23540 ?        Sl   10:12    0:00 /usr/bin/
nautilus --no-default-window
[...]

```

To check how many **sshd** processes are running, use the option **-p** together with the command **pidof**, which lists the process IDs of the given processes.

```
>ps -p $(pidof sshd)
  PID TTY      STAT   TIME COMMAND
 1545 ?        Ss      0:00 /usr/sbin/sshd -D
 4608 ?        Ss      0:00 sshd: root@pts/0

```

The process list can be formatted according to your needs. The option **L** returns a list of all keywords. Enter the following command to issue a list of all processes sorted by memory usage:

```
>ps ax --format pid,rss,cmd --sort rss
  PID  RSS  CMD
  PID  RSS  CMD
    2     0 [kthreadd]
    3     0 [ksoftirqd/0]
    4     0 [kworker/0:0]
    5     0 [kworker/0:0H]
    6     0 [kworker/u2:0]
    7     0 [migration/0]
    8     0 [rcu_bh]
[...]
12518 22996 /usr/lib/gnome-settings-daemon-3.0/gnome-settings-daemon
12626 23540 /usr/bin/nautilus --no-default-window
12305 32188 /usr/bin/Xorg :0 -background none -verbose
12594 164900 /usr/bin/gnome-shell

```

Useful **ps** calls

ps aux --sort COLUMN

Sort the output by **COLUMN**. Replace **COLUMN** with

pmem for physical memory ratio

pcpu for CPU ratio

rss for resident set size (non-swapped physical memory)

```
ps axo pid,%cpu,rss,vsz,args,wchan
```

Shows every process, their PID, CPU usage ratio, memory size (resident and virtual), name, and their syscall.

ps axfo pid,args

Show a process tree.

2.3. Process tree: pstree

The command **ps tree** produces a list of processes in the form of a tree:

[illegible]

The parameter `-p` adds the process ID to a given name. To have the command lines displayed as well, use the `-a` parameter:

2.4. Table of processes: top

The command **top** (an abbreviation of “table of processes”) displays a list of processes that is refreshed every two seconds. To stop the program, press **Q**. The parameter **-n 1** stops the program after a single display of the process list. The following is an example output of the command **top -n 1**:

```
>top -n 1
Tasks: 128 total, 1 running, 127 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.4 us, 1.2 sy, 0.0 ni, 96.3 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem: 1535508 total, 699948 used, 835560 free, 880 buffers
KiB Swap: 1541116 total, 0 used, 1541116 free. 377000 cached Mem

  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0 116292   4660   2028 S   0.000 0.303   0:04.45 systemd
    2 root        20   0     0      0      0 S   0.000 0.000   0:00.00 kthreadd
    3 root        20   0     0      0      0 S   0.000 0.000   0:00.07 ksoftirqd+
    5 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 kworker/0+
    6 root        20   0     0      0      0 S   0.000 0.000   0:00.00 kworker/u+
    7 root        rt    0     0      0      0 S   0.000 0.000   0:00.00 migration+
    8 root        20   0     0      0      0 S   0.000 0.000   0:00.00 rcu_bh
    9 root        20   0     0      0      0 S   0.000 0.000   0:00.24 rcu_sched
   10 root        rt    0     0      0      0 S   0.000 0.000   0:00.01 watchdog/0
   11 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 khelper
   12 root        20   0     0      0      0 S   0.000 0.000   0:00.00 kdevtmpfs
   13 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 netns
   14 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 writeback
   15 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 kintegrit+
   16 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 bioset
   17 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 crypto
   18 root         0 -20     0      0      0 S   0.000 0.000   0:00.00 kblockd
```

By default the output is sorted by CPU usage (column *%CPU*, shortcut **Shift-P**). Use the following key combinations to change the sort field:

Shift-M : Resident Memory (*RES*)

Shift-N : Process ID (*PID*)

Shift-T : Time (*TIME+*)

To use any other field for sorting, press **F** and select a field from the list. To toggle the sort order, Use **Shift-R**.

The parameter **-U *UID*** monitors only the processes associated with a particular user. Replace *UID* with the user ID of the user. Use **top -U \$(id -u)** to show processes of the current user

2.5. IBM Z hypervisor monitor: **hyptop**

hyptop provides a dynamic real-time view of an IBM Z hypervisor environment, using the kernel infrastructure through debugfs. It works with either the z/VM or the LPAR hypervisor. Depending on the available data it, for example, shows CPU and memory consumption of active LPARs or z/VM guests. It provides a curses based user interface similar to the **top** command. **hyptop** provides two windows:

- **sys_list**: Lists systems that the current hypervisor is running
- **sys**: Shows one system in more detail

You can run **hyptop** in interactive mode (default) or in batch mode with the **-b** option. Help in the interactive mode is available by pressing **?** after **hyptop** is started.

Output for the **sys_list** window under LPAR:

```

12:30:48 | CPU-T: IFL(18) CP(3) UN(3)           ?=help
system  #cpu    cpu    mgm    Cpu+   Mgm+   online
(str)   (#)    (%)    (%)    (hm)   (hm)   (dhm)
H05LP30  10  461.14  10.18  1547:41  8:15  11:05:59
H05LP33   4  133.73   7.57   220:53   6:12  11:05:54
H05LP50   4   99.26   0.01   146:24   0:12  10:04:24
H05LP02   1   99.09   0.00   269:57   0:00  11:05:58
TRX2CFA   1    2.14   0.03    3:24   0:04  11:06:01
H05LP13   6    1.36   0.34    4:23   0:54  11:05:56
TRX1      19    1.22   0.14   13:57   0:22  11:06:01
TRX2      20    1.16   0.11   26:05   0:25  11:06:00
H05LP55   2    0.00   0.00    0:22   0:00  11:05:52
H05LP56   3    0.00   0.00    0:00   0:00  11:05:52
          413  823.39  23.86  3159:57  38:08  11:06:01

```

Output for the "sys_list" window under z/VM:

```

12:32:21 | CPU-T: UN(16)                       ?=help
system  #cpu    cpu    Cpu+   online  memuse  memmax  wcur
(str)   (#)    (%)    (hm)   (dhm)   (GiB)   (GiB)   (#)
T6360004  6  100.31  959:47  53:05:20  1.56    2.00    100
T6360005  2   0.44   1:11   3:02:26  0.42    0.50    100
T6360014  2   0.27   0:45  10:18:41  0.54    0.75    100
DTCVSW1   1   0.00   0:00  53:16:42  0.01    0.03    100
T6360002  6   0.00  166:26  40:19:18  1.87    2.00    100
OPERATOR   1   0.00   0:00  53:16:42  0.00    0.03    100
T6360008  2   0.00   0:37  30:22:55  0.32    0.75    100
T6360003  6   0.00  3700:57  53:03:09  4.00    4.00    100
NSLCF1     1   0.00   0:02  53:16:41  0.03    0.25    500
EREP       1   0.00   0:00  53:16:42  0.00    0.03    100
PERFSVM    1   0.00   0:53   2:21:12  0.04    0.06     0
TCPIP      1   0.00   0:01  53:16:42  0.01    0.12   3000
DATAMOVE   1   0.00   0:05  53:16:42  0.00    0.03    100
DIRMAINT   1   0.00   0:04  53:16:42  0.01    0.03    100
DTCVSW2    1   0.00   0:00  53:16:42  0.01    0.03    100
RACFVM     1   0.00   0:00  53:16:42  0.01    0.02    100
          75  101.57  5239:47  53:16:42  15.46   22.50   3000

```

Output for the sys window under LPAR:

```

14:08:41 | H05LP30 | CPU-T: IFL(18) CP(3) UN(3)           ? = help
cpuid    type    cpu    mgm  visual.
(#)      (str)   (%)    (%)  (vis)
0         IFL  96.91  1.96  #####
1         IFL  81.82  1.46  #####
2         IFL  88.00  2.43  #####
3         IFL  92.27  1.29  #####
4         IFL  83.32  1.05  #####
5         IFL  92.46  2.59  #####
6         IFL   0.00  0.00  |
7         IFL   0.00  0.00  |
8         IFL   0.00  0.00  |
9         IFL   0.00  0.00  |
          534.79  10.78

```

Output for the sys window under z/VM:

```

15:46:57 | T6360003 | CPU-T: UN(16)                       ? = help
cpuid    cpu  visual
(#)      (%)  (vis)
0        548.72 |#####|
          548.72

```


2.6. A top-like I/O monitor: **iostat**

The **iostat** utility displays a table of I/O usage by processes or threads.



Installing **iostat**

iostat is not installed by default. You need to install it manually with **zypper** in **iostat** as root.

iostat displays columns for the I/O bandwidth read and written by each process during the sampling period. It also displays the percentage of time the process spent while swapping in and while waiting on I/O. For each process, its I/O priority (class/level) is shown. In addition, the total I/O bandwidth read and written during the sampling period is displayed at the top of the interface.

- The **←** and **→** keys change the sorting.
- **R** reverses the sort order.
- **O** toggles between showing all processes and threads (default view) and showing only those doing I/O. (This function is similar to adding **--only** on command line.)
- **P** toggles between showing threads (default view) and processes. (This function is similar to **--only**.)
- **A** toggles between showing the current I/O bandwidth (default view) and accumulated I/O operations since **iostat** was started. (This function is similar to **--accumulated**.)
- **I** lets you change the priority of a thread or a process's threads.
- **Q** quits **iostat**.
- Pressing any other key forces a refresh.

Following is an example output of the command **iostat --only**, while **find** and **emacs** are running:

```
#iostat --only
Total DISK READ: 50.61 K/s | Total DISK WRITE: 11.68 K/s
  TID  PRI0  USER      DISK READ  DISK WRITE  SWAPIN      IO>    COMMAND
 3416 be/4  tux        50.61 K/s    0.00 B/s    0.00 %    4.05 % find /
 275  be/3  root        0.00 B/s    3.89 K/s    0.00 %    2.34 % [jbd2/sda2-8]
 5055 be/4  tux         0.00 B/s    3.89 K/s    0.00 %    0.04 % emacs
```

iostat can be also used in a batch mode (**-b**) and its output stored in a file for later analysis. For a complete set of options, see the manual page (**man 8 iostat**).

2.7. Modify a process's niceness: **nice** and **renice**

The kernel determines which processes require more CPU time than others by the process's nice level, also called niceness. The higher the “nice” level of a process is, the less CPU time it takes

from other processes. Nice levels range from -20 (the least “nice” level) to 19. Negative values can only be set by root.

Adjusting the niceness level is useful when running a non time-critical process that lasts long and uses large amounts of CPU time. For example, compiling a kernel on a system that also performs other tasks. Making such a process “nicer” ensures that the other tasks, for example, a Web server, have a higher priority.

Calling **nice** without any parameters prints the current niceness:

```
>nice
0
```

Running **nice***COMMAND* increments the current nice level for the given command by 10. Using **nice** -n *LEVELCOMMAND* lets you specify a new niceness relative to the current one.

To change the niceness of a running process, use **renice***PRIORITY*-p *PROCESS_ID*, for example:

```
>renice +5 3266
```

To restore the niceness of all processes owned by a specific user, use the option -u *USER*. Process groups are reniced by the option -g *PROCESS_GROUP_ID*.

2.4. Memory

2.1. Memory usage: **free**

The utility **free** examines RAM and swap usage. Details of both free and used memory and swap areas are shown:

```
>free
              total        used        free      shared    buffers     cached
Mem:      32900500    32703448    197052         0      255668     5787364
-/+ buffers/cache:    26660416    6240084
Swap:      2046972       304680    1742292
```

The options -b, -k, -m, -g show the output in bytes, KB, MB, or GB, respectively. The parameter -s *delay* ensures that the display is refreshed every *DELAY* seconds. For example, **free -s 1.5** produces an update every 1.5 seconds.

2.2. Detailed memory usage: **/proc/meminfo**

Use **/proc/meminfo** to get more detailed information on memory usage than with **free**. In fact, **free** uses a certain amount of data from this file. See an example output from a 64-bit system below, which differs slightly on 32-bit systems because of different memory management:

```

MemTotal:      1942636 kB
MemFree:       1294352 kB
MemAvailable:  1458744 kB
Buffers:       876 kB
Cached:        278476 kB
SwapCached:    0 kB
Active:        368328 kB
Inactive:      199368 kB
Active(anon):  288968 kB
Inactive(anon): 10568 kB
Active(file):  79360 kB
Inactive(file): 188800 kB
Unevictable:   80 kB
Mlocked:       80 kB
SwapTotal:     2103292 kB
SwapFree:      2103292 kB
Dirty:         44 kB
Writeback:     0 kB
AnonPages:     288592 kB
Mapped:        70444 kB
Shmem:         11192 kB
Slab:          40916 kB
SReclaimable:  17712 kB
SUnreclaim:    23204 kB
KernelStack:   2000 kB
PageTables:    10996 kB
NFS_Unstable:  0 kB
Bounce:        0 kB
WritebackTmp:  0 kB
CommitLimit:   3074608 kB
Committed_AS:  1407208 kB
VmallocTotal:  34359738367 kB
VmallocUsed:    145996 kB
VmallocChunk:   34359588844 kB
HardwareCorrupted: 0 kB
AnonHugePages: 86016 kB
HugePages_Total: 0
HugePages_Free: 0
HugePages_Rsvd: 0
HugePages_Surp: 0
Hugepagesize:  2048 kB
DirectMap4k:   79744 kB
DirectMap2M:   2017280 kB

```

These entries stand for the following:

MemTotal

Total amount of RAM.

MemFree

Amount of unused RAM.

MemAvailable

Estimate of how much memory is available for starting new applications without swapping.

Buffers

File buffer cache in RAM containing file system metadata.

Cached

Page cache in RAM. This excludes buffer cache and swap cache, but includes *Shmem* memory.

SwapCached

Page cache for swapped-out memory.

Active, Active(anon), Active(file)

Recently used memory that is not reclaimed unless necessary or on explicit request. *Active* is the sum of *Active(anon)* and *Active(file)*:

- *Active(anon)* tracks swap-backed memory. This includes private and shared anonymous mappings and private file pages after copy-on-write.
- *Active(file)* tracks other file system backed memory.

Inactive, Inactive(anon), Inactive(file)

Less recently used memory that is generally reclaimed first. *Inactive* is the sum of *Inactive(anon)* and *Inactive(file)*:

- *Inactive(anon)* tracks swap backed memory. This includes private and shared anonymous mappings and private file pages after copy-on-write.
- *Inactive(file)* tracks other file system backed memory.

Unevictable

Amount of memory that cannot be reclaimed (for example, because it is *Mlocked* or used as a RAM disk).

Mlocked

Amount of memory that is backed by the `mlock` system call. `mlock` allows processes to define which part of physical RAM their virtual memory should be mapped to. However, `mlock` does not guarantee this placement.

SwapTotal

Amount of swap space.

SwapFree

Amount of unused swap space.

Dirty

Amount of memory waiting to be written to disk, because it contains changes compared to the backing storage. Dirty data can be explicitly synchronized either by the application or by the kernel after a short delay. A large amount of dirty data may take considerable time to write to disk resulting in stalls. The total amount of dirty data that can exist at any time can be controlled with the **sysctl** parameters `vm.dirty_ratio` or `vm.dirty_bytes` (refer to the section called “Writeback” for more details).

Writeback

Amount of memory that is currently being written to disk.

Mapped

Memory claimed with the `mmap` system call.

Shmem

Memory shared between groups of processes, such as IPC data, tmpfs data, and shared anonymous memory.

Slab

Memory allocation for internal data structures of the kernel.

SReclaimable

Slab section that can be reclaimed, such as caches (inode, dentry, etc.).

SUnreclaim

Slab section that cannot be reclaimed.

KernelStack

Amount of kernel space memory used by applications (through system calls).

PageTables

Amount of memory dedicated to page tables of all processes.

NFS_Unstable

NFS pages that have already been sent to the server, but are not yet committed there.

Bounce

Memory used for bounce buffers of block devices.

WritebackTmp

Memory used by FUSE for temporary writeback buffers.

CommitLimit

Amount of memory available to the system based on the overcommit ratio setting. This is only enforced if strict overcommit accounting is enabled.

Committed_AS

An approximation of the total amount of memory (RAM and swap) that the current workload would need in the worst case.

VmallocTotal

Amount of allocated kernel virtual address space.

VmallocUsed

Amount of used kernel virtual address space.

VmallocChunk

The largest contiguous block of available kernel virtual address space.

HardwareCorrupted

Amount of failed memory (can only be detected when using ECC RAM).

AnonHugePages

Anonymous hugepages that are mapped into user space page tables. These are allocated transparently for processes without being specifically requested, therefore they are also known as *transparent hugepages* (THP).

HugePages_Total

Number of preallocated hugepages for use by SHM_HUGETLB and MAP_HUGETLB or through the `hugetlbfs` file system, as defined in `/proc/sys/vm/nr_hugepages`.

HugePages_Free

Number of hugepages available.

HugePages_Rsvd

Number of hugepages that are committed.

HugePages_Surp

Number of hugepages available beyond *HugePages_Total* (“surplus”), as defined in `/proc/sys/vm/nr_overcommit_hugepages`.

Hugepagesize

Size of a hugepage—on AMD64/Intel 64 the default is 2048 KB.

DirectMap4k etc.

Amount of kernel memory that is mapped to pages with a given size (in the example: 4 kB).

2.3. Process memory usage: smaps

Exactly determining how much memory a certain process is consuming is not possible with standard tools like **top** or **ps**. Use the smaps subsystem, introduced in kernel 2.6.14, if you need exact data. It can be found at `/proc/PID/smaps` and shows you the number of clean and dirty memory pages the process with the ID *PID* is using at that time. It differentiates between shared and private memory, so you can see how much memory the process is using without including memory shared with other processes. For more information see `/usr/src/linux/Documentation/filesystems/proc.txt` (requires the package `kernel-source` to be installed).

smaps is expensive to read. Therefore it is not recommended to monitor it regularly, but only when closely monitoring a certain process.

2.4. numaTOP

numaTOP is a tool for NUMA (Non-uniform Memory Access) systems. The tool helps to identify NUMA-related performance bottlenecks by providing real-time analysis of a NUMA system.

Generally speaking, numaTOP allows you to identify and investigate processes and threads with poor locality (that is poor ratio of local versus remote memory usage) by analyzing the number of Remote Memory Accesses (RMA), the number of Local Memory Accesses (LMA), and the RMA/LMA ratio.

numaTOP is supported on PowerPC and the following Intel Xeon processors: 5500-series, 6500/7500-series, 5600-series, E7-x8xx-series, and E5-16xx/24xx/26xx/46xx-series.

numaTOP is available in the official software repositories, and you can install the tool using the **sudo zypper in numatop** command. To launch numaTOP, run the **numatop** command. To get an overview of numaTOP functionality and usage, use the **man numatop** command.

2.5. Networking



Traffic shaping

In case the network bandwidth is lower than expected, you should first check if any traffic shaping rules are active for your network segment.

2.1. Basic network diagnostics: **ip**

ip is a powerful tool to set up and control network interfaces. You can also use it to quickly view basic statistics about network interfaces of the system. For example, whether the interface is up or how many errors, dropped packets, or packet collisions there are.

If you run **ip** with no additional parameter, it displays a help output. To list all network interfaces, enter **ip addr show** (or abbreviated as **ip a**). **ip addr show up** lists only running network interfaces. **ip -s link show DEVICE** lists statistics for the specified interface only:

```
#ip -s link show br0
6: br0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode
DEFAULT
    link/ether 00:19:d1:72:d4:30 brd ff:ff:ff:ff:ff:ff
    RX: bytes  packets  errors  dropped overrun mcast
    6346104756 9265517    0      10860    0        0
    TX: bytes  packets  errors  dropped carrier collsns
    3996204683 3655523    0        0        0        0
```

ip can also show interfaces (link), routing tables (route), and much more—refer to **man 8 ip** for details.

```
#ip route
default via 192.168.2.1 dev eth1
192.168.2.0/24 dev eth0  proto kernel  scope link  src 192.168.2.100
192.168.2.0/24 dev eth1  proto kernel  scope link  src 192.168.2.101
192.168.2.0/24 dev eth2  proto kernel  scope link  src 192.168.2.102
```

```
#ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
mode DEFAULT group default qlen 1000
    link/ether 52:54:00:44:30:51 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
mode DEFAULT group default qlen 1000
    link/ether 52:54:00:a3:c1:fb brd ff:ff:ff:ff:ff:ff
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
mode DEFAULT group default qlen 1000
    link/ether 52:54:00:32:a4:09 brd ff:ff:ff:ff:ff:ff
```

2.2. Show the network usage of processes: **nethogs**

In certain cases, for example, if the network traffic suddenly becomes too high, it is desirable to quickly find out which applications are causing the traffic. **nethogs**, a tool with a design similar to **top**, shows incoming and outgoing traffic for all relevant processes:

PID	USER	PROGRAM	DEV	SENT	RECEIVED
27145	root	zypper	eth0	5.719	391.749 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30015		0.102	2.326 KB/sec
26635	tux	/usr/lib64/firefox/firefox	eth0	0.026	0.026 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30045		0.000	0.021 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30045		0.000	0.018 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30015		0.000	0.018 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30045		0.000	0.017 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30045		0.000	0.017 KB/sec
?	root	..0:113:80c0:8080:10:160:0:100:30045		0.069	0.000 KB/sec
?	root	unknown TCP		0.000	0.000 KB/sec
TOTAL				5.916	394.192 KB/sec

Like in **top**, **nethogs** features interactive commands:

M: cycle between display modes (kb/s, kb, b, mb)

R: sort by *RECEIVED*

S: sort by *SENT*

Q: quit

2.3. Ethernet cards in detail: ethtool

ethtool can display and change detailed aspects of your Ethernet network device. By default it prints the current setting of the specified device.

```
#ethtool eth0
Settings for eth0:
Supported ports: [ TP ]
Supported link modes:   10baseT/Half 10baseT/Full
                        100baseT/Half 100baseT/Full
                        1000baseT/Full
Supports auto-negotiation: Yes
Advertised link modes:  10baseT/Half 10baseT/Full
                        100baseT/Half 100baseT/Full
                        1000baseT/Full
Advertised pause frame use: No
[...]
Link detected: yes
```

The following table shows **ethtool** options that you can use to query the device for specific information:

Table 2.1. List of query options of ethtool

ethtool option	it queries the device for
-a	pause parameter information
-c	interrupt coalescing information
-g	Rx/Tx (receive/transmit) ring parameter information

ethtool option	it queries the device for
-i	associated driver information
-k	offload information
-S	NIC and driver-specific statistics

2.4. Show the network status: ss

ss is a tool to dump socket statistics and replaces the **netstat** command. To list all connections use **ss** without parameters:

```
#ss
Netid  State      Recv-Q Send-Q   Local Address:Port      Peer Address:Port
u_str  ESTAB      0      0           * 14082                * 14083
u_str  ESTAB      0      0           * 18582                * 18583
u_str  ESTAB      0      0           * 19449                * 19450
u_str  ESTAB      0      0   @/tmp/dbus-gmUUwXABPV 18784      * 18783
u_str  ESTAB      0      0   /var/run/dbus/system bus socket 19383 * 19382
u_str  ESTAB      0      0   @/tmp/dbus-gmUUwXABPV 18617      * 18616
u_str  ESTAB      0      0   @/tmp/dbus-58TPDv8qv 19352      * 19351
u_str  ESTAB      0      0           * 17658                * 17657
u_str  ESTAB      0      0           * 17693                * 17694
[...]
```

To show all network ports currently open, use the following command:

```
#ss -t
Netid  State      Recv-Q Send-Q   Local Address:Port      Peer Address:Port
nl      UNCONN     0      0           rtnl:4195117            *
nl      UNCONN     0      0   rtnl:wickedd-auto4/811  *
nl      UNCONN     0      0   rtnl:wickedd-dhcp4/813  *
nl      UNCONN     0      0           rtnl:4195121            *
nl      UNCONN     0      0           rtnl:4195115            *
nl      UNCONN     0      0   rtnl:wickedd-dhcp6/814  *
nl      UNCONN     0      0           rtnl:kernel             *
nl      UNCONN     0      0   rtnl:wickedd/817        *
nl      UNCONN     0      0           rtnl:4195118            *
nl      UNCONN     0      0   rtnl:nsd/706            *
nl      UNCONN     4352   0      tcpdiag:ss/2381         *
[...]
```

When displaying network connections, you can specify the socket type to display: TCP (-t) or UDP (-u) for example. The -p option shows the PID and name of the program to which each socket belongs.

The following example lists all TCP connections and the programs using these connections. The -a option make sure all established connections (listening and non-listening) are shown. The -p option shows the PID and name of the program to which each socket belongs.

```
#ss -t -a -p
State Recv-Q Send-Q Local Address:Port Peer Address:Port
LISTEN 0 128 *:ssh *:users:(("sshd",
1551,3))
LISTEN 0 100 127.0.0.1:smtp *:users:
(("master",1704,13))
ESTAB 0 132 10.120.65.198:ssh 10.120.4.150:55715 users:(("sshd",
2103,5))
LISTEN 0 128 :::ssh :::users:(("sshd",
1551,4))
LISTEN 0 100 ::1:smtp :::users:
(("master",1704,14))
```

2.6. The /proc file system

The /proc file system is a pseudo file system in which the kernel reserves important information in the form of virtual files. For example, display the CPU type with this command:

```
>cat /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 30
model name     : Intel(R) Core(TM) i5 CPU           750  @ 2.67GHz
stepping       : 5
microcode      : 0x6
cpu MHz        : 1197.000
cache size     : 8192 KB
physical id    : 0
siblings       : 4
core id        : 0
cpu cores      : 4
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 11
wp             : yes
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx rdtscp lm
constant_tsc arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc
aperfperf_pni dtes64 monitor ds_cpl vmx smx est tm2 ssse3 cx16 xtpr pdcm sse4_1
sse4_2 popcnt lahf_lm ida dtherm tpr_shadow vnmi flexpriority ept vpid
bogomips      : 5333.85
clflush size   : 64
cache_alignment : 64
address sizes  : 36 bits physical, 48 bits virtual
power management:
[...]
```



Detailed processor information

Detailed information about the processor on the AMD64/Intel 64 architecture is also available by running **x86info**.

Query the allocation and use of interrupts with the following command:

```
>cat /proc/interrupts
```

	CPU0	CPU1	CPU2	CPU3		
0:	121	0	0	0	IO-APIC-edge	timer
8:	0	0	0	1	IO-APIC-edge	rtc0
9:	0	0	0	0	IO-APIC-fasteoi	acpi
16:	0	11933	0	0	IO-APIC-fasteoi	ehci_hcd:+
18:	0	0	0	0	IO-APIC-fasteoi	i801_smbus
19:	0	117978	0	0	IO-APIC-fasteoi	ata_piix,+
22:	0	0	3275185	0	IO-APIC-fasteoi	enp5s1
23:	417927	0	0	0	IO-APIC-fasteoi	ehci_hcd:+
40:	2727916	0	0	0	HPET_MSI-edge	hpet2
41:	0	2749134	0	0	HPET_MSI-edge	hpet3
42:	0	0	2759148	0	HPET_MSI-edge	hpet4
43:	0	0	0	2678206	HPET_MSI-edge	hpet5
45:	0	0	0	0	PCI-MSI-edge	aerdrv, P+
46:	0	0	0	0	PCI-MSI-edge	PCIE PME,+
47:	0	0	0	0	PCI-MSI-edge	PCIE PME,+
48:	0	0	0	0	PCI-MSI-edge	PCIE PME,+
49:	0	0	0	387	PCI-MSI-edge	snd_hda_i+
50:	933117	0	0	0	PCI-MSI-edge	nvidia_
NMI:	2102	2023	2031	1920	Non-maskable interrupts	
LOC:	92	71	57	41	Local timer interrupts	
SPU:	0	0	0	0	Spurious interrupts	
PMI:	2102	2023	2031	1920	Performance monitoring int+	
IWI:	47331	45725	52464	46775	IRQ work interrupts	
RTR:	2	0	0	0	APIC ICR read retries	
RES:	472911	396463	339792	323820	Rescheduling interrupts	
CAL:	48389	47345	54113	50478	Function call interrupts	
TLB:	28410	26804	24389	26157	TLB shutdowns	
TRM:	0	0	0	0	Thermal event interrupts	
THR:	0	0	0	0	Threshold APIC interrupts	
MCE:	0	0	0	0	Machine check exceptions	
MCP:	40	40	40	40	Machine check polls	
ERR:	0					
MIS:	0					

The address assignment of executables and libraries is contained in the maps file:

```
>cat /proc/self/maps
```

08048000-0804c000	r-xp	00000000	03:03	17753	/bin/cat
0804c000-0804d000	rw-p	00004000	03:03	17753	/bin/cat
0804d000-0806e000	rw-p	0804d000	00:00	0	[heap]
b7d27000-b7d5a000	r--p	00000000	03:03	11867	/usr/lib/locale/en_GB.utf8/
b7d5a000-b7e32000	r--p	00000000	03:03	11868	/usr/lib/locale/en_GB.utf8/
b7e32000-b7e33000	rw-p	b7e32000	00:00	0	
b7e33000-b7f45000	r-xp	00000000	03:03	8837	/lib/libc-2.3.6.so
b7f45000-b7f46000	r--p	00112000	03:03	8837	/lib/libc-2.3.6.so
b7f46000-b7f48000	rw-p	00113000	03:03	8837	/lib/libc-2.3.6.so
b7f48000-b7f4c000	rw-p	b7f48000	00:00	0	
b7f52000-b7f53000	r--p	00000000	03:03	11842	/usr/lib/locale/en_GB.utf8/
[...]					
b7f5b000-b7f61000	r--s	00000000	03:03	9109	/usr/lib/gconv/gconv-module
b7f61000-b7f62000	r--p	00000000	03:03	9720	/usr/lib/locale/en_GB.utf8/
b7f62000-b7f76000	r-xp	00000000	03:03	8828	/lib/ld-2.3.6.so
b7f76000-b7f78000	rw-p	00013000	03:03	8828	/lib/ld-2.3.6.so
bfd61000-bfd76000	rw-p	bfd61000	00:00	0	[stack]
ffffe000-fffff000	---p	00000000	00:00	0	[vdso]

A lot more information can be obtained from the /proc file system. Some important files and their contents are:

/proc/devices

Available devices

/proc/modules

Kernel modules loaded

/proc/cmdline

Kernel command line

/proc/meminfo

Detailed information about memory usage

/proc/config.gz

gzip-compressed configuration file of the kernel currently running

/proc/PID/

Find information about processes currently running in the `/proc/NNN` directories, where `NNN` is the process ID (PID) of the relevant process. Every process can find its own characteristics in `/proc/self/`.

Further information is available in the text file `/usr/src/linux/Documentation/filesystems/proc.txt` (this file is available when the package `kernel-source` is installed).

2.1. procinfo

Important information from the `/proc` file system is summarized by the command **procinfo**:

```

>procinfo
Linux 3.11.10-17-desktop (geeko@buildhost) (gcc 4.8.1 20130909) #1 4CPU
[jupiter.example.com]

Memory:      Total      Used      Free      Shared      Buffers      Cached
Mem:         8181908    8000632    181276      0         85472      2850872
Swap:        10481660    1576      10480084

Bootup: Mon Jul 28 09:54:13 2014      Load average: 1.61 0.85 0.74 2/904 25949

user  :      1:54:41.84  12.7%  page in :      2107312  disk 1:      52212r
20199w
nice  :      0:00:00.46   0.0%  page out:      1714461  disk 2:      19387r
10928w
system:      0:25:38.00   2.8%  page act:      466673  disk 3:       548r
10w
IOwait:      0:04:16.45   0.4%  page dea:      272297
hw irq:      0:00:00.42   0.0%  page flt:    105754526
sw irq:      0:01:26.48   0.1%  swap in :           0
idle  :     12:14:43.65  81.5%  swap out:       394
guest  :      0:02:18.59   0.2%
uptime:      3:45:22.24      context :    99809844

irq 0:      121 timer                irq 41:    3238224 hpet3
irq 8:         1 rtc0                irq 42:    3251898 hpet4
irq 9:         0 acpi                irq 43:    3156368 hpet5
irq 16:    14589 ehci_hcd:usb1        irq 45:         0 aerdrv, PCIe PME
irq 18:         0 i801_smbus          irq 46:         0 PCIe PME, pciehp
irq 19:    124861 ata_piix, ata_piix, f irq 47:         0 PCIe PME, pciehp
irq 22:    3742817 enp5s1             irq 48:         0 PCIe PME, pciehp
irq 23:    479248 ehci_hcd:usb2        irq 49:       387 snd_hda_intel
irq 40:    3216894 hpet2              irq 50:    1088673 nvidia

```

To see all the information, use the parameter `-a`. The parameter `-nN` produces updates of the information every *N* seconds. In this case, stop the program by pressing `Q`.

By default, the cumulative values are displayed. The parameter `-d` produces the differential values. **procinfo -dn5** displays the values that have changed in the last five seconds:

2.2. System control parameters: `/proc/sys/`

System control parameters are used to modify the Linux kernel parameters at runtime. They reside in `/proc/sys/` and can be viewed and modified with the **sysctl** command. To list all parameters, run **sysctl -a**. A single parameter can be listed with **sysctl***PARAMETER_NAME*.

Parameters are grouped into categories and can be listed with **sysctl***CATEGORY* or by listing the contents of the respective directories. The most important categories are listed below. The links to further readings require the installation of the package `kernel-source`.

sysctl dev (`/proc/sys/dev/`)

Device-specific information.

sysctl fs (`/proc/sys/fs/`)

Used file handles, quotas, and other file system-oriented parameters. For details see `/usr/src/linux/Documentation/sysctl/fs.txt`.

sysctl kernel (/proc/sys/kernel/)

Information about the task scheduler, system shared memory, and other kernel-related parameters. For details see `/usr/src/linux/Documentation/sysctl/kernel.txt`

sysctl net (/proc/sys/net/)

Information about network bridges, and general network parameters (mainly the `ipv4/` subdirectory). For details see `/usr/src/linux/Documentation/sysctl/net.txt`

sysctl vm (/proc/sys/vm/)

Entries in this path relate to information about the virtual memory, swapping, and caching. For details see `/usr/src/linux/Documentation/sysctl/vm.txt`

To set or change a parameter for the current session, use the command **sysctl-w**`PARAMETER=VALUE`. To permanently change a setting, add a line `PARAMETER=VALUE` to `/etc/sysctl.conf`.

2.7. Hardware information

2.1. PCI resources: `lspci`



Accessing PCI configuration.

Most operating systems require root user privileges to grant access to the computer's PCI configuration.

The command **lspci** lists the PCI resources:

```
#lspci
00:00.0 Host bridge: Intel Corporation 82845G/GL[Brookdale-G]/GE/PE \
    DRAM Controller/Host-Hub Interface (rev 01)
00:01.0 PCI bridge: Intel Corporation 82845G/GL[Brookdale-G]/GE/PE \
    Host-to-AGP Bridge (rev 01)
00:1d.0 USB Controller: Intel Corporation 82801DB/DBL/DBM \
    (ICH4/ICH4-L/ICH4-M) USB UHCI Controller #1 (rev 01)
00:1d.1 USB Controller: Intel Corporation 82801DB/DBL/DBM \
    (ICH4/ICH4-L/ICH4-M) USB UHCI Controller #2 (rev 01)
00:1d.2 USB Controller: Intel Corporation 82801DB/DBL/DBM \
    (ICH4/ICH4-L/ICH4-M) USB UHCI Controller #3 (rev 01)
00:1d.7 USB Controller: Intel Corporation 82801DB/DBM \
    (ICH4/ICH4-M) USB2 EHCI Controller (rev 01)
00:1e.0 PCI bridge: Intel Corporation 82801 PCI Bridge (rev 81)
00:1f.0 ISA bridge: Intel Corporation 82801DB/DBL (ICH4/ICH4-L) \
    LPC Interface Bridge (rev 01)
00:1f.1 IDE interface: Intel Corporation 82801DB (ICH4) IDE \
    Controller (rev 01)
00:1f.3 SMBus: Intel Corporation 82801DB/DBL/DBM (ICH4/ICH4-L/ICH4-M) \
    SMBus Controller (rev 01)
00:1f.5 Multimedia audio controller: Intel Corporation 82801DB/DBL/DBM \
    (ICH4/ICH4-L/ICH4-M) AC'97 Audio Controller (rev 01)
01:00.0 VGA compatible controller: Matrox Graphics, Inc. G400/G450 (rev 85)
02:08.0 Ethernet controller: Intel Corporation 82801DB PRO/100 VE (LOM) \
    Ethernet Controller (rev 81)
```

Using `-v` results in a more detailed listing:

```
#lspci -v
[...]
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet \
Controller (rev 02)
    Subsystem: Intel Corporation PRO/1000 MT Desktop Adapter
    Flags: bus master, 66MHz, medium devsel, latency 64, IRQ 19
    Memory at f0000000 (32-bit, non-prefetchable) [size=128K]
    I/O ports at d010 [size=8]
    Capabilities: [dc] Power Management version 2
    Capabilities: [e4] PCI-X non-bridge device
    Kernel driver in use: e1000
    Kernel modules: e1000
```

Information about device name resolution is obtained from the file `/usr/share/pci.ids`. PCI IDs not listed in this file are marked “Unknown device.”

The parameter `-vv` produces all the information that could be queried by the program. To view the pure numeric values, use the parameter `-n`.

2.2. USB devices: `lsusb`

The command `lsusb` lists all USB devices. With the option `-v`, print a more detailed list. The detailed information is read from the directory `/proc/bus/usb/`. The following is the output of `lsusb` with these USB devices attached: hub, memory stick, hard disk and mouse.


```
#lsusb
Bus 004 Device 007: ID 0ea0:2168 Ours Technology, Inc. Transcend JetFlash \
2.0 / Astone USB Drive
Bus 004 Device 006: ID 04b4:6830 Cypress Semiconductor Corp. USB-2.0 IDE \
Adapter
Bus 004 Device 005: ID 05e3:0605 Genesys Logic, Inc.
Bus 004 Device 001: ID 0000:0000
Bus 003 Device 001: ID 0000:0000
Bus 002 Device 001: ID 0000:0000
Bus 001 Device 005: ID 046d:c012 Logitech, Inc. Optical Mouse
Bus 001 Device 001: ID 0000:0000
```

2.3. Monitoring and tuning the thermal subsystem: **tmon**

tmon is a tool to help visualize, tune, and test the complex thermal subsystem. When started without parameters, **tmon** runs in monitoring mode:

```

┌─── THERMAL ZONES (SENSORS) ───┐
Thermal Zones:                  acpitz00
Trip Points:                    PC
└────────────────────────────────┘

┌─── COOLING DEVICES ───┐
ID  Cooling Dev  Cur  Max  Thermal Zone Binding
00   Processor    0    3
01   Processor    0    3
02   Processor    0    3
03   Processor    0    3
04 intel_powerc  -1   50
└────────────────────────────────┘

acpitz 0:[ 8][>>>>>>>>>P9
          10      20      30      40
          C31

┌─── CONTROLS ───┐
PID gain: kp=0.36 ki=5.00 kd=0.19 Output 0.00
Target Temp: 65.0C, Zone: 0, Control Device: None
└────────────────────────────────┘

Ctrl-c - Quit  TAB - Tuning

```

For detailed information on how to interpret the data, how to log thermal data and how to use **tmon** to test and tune cooling devices and sensors, refer to the man page: **man 8 tmon**. The package **tmon** is not installed by default.

2.4. MCELog: machine check exceptions (MCE)



Availability

This tool is only available on AMD64/Intel 64 systems.

The **mcelog** package logs and parses/translates Machine Check Exceptions (MCE) on hardware errors, including I/O, CPU, and memory errors. Additionally, **mcelog** handles predictive bad page offlining and automatic core offlining when cache errors happen. Formerly this was managed by a cron job executed hourly. Now hardware errors are immediately processed by an **mcelog** daemon.



Support for AMD scalable MCA

SUSE Linux Enterprise Server supports AMD's Scalable Machine Check Architecture (Scalable MCA). Scalable MCA improves hardware error reporting in AMD Zen processors. It expands information logged in MCA banks for improved error handling and better diagnosability.

mcelog captures MCA messages (rasdaemon and dmesg also capture MCA messages). See section 3.1, *Machine Check Architecture of Processor Programming Reference (PPR) for AMD Family 17h Model 01h, Revision B1 Processors* for detailed information, https://developer.amd.com/wordpress/media/2017/11/54945_PPR_Family_17h_Models_00h-0Fh.pdf.

mcelog is configured in `/etc/mcelog/mcelog.conf`. Configuration options are documented in **man mcelog**, and at <https://mcelog.org/>. The following example shows only changes to the default file:

```
daemon = yes
filter = yes
filter-memory-errors = yes
no-syslog = yes
logfile = /var/log/mcelog
run-credentials-user = root
run-credentials-group = nobody
client-group = root
socket-path = /var/run/mcelog-client
```

The mcelog service is not enabled by default. The service can either be enabled and started via the YaST system services editor, or via command line:

```
#systemctl enable mcelog
#systemctl start mcelog
```

2.5. AMD64/Intel 64: dmidecode: DMI table decoder

dmidecode shows the machine's DMI table containing information such as serial numbers and BIOS revisions of the hardware.

```
#dmidecode
# dmidecode 2.12
SMBIOS 2.5 present.
27 structures occupying 1298 bytes.
Table at 0x000EB250.

Handle 0x0000, DMI type 4, 35 bytes
Processor Information
    Socket Designation: J1PR
    Type: Central Processor
    Family: Other
    Manufacturer: Intel(R) Corporation
    ID: E5 06 01 00 FF FB EB BF
    Version: Intel(R) Core(TM) i5 CPU           750  @ 2.67GHz
    Voltage: 1.1 V
    External Clock: 133 MHz
    Max Speed: 4000 MHz
    Current Speed: 2667 MHz
    Status: Populated, Enabled
    Upgrade: Other
    L1 Cache Handle: 0x0004
    L2 Cache Handle: 0x0003
    L3 Cache Handle: 0x0001
    Serial Number: Not Specified
    Asset Tag: Not Specified
    Part Number: Not Specified

[...]
```

2.6. POWER: list hardware

lshw extracts and displays the hardware configuration of the machine.

2.8. Files and file systems

For file system-specific information, refer to Storage Administration Guide in “[Storage Administration Guide](#)”.

2.1. Determine the file type: **file**

The command **file** determines the type of a file or a list of files by checking `/usr/share/misc/magic`.

```
>file /usr/bin/file
/usr/bin/file: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), \
    for GNU/Linux 2.6.4, dynamically linked (uses shared libs), stripped
```

The parameter `-f LIST` specifies a file with a list of file names to examine. The `-z` allows **file** to look inside compressed files:

```
>file /usr/share/man/man1/file.1.gz
/usr/share/man/man1/file.1.gz: gzip compressed data, from Unix, max compression
>file -z /usr/share/man/man1/file.1.gz
/usr/share/man/man1/file.1.gz: troff or preprocessor input text \
    (gzip compressed data, from Unix, max compression)
```

The parameter `-i` outputs a mime type string rather than the traditional description.

```
>file -i /usr/share/misc/magic
/usr/share/misc/magic: text/plain charset=utf-8
```

2.2. File systems and their usage: **mount**, **df** and **du**

The command **mount** shows which file system (device and type) is mounted at which mount point:

```
#mount
/dev/sda2 on / type ext4 (rw,acl,user_xattr)
proc on /proc type proc (rw)
sysfs on /sys type sysfs (rw)
debugfs on /sys/kernel/debug type debugfs (rw)
devtmpfs on /dev type devtmpfs (rw,mode=0755)
tmpfs on /dev/shm type tmpfs (rw,mode=1777)
devpts on /dev/pts type devpts (rw,mode=0620,gid=5)
/dev/sda3 on /home type ext3 (rw)
securityfs on /sys/kernel/security type securityfs (rw)
fusectl on /sys/fs/fuse/connections type fusectl (rw)
gvfs-fuse-daemon on /home/tux/.gvfs type fuse.gvfs-fuse-daemon \
(rw,nosuid,nodev,user=tux)
```

Obtain information about total usage of the file systems with the command **df**. The parameter **-h** (or **--human-readable**) transforms the output into a form understandable for common users.

```
>df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        20G   5,9G   13G   32% /
devtmpfs         1,6G   236K   1,6G    1% /dev
tmpfs            1,6G   668K   1,6G    1% /dev/shm
/dev/sda3       208G    40G   159G   20% /home
```

Display the total size of all the files in a given directory and its subdirectories with the command **du**. The parameter **-s** suppresses the output of detailed information and gives only a total for each argument. **-h** again transforms the output into a human-readable form:

```
>du -sh /opt
192M    /opt
```

2.3. Additional information about ELF binaries

Read the content of binaries with the **readelf** utility. This even works with ELF files that were built for other hardware architectures:

```
>readelf --file-header /bin/ls
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00
  Class:                                ELF64
  Data:                                      2's complement, little endian
  Version:                               1 (current)
  OS/ABI:                                UNIX - System V
  ABI Version:                           0
  Type:                                  EXEC (Executable file)
  Machine:                               Advanced Micro Devices X86-64
  Version:                               0x1
  Entry point address:                   0x402540
  Start of program headers:              64 (bytes into file)
  Start of section headers:              95720 (bytes into file)
  Flags:                                  0x0
  Size of this header:                   64 (bytes)
  Size of program headers:               56 (bytes)
  Number of program headers:              9
  Size of section headers:               64 (bytes)
  Number of section headers:              32
  Section header string table index:     31
```

2.4. File properties: **stat**

The command **stat** displays file properties:

```
>stat /etc/profile
  File: '/etc/profile'
  Size: 9662          Blocks: 24          IO Block: 4096   regular file
Device: 802h/2050d   Inode: 132349       Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2009-03-20 07:51:17.000000000 +0100
Modify: 2009-01-08 19:21:14.000000000 +0100
Change: 2009-03-18 12:55:31.000000000 +0100
```

The parameter `--file-system` produces details of the properties of the file system in which the specified file is located:

```
>stat /etc/profile --file-system
  File: "/etc/profile"
   ID: d4fb76e70b4d1746 Namelen: 255      Type: ext2/ext3
Block size: 4096      Fundamental block size: 4096
Blocks: Total: 2581445   Free: 1717327   Available: 1586197
Inodes: Total: 655776    Free: 490312
```

2.9. User information

2.1. User accessing files: **fuser**

It can be useful to determine what processes or users are currently accessing certain files. Suppose, for example, you want to unmount a file system mounted at `/mnt`. **umount** returns "device is busy." The command **fuser** can then be used to determine what processes are accessing the device:

```
>fuser -v /mnt/*

      USER      PID ACCESS COMMAND
/mnt/notes.txt  tux      26597 f....  less
```

Following termination of the **less** process, which was running on another terminal, the file system can successfully be unmounted. When used with **-k** option, **fuser** stops processes accessing the file as well.

2.2. Who is doing what: **w**

With the command **w**, find out who is logged in to the system and what each user is doing. For example:

```
>w
 16:00:59 up 1 day,  2:41,  3 users,  load average: 0.00, 0.01, 0.05
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU WHAT
tux       :0        console         Wed13    ?xdm?  8:15   0.03s /usr/lib/gdm/gd
tux       console  :0              Wed13    26:41m 0.00s  0.03s /usr/lib/gdm/gd
tux       pts/0    :0              Wed13    20:11  0.10s  2.89s /usr/lib/gnome-
```

If any users of other systems have logged in remotely, the parameter **-f** shows the computers from which they have established the connection.

2.10. Time and date

2.1. Time measurement with **time**

Determine the time spent by commands with the **time** utility. This utility is available in two versions: as a Bash built-in and as a program (**/usr/bin/time**).

```
>time find . > /dev/null

real    0m4.051s❶
user    0m0.042s❷
sys     0m0.205s❸
```

- ❶ The real time that elapsed from the command's start-up until it finished.
- ❷ CPU time of the user as reported by the **times** system call.
- ❸ CPU time of the system as reported by the **times** system call.

The output of **/usr/bin/time** is much more detailed. It is recommended to run it with the **-v** switch to produce human-readable output.

```

/usr/bin/time -v find . > /dev/null
Command being timed: "find ."
User time (seconds): 0.24
System time (seconds): 2.08
Percent of CPU this job got: 25%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:09.03
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 2516
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 0
Minor (reclaiming a frame) page faults: 1564
Voluntary context switches: 36660
Involuntary context switches: 496
Swaps: 0
File system inputs: 0
File system outputs: 0
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0

```

2.11. Graph your data: RRDtool

There are a lot of data in the world around you, which can be easily measured in time. For example, changes in the temperature, or the number of data sent or received by your computer's network interface. RRDtool can help you store and visualize such data in detailed and customizable graphs.

RRDtool is available for most Unix platforms and Linux distributions. SUSE® Linux Enterprise Server ships RRDtool as well. Install it either with YaST or by entering

zypperinstallrrdtool in the command line as root.



Bindings

There are Perl, Python, Ruby, and PHP bindings available for RRDtool, so that you can write your own monitoring scripts in your preferred scripting language.

2.1. How RRDtool works

RRDtool is an abbreviation of *Round Robin Database tool*. *Round Robin* is a method for manipulating with a constant amount of data. It uses the principle of a circular buffer, where there is no end nor beginning to the data row which is being read. RRDtool uses Round Robin Databases to store and read its data.

As mentioned above, RRDtool is designed to work with data that change in time. The ideal case is a sensor which repeatedly reads measured data (like temperature, speed etc.) in constant periods

of time, and then exports them in a given format. Such data is perfectly ready for RRDtool, and it is easy to process them and create the desired output.

Sometimes it is not possible to obtain the data automatically and regularly. Their format needs to be pre-processed before it is supplied to RRDtool, and often you need to manipulate RRDtool even manually.

The following is a simple example of basic RRDtool usage. It illustrates all three important phases of the usual RRDtool workflow: *creating* a database, *updating* measured values, and *viewing* the output.

2.2. A practical example

Suppose we want to collect and view information about the memory usage in the Linux system as it changes in time. To make the example more vivid, we measure the currently free memory over a period of 40 seconds in 4-second intervals. Three applications that generally consume a lot of system memory are started and closed: the Firefox Web browser, the Evolution e-mail client, and the Eclipse development framework.

2.1. Collecting data

RRDtool is frequently used to measure and visualize network traffic. In such case, the Simple Network Management Protocol (SNMP) is used. This protocol can query network devices for relevant values of their internal counters. Exactly these values are to be stored with RRDtool. For more information on SNMP, see <http://www.net-snmp.org/>.

Our situation is different—we need to obtain the data manually. A helper script **free_mem.sh** repetitively reads the current state of free memory and writes it to the standard output.

```
>cat free_mem.sh
INTERVAL=4
for steps in {1..10}
do
    DATE=`date +%s`
    FREEMEM=`free -b | grep "Mem" | awk '{ print $4 }'`
    sleep $INTERVAL
    echo "rrdtool update free_mem.rrd $DATE:$FREEMEM"
done
```

- The time interval is set to 4 seconds, and is implemented with the **sleep** command.
- RRDtool accepts time information in a special format - so called *Unix time*. It is defined as the number of seconds since the midnight of January 1, 1970 (UTC). For example, 1272907114 represents 2010-05-03 17:18:34.
- The free memory information is reported in bytes with **free-b**. Prefer to supply basic units (bytes) instead of multiple units (like kilobytes).
- The line with the **echo ...** command contains the future name of the database file (`free_mem.rrd`), and together creates a command line for updating RRDtool values.

After running **free_mem.sh**, you see an output similar to this:

```
>sh free_mem.sh
rrdtool update free_mem.rrd 1272974835:1182994432
rrdtool update free_mem.rrd 1272974839:1162817536
rrdtool update free_mem.rrd 1272974843:1096269824
rrdtool update free_mem.rrd 1272974847:1034219520
rrdtool update free_mem.rrd 1272974851:909438976
rrdtool update free_mem.rrd 1272974855:832454656
rrdtool update free_mem.rrd 1272974859:829120512
rrdtool update free_mem.rrd 1272974863:1180377088
rrdtool update free_mem.rrd 1272974867:1179369472
rrdtool update free_mem.rrd 1272974871:1181806592
```

It is convenient to redirect the command's output to a file with

```
sh free_mem.sh > free_mem_updates.log
```

to simplify its future execution.

2.2. Creating the database

Create the initial Robin Round database for our example with the following command:

```
>rrdtool create free_mem.rrd --start 1272974834 --step=4 \
DS:memory:GAUGE:600:U:U RRA:AVERAGE:0.5:1:24
```

Points to notice

- This command creates a file called `free_mem.rrd` for storing our measured values in a Round Robin type database.
- The `--start` option specifies the time (in Unix time) when the first value is added to the database. In this example, it is one less than the first time value of the **free_mem.sh** output (1272974835).
- The `--step` specifies the time interval in seconds with which the measured data is supplied to the database.
- The `DS:memory:GAUGE:600:U:U` part introduces a new data source for the database. It is called *memory*, its type is *gauge*, the maximum number between two updates is 600 seconds, and the *minimal* and *maximal* value in the measured range are unknown (U).
- `RRA:AVERAGE:0.5:1:24` creates Round Robin archive (RRA) whose stored data is processed with the *consolidation functions* (CF) that calculates the *average* of data points. 3 arguments of the consolidation function are appended to the end of the line.

If no error message is displayed, then `free_mem.rrd` database is created in the current directory:

```
>ls -l free_mem.rrd
-rw-r--r-- 1 tux users 776 May  5 12:50 free_mem.rrd
```

2.3. Updating database values

After the database is created, you need to fill it with the measured data. In *the section called “Collecting data”*, we already prepared the file `free_mem_updates.log` which consists of **rrdtool update** commands. These commands do the update of database values for us.

```
>sh free_mem_updates.log; ls -l free_mem.rrd
-rw-r--r-- 1 tux users 776 May 5 13:29 free_mem.rrd
```

As you can see, the size of `free_mem.rrd` remained the same even after updating its data.

2.4. Viewing measured values

We have already measured the values, created the database, and stored the measured value in it. Now we can play with the database, and retrieve or view its values.

To retrieve all the values from our database, enter the following on the command line:

```
>rrdtool fetch free_mem.rrd AVERAGE --start 1272974830 \
--end 1272974871
memory
1272974832: nan
1272974836: 1.1729059840e+09
1272974840: 1.1461806080e+09
1272974844: 1.0807572480e+09
1272974848: 1.0030243840e+09
1272974852: 8.9019289600e+08
1272974856: 8.3162112000e+08
1272974860: 9.1693465600e+08
1272974864: 1.1801251840e+09
1272974868: 1.1799787520e+09
1272974872: nan
```

Points to notice

- AVERAGE fetches the average value points from the database, because only one data source is defined (*the section called “Creating the database”*) with AVERAGE processing and no other function is available.
- The first line of the output prints the name of the data source as defined in *the section called “Creating the database”*.
- The left results column represents individual points in time, while the right one represents corresponding measured average values in scientific notation.
- The nan in the last line stands for “not a number”.

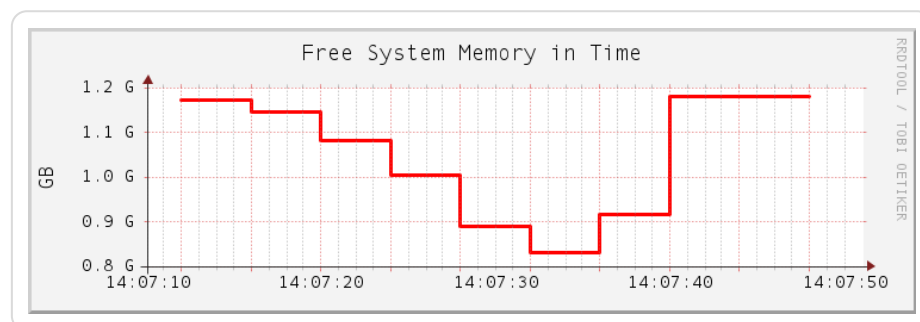
Now a graph representing the values stored in the database is drawn:

```
>rrdtool graph free_mem.png \
--start 1272974830 \
--end 1272974871 \
--step=4 \
DEF:free_memory=free_mem.rrd:memory:AVERAGE \
LINE2:free_memory#FF0000 \
--vertical-label "GB" \
--title "Free System Memory in Time" \
--zoom 1.5 \
--x-grid SECOND:1:SECOND:4:SECOND:10:0:%X
```

Points to notice

- `free_mem.png` is the file name of the graph to be created.
- `--start` and `--end` limit the time range within which the graph is drawn.
- `--step` specifies the time resolution (in seconds) of the graph.
- The `DEF:...` part is a data definition called *free_memory*. Its data is read from the `free_mem.rrd` database and its data source called *memory*. The *average* value points are calculated, because no others were defined in *the section called "Creating the database"*.
- The `LINE...` part specifies properties of the line to be drawn into the graph. It is 2 pixels wide, its data come from the *free_memory* definition, and its color is red.
- `--vertical-label` sets the label to be printed along the y axis, and `--title` sets the main label for the whole graph.
- `--zoom` specifies the zoom factor for the graph. This value must be greater than zero.
- `--x-grid` specifies how to draw grid lines and their labels into the graph. Our example places them every second, while major grid lines are placed every 4 seconds. Labels are placed every 10 seconds under the major grid lines.

Figure 2.1. Example graph created with RRDtool



2.3. More information

RRDtool is a complex tool with a lot of subcommands and command-line options. Some are easy to understand, but to make them produce the results you want and fine-tune them according to your liking may require a lot of effort.

Apart from RRDtool's man page (**man 1 rrdtool**) which gives you only basic information, you should have a look at the [RRDtool home page](#). There is a detailed [documentation](#) of the **rrdtool**

command and all its sub-commands. There are also several [tutorials](#) to help you understand the common RRDtool workflow.

If you are interested in monitoring network traffic, have a look at [MRTG \(Multi Router Traffic Grapher\)](#). MRTG can graph the activity of many network devices. It can use RRDtool.

Chapter 3. System log files

System log file analysis is one of the most important tasks when analyzing the system. In fact, looking at the system log files should be the first thing to do when maintaining or troubleshooting a system. SUSE Linux Enterprise Server automatically logs almost everything that happens on the system in detail. Since the move to `systemd`, kernel messages and messages of system services registered with `systemd` are logged in `systemd` journal (see Chapter 21, **journalctl**: query the `systemd` journal in “[Administration Guide](#)”). Other log files (mainly those of system applications) are written in plain text and can be easily read using an editor or pager. It is also possible to parse them using scripts. This allows you to filter their content.

3.1. System log files in `/var/log/`

System log files are always located under the `/var/log` directory. The following list presents an overview of all system log files from SUSE Linux Enterprise Server present after a default installation. Depending on your installation scope, `/var/log` also contains log files from other services and applications not listed here. Some files and directories described below are “placeholders” and are only used, when the corresponding application is installed. Most log files are only visible for the user `root`.

apparmor/

AppArmor log files. For more information about AppArmor, see Part V, “Confining privileges with AppArmor” in “[Security and Hardening Guide](#)”.

audit/

Logs from the audit framework. See Part VII, “The Linux Audit Framework” in “[Security and Hardening Guide](#)” for details.

ConsoleKit/

Logs of the ConsoleKit daemon (daemon for tracking what users are logged in and how they interact with the computer).

cups/

Access and error logs of the Common Unix Printing System (cups).

firewall

Firewall logs.

gdm/

Log files from the GNOME display manager.

krb5/

Log files from the Kerberos network authentication system.

lastlog

A database containing information on the last login of each user. Use the command **lastlog** to view. See **man 8 lastlog** for more information.

localmessages

Log messages of certain boot scripts, for example, the log of the DHCP client.

mail*

Mail server (postfix, sendmail) logs.

messages

This is the default place where all kernel and system log messages go and should be the first place (along with `/var/log/warn`) to look at in case of problems.

NetworkManager

NetworkManager log files.

news/

Log messages from a news server.

chrony/

Logs from the Network Time Protocol daemon (chrony).

pk_backend_zypp*

PackageKit (with libzypp back-end) log files.

samba/

Log files from Samba, the Windows SMB/CIFS file server.

warn

Log of all system warnings and errors. This should be the first place (along with the output of the `systemd journal`) to look in case of problems.

wtmp

Database of all login/logout activities, and remote connections. Use the command **last** to view. See **man 1 last** for more information.

Xorg.NUMBER.log

X.Org start-up log file. Refer to these files in case you have problems starting X.Org.

The *NUMBER* in the file name is the display number. For example, the default `Xorg.0.log` is the log for display number 0, and `Xorg.1.log` is the log for display number 1. Copies from previous X.Org starts are named as `Xorg.NUMBER.log.old`.



Note

The X.Org log files are available in the `/var/log/` directory only if you start an X.Org session as `root`. If you start an X.Org session as any other user, you can locate the log files in the `~/.local/share/xorg/` directory.

YaST2/

All YaST log files.

zypp/

libzypp log files. Refer to these files for the package installation history.

zypper.log

Logs from the command line installer **zypper**.

3.2. Viewing and parsing log files

To view log files, you can use any text editor. There is also a simple YaST module for viewing the system log available in the YaST control center under *Miscellaneous > System Log*.

For viewing log files in a text console, use the commands **less** or **more**. Use **head** and **tail** to view the beginning or end of a log file. To view entries appended to a log file in real-time use **tail -f**. For information about how to use these tools, see their man pages.

To search for strings or regular expressions in log files use **grep**. **awk** is useful for parsing and rewriting log files.

3.3. Managing log files with logrotate

Log files under `/var/log` grow on a daily basis and quickly become large. **logrotate** is a tool that helps you manage log files and their growth. It allows automatic rotation, removal,

compression, and mailing of log files. Log files can be handled periodically (daily, weekly, or monthly) or when exceeding a particular size.

logrotate is run daily by **systemd**, and thus modifies log files only once a day. However, exceptions occur when a log file is modified because of its size, if **logrotate** is run multiple times a day, or if `--force` is enabled. Use `/var/lib/misc/logrotate.status` to find out when a particular file was last rotated.

The main configuration file of **logrotate** is `/etc/logrotate.conf`. System packages and programs that produce log files (for example, `apache2`) put their own configuration files in the `/etc/logrotate.d/` directory. The content of `/etc/logrotate.d/` is included via `/etc/logrotate.conf`.

Example 3.1. Example for `/etc/logrotate.conf`

```
# see "man logrotate" for details
# rotate log files weekly
weekly

# keep 4 weeks worth of backlogs
rotate 4

# create new (empty) log files after rotating old ones
create

# use date as a suffix of the rotated file
dateext

# uncomment this if you want your log files compressed
#compress

# comment these to switch compression to use gzip or another
# compression scheme
compresscmd /usr/bin/bzip2
uncompresscmd /usr/bin/bunzip2

# RPM packages drop log rotation information into this directory
include /etc/logrotate.d
```

Avoid permission conflicts



The `create` option pays heed to the modes and ownership of files specified in `/etc/permissions*`. If you modify these settings, make sure no conflicts arise.

3.4. Monitoring log files with **logwatch**

logwatch is a customizable, pluggable log-monitoring script. It parses system logs, extracts the important information and presents them in a human readable manner. To use **logwatch**, install the **logwatch** package.

logwatch can either be used at the command line to generate on-the-fly reports, or via `cron` to regularly create custom reports. Reports can either be printed on the screen, saved to a file, or be

mailed to a specified address. The latter is especially useful when automatically generating reports via `cron`.

On the command line, you can tell **logwatch** for which service and time span to generate a report and how much detail should be included:

```
# Detailed report on all kernel messages from yesterday
logwatch --service kernel --detail High --range Yesterday --print

# Low detail report on all sshd events recorded (incl. archived logs)
logwatch --service sshd --detail Low --range All --archives --print

# Mail a report on all smartd messages from May 5th to May 7th to root@localhost
logwatch --service smartd --range 'between 5/5/2005 and 5/7/2005' \
--mailto root@localhost --print
```

The `--range` option has got a complex syntax—see **logwatch --range help** for details. A list of all services that can be queried is available with the following command:

```
>ls /usr/share/logwatch/default.conf/services/ | sed 's/\.conf//g'
```

logwatch can be customized to great detail. However, the default configuration should be sufficient. The default configuration files are located under `/usr/share/logwatch/default.conf/`. Never change them because they would get overwritten again with the next update. Rather place custom configuration in `/etc/logwatch/conf/` (you may use the default configuration file as a template, though). A detailed HOWTO on customizing **logwatch** is available at `/usr/share/doc/packages/logwatch/HOWTO-Customize-LogWatch`. The following configuration files exist:

logwatch.conf

The main configuration file. The default version is extensively commented. Each configuration option can be overwritten on the command line.

ignore.conf

Filter for all lines that should globally be ignored by **logwatch**.

services/*.conf

The service directory holds configuration files for each service you can generate a report for.

logfiles/*.conf

Specifications on which log files should be parsed for each service.

3.5. Configuring mail forwarding for root

System daemons, cron jobs, systemd timers, and other applications can generate messages and send them to the `root` user of the system. By default, each user account owns a local mailbox and will be notified about new mail messages upon login.

These messages can contain security relevant reports and incidents that require a quick response by the system administrator. To get notified about these messages in a timely fashion, it is strongly recommended to forward these mails to a dedicated remote e-mail account that is regularly checked.

Procedure 3.1. Configure mail forwarding for the **root** user

To forward mail for the **root** user, perform the following steps:

1. Install the `yast2-mail` package:

```
#zypper in yast2-mail
```

2. Run the interactive YaST mail configuration:

```
#yast mail
```

3. Choose *Permanent* as *Connection type* and proceed with *Next*.
4. Enter the address of the *Outgoing mail server*. If necessary, configure *Authentication*. It is strongly recommended to *Enforce TLS encryption* to prevent potentially sensitive system data from being sent unencrypted over the network. Proceed with *Next*.
5. Enter the e-mail address to *Forward root's mail to* and *Finish* the configuration.

Do not accept remote SMTP connections



Do *not* enable *Accept remote SMTP connections*, otherwise the local machine will act as a mail relay.

6. Send a message to test whether mail forwarding works correctly:

```
>mail root
subject: test
test
.
```

7. Use the **mailq** command to verify that the test message has been sent. Upon success, the queue should be empty. The message should be received by the dedicated mail address configured previously.

Depending on the number of managed machines and the number of persons who need to be informed about system events, different e-mail address models can be established:

- Collect messages from different systems in an e-mail account that is only accessed by a single person.
- Collect messages from different systems in a group e-mail account (aliases or mailing list) that can be accessed by all relevant persons.
- Create separate e-mail accounts for each system.

It is crucial that administrators regularly check the related e-mail accounts. To facilitate this effort and identify important events, avoid sending unnecessary information. Configure applications to only send relevant information.

3.6. Forwarding log messages to a central syslog server

System log data can be forwarded from individual systems to a central syslog server on the network. This allows administrators to get an overview of events on all hosts, and prevents attackers that succeed in taking over a system from manipulating system logs to cover their tracks.

Setting up a central syslog server consists of two parts. First you configure the central log server, then the clients for remote logging.

3.1. Set up the central syslog server

Procedure 3.2. Configure the central rsyslog server

To set up a central syslog server, perform the following steps:

1. Edit the configuration file `/etc/rsyslog.d/remote.conf`.
2. Uncomment the following lines in the UDP Syslog Server or TCP Syslog Server section of the configuration file. Assign an IP address and port for `rsyslogd`.

TCP example:

```
$ModLoad imtcp.so
$UDPServerAddress IP❶
$InputTCPServerRun PORT❷
```

UDP example:

```
$ModLoad imudp.so
$UDPServerAddress IP❶
$UDPServerRun PORT❷
```

- ❶ IP address of the interface for `rsyslogd` to listen on. If no address is given, the daemon listens on all interfaces.
- ❷ Port for `rsyslogd` to listen on. Select a privileged port below 1024. The default is 514.

TCP versus UDP protocol



Traditionally syslog uses the UDP protocol to transmit log messages over the network. This involves less overhead, but lacks reliability. Log messages can get lost under high load.

The TCP protocol is more reliable and should be preferred over UDP.



UDPServerAddress with TCP

The `$UDPServerAddress` configuration parameter in the TCP example is no error. Despite its name it is used for both TCP and UDP.

3. Save the file.
4. Restart the `rsyslog` service:

```
>sudo systemctl restart rsyslog.service
```

5. Open the respective port in the firewall. For `firewalld` with TCP on port 514 run:

```
>sudo firewall-cmd --add-port 514/tcp --permanent>sudo firewall-cmd --reload
```

You have now configured the central syslog server. Next, configure clients for remote logging.

3.2. Set up the client machines

Procedure 3.3. Configure a *rsyslog* instance for remote logging

To configure a machine for remote logging to a central syslog server, perform the following steps:

1. Edit the configuration file `/etc/rsyslog.d/remote.conf`.
2. Uncomment the appropriate line (TCP or UDP) and replace `remote-host` with the address of the central syslog server set up in *the section called "Set up the central syslog server"*.

TCP example:

```
# Remote Logging using TCP for reliable delivery
# remote host is: name/ip:port, e.g. 192.168.0.1:514, port optional
*. * @remote-host
```

UDP example:

```
# Remote Logging using UDP
# remote host is: name/ip:port, e.g. 192.168.0.1:514, port optional
*. * @remote-host
```

3. Save the file.
4. Restart the `rsyslog` service:

```
>sudo systemctl restart rsyslog.service
```

5. Verify the proper function of the syslog forwarding:

```
>logger "hello world"
```

The log message `hello world` should now appear on the central syslog server.

You have now configured a system for remote logging to your central syslog server. Repeat this procedure for all systems that should log remotely.

3.3. More information

This basic setup does not include encryption and is only suitable for trusted internal networks. TLS encryption is strongly recommended, but requires a certificate infrastructure.

In this configuration, all messages from remote hosts are treated the same on the central syslog server. Consider filtering messages into separate files by remote host or classify them by message category.

For more information about encryption, filtering, and other advanced topics, consult the RSyslog documentation at <https://www.rsyslog.com/doc/master/index.html#manual>.

3.7. Using **logger** to make system log entries

logger is a tool for making entries in the system log. It provides a shell command interface to the rsyslogd system log module. For example, the following line outputs its message in `/var/log/` messages or directly in the journal (if no logging facility is running):

```
>logger -t Test "This message comes from $USER"
```

Depending on the current user and host name, the log contains a line similar to this:

```
Sep 28 13:09:31 venus Test: This message comes from tux
```

Part III. Kernel monitoring

- 4 **SystemTap—filtering and analyzing system data 62**
- 5 **Kernel probes 75**
- 6 **Hardware-based performance monitoring with Perf 79**
- 7 **OProfile—system-wide profiler 83**
- 8 **Dynamic debug—kernel debugging messages 88**

Chapter 4. SystemTap—filtering and analyzing system data

SystemTap provides a command line interface and a scripting language to examine the activities of a running Linux system, particularly the kernel, in fine detail. SystemTap scripts are written in the SystemTap scripting language, are then compiled to C-code kernel modules and inserted into the kernel. The scripts can be designed to extract, filter and summarize data, thus allowing the diagnosis of complex performance problems or functional problems. SystemTap provides information similar to the output of tools like **netstat**, **ps**, **top**, and **iostat**. However, more filtering and analysis options can be used for the collected information.

4.1. Conceptual overview

Each time you run a SystemTap script, a SystemTap session is started. Several passes are done on the script before it is allowed to run. Then, the script is compiled into a kernel module and loaded. If the script has been executed before and no system components have changed (for example, different compiler or kernel versions, library paths, or script contents), SystemTap does not compile the script again. Instead, it uses the *.c and *.ko data stored in the SystemTap cache (~/.systemtap).

The module is unloaded when the tap has finished running. For an example, see the test run in *the section called “Installation and setup”* and the respective explanation.

4.1. SystemTap scripts

SystemTap usage is based on SystemTap scripts (*.stp). They tell SystemTap which type of information to collect, and what to do once that information is collected. The scripts are written in the SystemTap scripting language that is similar to AWK and C. For the language definition, see <https://sourceware.org/systemtap/langref.pdf>. A lot of useful example scripts are available from <https://www.sourceware.org/systemtap/examples/>.

The essential idea behind a SystemTap script is to name events, and to give them handlers. When SystemTap runs the script, it monitors for certain events. When an event occurs, the Linux kernel runs the handler as a sub-routine, then resumes. Thus, events serve as the triggers for handlers to run. Handlers can record specified data and print it in a certain manner.

The SystemTap language only uses a few data types (integers, strings, and associative arrays of these), and full control structures (blocks, conditionals, loops, functions). It has a lightweight punctuation (semicolons are optional) and does not need detailed declarations (types are inferred and checked automatically).

For more information about SystemTap scripts and their syntax, refer to *the section called “Script syntax”* and to the **stapprobes** and **stapfuncs** man pages, that are available with the systemtap-docs package.

4.2. Tapsets

Tapsets are a library of pre-written probes and functions that can be used in SystemTap scripts. When a user runs a SystemTap script, SystemTap checks the script's probe events and handlers against the tapset library. SystemTap then loads the corresponding probes and functions before translating the script to C. Like SystemTap scripts themselves, tapsets use the file name extension `*.stp`.

However, unlike SystemTap scripts, tapsets are not meant for direct execution. They constitute the library from which other scripts can pull definitions. Thus, the tapset library is an abstraction layer designed to make it easier for users to define events and functions. Tapsets provide aliases for functions that users could want to specify as an event. Knowing the proper alias is often easier than remembering specific kernel functions that may vary between kernel versions.

4.3. Commands and privileges

The main commands associated with SystemTap are **stap** and **staprun**. To execute them, you either need root privileges or must be a member of the `stapdev` or `stapusr` group.

stap

SystemTap front-end. Runs a SystemTap script (either from file, or from standard input). It translates the script into C code, compiles it, and loads the resulting kernel module into a running Linux kernel. Then, the requested system trace or probe functions are performed.

staprun

SystemTap back-end. Loads and unloads kernel modules produced by the SystemTap front-end.

For a list of options for each command, use `--help`. For details, refer to the **stap** and the **staprun** man pages.

To avoid giving root access to users solely to enable them to work with SystemTap, use one of the following SystemTap groups. They are not available by default on SUSE Linux Enterprise Server, but you can create the groups and modify the access rights accordingly. Also, adjust the permissions of the **staprun** command if the security implications are appropriate for your environment.

stapdev

Members of this group can run SystemTap scripts with **stap**, or run SystemTap instrumentation modules with **staprun**. As running **stap** involves compiling scripts into kernel modules and loading them into the kernel, members of this group still have effective root access.

stapusr

Members of this group are only allowed to run SystemTap instrumentation modules with **staprun**. In addition, they can only run those modules from `/lib/modules/KERNEL_VERSION/systemtap/`. This directory must be owned by root and must only be writable for the root user.

4.4. Important files and directories

The following list gives an overview of the SystemTap main files and directories.

`/lib/modules/KERNEL_VERSION/systemtap/`

Holds the SystemTap instrumentation modules.

`/usr/share/systemtap/tapset/`

Holds the standard library of tapsets.

`/usr/share/doc/packages/systemtap/examples`

Holds several example SystemTap scripts for different purposes. Only available if the `systemtap-docs` package is installed.

`~/.systemtap/cache`

Data directory for cached SystemTap files.

`/tmp/stap*`

Temporary directory for SystemTap files, including translated C code and kernel object.

4.2. Installation and setup

As SystemTap needs information about the kernel, some additional kernel-related packages must be installed. For each kernel you want to probe with SystemTap, you need to install a set of the following packages. This set should exactly match the kernel version and flavor (indicated by * in the overview below).

Repository for packages with debugging information



If you subscribed your system for online updates, you can find “debuginfo” packages in the `*-Debuginfo-Updates` online installation repository relevant for SUSE Linux Enterprise Server15 SP7. Use YaST to enable the repository.

For the classic SystemTap setup, install the following packages (using either YaST or **zypper**).

- systemtap
- systemtap-server
- systemtap-docs (optional)
- kernel-**-base*
- kernel-**-debuginfo*
- kernel-**-devel*
- kernel-source-***
- gcc

To get access to the man pages and to a helpful collection of example SystemTap scripts for different purposes, additionally install the systemtap-docs package.

To check if all packages are correctly installed on the machine and if SystemTap is ready to use, execute the following command as root.

```
#stap -v -e 'probe vfs.read {printf("read performed\n"); exit()}'
```

It probes the currently used kernel by running a script and returning an output. If the output is similar to the following, SystemTap is successfully deployed and ready to use:

```
Pass ❶: parsed user script and 59 library script(s) in 80usr/0sys/214real ms.
Pass ❷: analyzed script: 1 probe(s), 11 function(s), 2 embed(s), 1 global(s) in
140usr/20sys/412real ms.
Pass ❸: translated to C into
"/tmp/stapDwEk76/stap_1856e21ealc246da85ad8c66b4338349_4970.c" in 160usr/0sys/
408real ms.
Pass ❹: compiled C into "stap_1856e21ealc246da85ad8c66b4338349_4970.ko" in
2030usr/360sys/10182real ms.
Pass ❺: starting run.
read performed
Pass ❺: run completed in 10usr/20sys/257real ms.
```

- ❶ Checks the script against the existing tapset library in `/usr/share/systemtap/tapset/` for any tapsets used. Tapsets are scripts that form a library of pre-written probes and functions that can be used in SystemTap scripts.
- ❷ Examines the script for its components.
- ❸ Translates the script to C. Runs the system C compiler to create a kernel module from it. Both the resulting C code (`*.c`) and the kernel module (`*.ko`) are stored in the SystemTap cache, `~/.systemtap`.
- ❹ Loads the module and enables all the probes (events and handlers) in the script by hooking into the kernel. The event being probed is a Virtual File System (VFS) read. As the event occurs on

any processor, a valid handler is executed (prints the text read performed) and closed with no errors.

- ⑥ After the SystemTap session is terminated, the probes are disabled, and the kernel module is unloaded.

In case any error messages appear during the test, check the output for hints about any missing packages and make sure they are installed correctly. Rebooting and loading the appropriate kernel may also be needed.

4.3. Script syntax

SystemTap scripts consist of the following two components:

SystemTap events (probe points)

Name the kernel events at the associated handler should be executed. Examples for events are entering or exiting a certain function, a timer expiring, or starting or terminating a session.

SystemTap handlers (probe body)

Series of script language statements that specify the work to be done whenever a certain event occurs. This normally includes extracting data from the event context, storing them into internal variables, or printing results.

An event and its corresponding handler is collectively called a probe. SystemTap events are also called probe points. A probe's handler is also called a probe body.

Comments can be inserted anywhere in the SystemTap script in different styles: using either #, /* */, or // as marker.

4.1. Probe format

A SystemTap script can have multiple probes. They must be written in the following format:

```
probe EVENT {STATEMENTS}
```

Each probe has a corresponding statement block. This statement block must be enclosed in { } and contains the statements to be executed per event.

Example 4.1. Simple SystemTap script

The following example shows a simple SystemTap script.

```

probe❶ begin❷
{❸
    printf❹ ("hello world\n")❺
    exit()❻
}❼

```

- ❶ Start of the probe.
- ❷ Event begin (the start of the SystemTap session).
- ❸ Start of the handler definition, indicated by {.
- ❹ First function defined in the handler: the `printf` function.
- ❺ String to be printed by the `printf` function, followed by a line break (`/n`).
- ❻ Second function defined in the handler: the `exit()` function. Note that the SystemTap script will continue to run until the `exit()` function executes. If you want to stop the execution of the script before, stop it manually by pressing `Ctrl-C`.
- ❼ End of the handler definition, indicated by }.

The event `begin` (the start of the SystemTap session) triggers the handler enclosed in `{ }`. Here, that is the `printf` function. In this case, it prints `hello world` followed by a new line. Then, the script exits.

If your statement block holds several statements, SystemTap executes these statements in sequence—you do not need to insert special separators or terminators between multiple statements. A statement block can also be nested within another statement blocks. Generally, statement blocks in SystemTap scripts use the same syntax and semantics as in the C programming language.

4.2. SystemTap events (probe points)

SystemTap supports several built-in events.

The general event syntax is a dotted-symbol sequence. This allows a breakdown of the event namespace into parts. Each component identifier may be parameterized by a string or number literal, with a syntax like a function call. A component may include a `*` character, to expand to other matching probe points. A probe point may be followed by a `?` character, to indicate that it is optional, and that no error should result if it fails to expand. Alternately, a probe point may be followed by a `!` character to indicate that it is both optional and sufficient.

SystemTap supports multiple events per probe—they need to be separated by a comma (`,`). If multiple events are specified in a single probe, SystemTap will execute the handler when any of the specified events occur.

The events can be classified into the following categories:

- Synchronous events: Occur when any process executes an instruction at a particular location in kernel code. This gives other events a reference point (instruction address) from which more contextual data may be available.

An example for a synchronous event is `vfs.FILE_OPERATION`: The entry to the `FILE_OPERATION` event for Virtual File System (VFS). For example, in the section called “Installation and setup”, read is the `FILE_OPERATION` event used for VFS.

- Asynchronous events: Not tied to a particular instruction or location in code. This family of probe points consists mainly of counters, timers, and similar constructs.

Examples for asynchronous events are: `begin` (start of a SystemTap session—when a SystemTap script is run), `end` (end of a SystemTap session), or timer events. Timer events specify a handler to be executed periodically, like example `timer.s(SECONDS)`, or `timer.ms(MILLISECONDS)`.

When used together with other probes that collect information, timer events allow you to print periodic updates and see how that information changes over time.

Example 4.2. Probe with timer event

For example, the following probe would print the text “hello world” every 4 seconds:

```
probe timer.s(4)
{
    printf("hello world\n")
}
```

For detailed information about supported events, refer to the **stapprobes** man page. The See Also section of the man page also contains links to other man pages that discuss supported events for specific subsystems and components.

4.3. SystemTap handlers (probe body)

Each SystemTap event is accompanied by a corresponding handler defined for that event, consisting of a statement block.

4.1. Functions

If you need the same set of statements in multiple probes, you can place them in a function for easy reuse. Functions are defined by the keyword `function` followed by a name. They take any number of string or numeric arguments (by value) and may return a single string or number.

```
function FUNCTION_NAME(ARGUMENTS) {STATEMENTS}
probe EVENT {FUNCTION_NAME(ARGUMENTS)}
```

The statements in `FUNCTION_NAME` are executed when the probe for `EVENT` executes. The `ARGUMENTS` are optional values passed into the function.

Functions can be defined anywhere in the script. They may take any

One of the functions frequently needed is already introduced in *Example 4.1*, “*Simple SystemTap script*”: the `printf` function for printing data in a formatted way. When using the `printf` function, you can specify how arguments should be printed by using a format string. The format string is included in quotation marks and can contain further format specifiers, introduced by a `%` character.

Which format strings to use depends on your list of arguments. Format strings can have multiple format specifiers—each matching a corresponding argument. Multiple arguments can be separated by a comma.

Example 4.3. `printf` Function with format specifiers

```
printf ("❶%s❷(%d❸) open\n❹", execname(), pid())
```

- ❶ Start of the format string, indicated by `"`.
- ❷ String format specifier.
- ❸ Integer format specifier.
- ❹ End of the format string, indicated by `"`.

The example above prints the current executable name (`execname()`) as a string and the process ID (`pid()`) as an integer in brackets. Then, a space, the word `open` and a line break follow:

```
[...]
vmware-guestd(2206) open
held(2360) open
[...]
```

Apart from the two functions `execname()` and `pid()` used in *Example 4.3*, “*printf Function with format specifiers*”, a variety of other functions can be used as `printf` arguments.

Among the most commonly used SystemTap functions are the following:

`tid()`

ID of the current thread.

`pid()`

Process ID of the current thread.

`uid()`

ID of the current user.

cpu()

Current CPU number.

execname()

Name of the current process.

gettimeofday_s()

Number of seconds since Unix epoch (January 1, 1970).

ctime()

Convert time into a string.

pp()

String describing the probe point currently being handled.

thread_indent()

Useful function for organizing print results. It (internally) stores an indentation counter for each thread (`tid()`). The function takes one argument, an indentation delta, indicating how many spaces to add or remove from the thread's indentation counter. It returns a string with some generic trace data along with an appropriate number of indentation spaces. The generic data returned includes a time stamp (number of microseconds since the initial indentation for the thread), a process name, and the thread ID itself. This allows you to identify what functions were called, who called them, and how long they took.

Call entries and exits often do not immediately precede each other (otherwise it would be easy to match them). In between a first call entry and its exit, usually other call entries and exits are made. The indentation counter helps you match an entry with its corresponding exit as it indents the next function call in case it is *not* the exit of the previous one.

For more information about supported SystemTap functions, refer to the **stapfuncs** man page.

4.2. Other basic constructs

Apart from functions, you can use other common constructs in SystemTap handlers, including variables, conditional statements (like `if/else`, `while` loops, `for` loops, arrays or command line arguments).

4.1. Variables

Variables may be defined anywhere in the script. To define one, simply choose a name and assign a value from a function or expression to it:

```
foo = gettimeofday( )
```

Then you can use the variable in an expression. From the type of values assigned to the variable, SystemTap automatically infers the type of each identifier (string or number). Any inconsistencies will be reported as errors. In the example above, `foo` would automatically be classified as a number and could be printed via `printf()` with the integer format specifier (`%d`).

However, by default the variables are local to the probe containing them. They are initialized, used and disposed of at each handler evocation. To share variables between probes, declare them global anywhere in the script. To do so, use the `global` keyword outside of the probes:

Example 4.4. Using global variables

```
global count_jiffies, count_ms
probe timer.jiffies(100) { count_jiffies ++ }
probe timer.ms(100) { count_ms ++ }
probe timer.ms(12345)
{
    hz=(1000*count_jiffies) / count_ms
    printf ("jiffies:ms ratio %d:%d=> CONFIG_HZ=%d\n",
        count_jiffies, count_ms, hz)
    exit ()
}
```

This example script computes the `CONFIG_HZ` setting of the kernel by using timers that count jiffies and milliseconds, then computing accordingly. (A jiffy is the duration of one tick of the system timer interrupt. It is not an absolute time interval unit, since its duration depends on the clock interrupt frequency of the particular hardware platform). With the `global` statement it is possible to use the variables `count_jiffies` and `count_ms` also in the probe `timer.ms(12345)`. With `++` the value of a variable is incremented by 1.

4.2. Conditional statements

There are several conditional statements that you can use in SystemTap scripts. The most common conditional statements are the following:

If/else statements

They are expressed in the following format:

```
if (CONDITION)① STATEMENT1②
else③ STATEMENT2④
```

The `if` statement compares an integer-valued expression to zero. If the condition expression is non-zero, the first statement is executed. If the condition expression is zero, the second statement is executed. The `else` clause (`and`) is optional. Both `and` can also be statement blocks.

While loops

They are expressed in the following format:

```
while (CONDITION)① STATEMENT②
```

While condition is non-zero, the statement is executed. can also be a statement block. It must change a value so condition will eventually be zero.

For loops

They are a shortcut for while loops and are expressed in the following format:

```
for (INITIALIZATION①; CONDITIONAL②; INCREMENT③) statement
```

The expression specified in is used to initialize a counter for the number of loop iterations and is executed before execution of the loop starts. The execution of the loop continues until the loop condition is false. (This expression is checked at the beginning of each loop iteration). The expression specified in is used to increment the loop counter. It is executed at the end of each loop iteration.

Conditional operators

The following operators can be used in conditional statements:

`==`:Is equal to

`!=`:Is not equal to

`>=`:Is greater than or equal to

`<=`:Is less than or equal to

4.4. Example script

If you have installed the `systemtap-docs` package, you can find several useful SystemTap example scripts in `/usr/share/doc/packages/systemtap/examples`.

This section describes a rather simple example script in more detail: `/usr/share/doc/packages/systemtap/examples/network/tcp_connections.stp`.

Example 4.5. Monitoring incoming TCP connections with `tcp_connections.stp`

```
#!/usr/bin/env stap

probe begin {
    printf("%6s %16s %6s %6s %16s\n",
           "UID", "CMD", "PID", "PORT", "IP_SOURCE")
}

probe kernel.function("tcp_accept").return?,
       kernel.function("inet_csk_accept").return? {
    sock = $return
    if (sock != 0)
        printf("%6d %16s %6d %6d %16s\n", uid(), execname(), pid(),
           inet_get_local_port(sock), inet_get_ip_source(sock))
}
```

This SystemTap script monitors the incoming TCP connections and helps to identify unauthorized or unwanted network access requests in real time. It shows the following information for each new incoming TCP connection accepted by the computer:

- User ID (UID)
- Command accepting the connection (CMD)
- Process ID of the command (PID)
- Port used by the connection (PORT)
- IP address from which the TCP connection originated (IP_SOURCE)

To run the script, execute

```
stap /usr/share/doc/packages/systemtap/examples/network/tcp_connections.stp
```

and follow the output on the screen. To manually stop the script, press **Ctrl-C**.

4.5. User space probing

For debugging user space applications (like DTrace can do), SUSE Linux Enterprise Server15 SP7 supports user space probing with SystemTap. Custom probe points can be inserted in any user space application. Thus, SystemTap lets you use both kernel space and user space probes to debug the behavior of the whole system.

To get the required utrace infrastructure and the uprobes kernel module for user space probing, you need to install the `kernel-trace` package in addition to the packages listed in *the section called “Installation and setup”*.

utrace implements a framework for controlling user space tasks. It provides an interface that can be used by various tracing “engines”, implemented as loadable kernel modules. The engines register callback functions for specific events, then attach to whichever thread they want to trace. As the callbacks are made from “safe” places in the kernel, this allows for great leeway in the kinds of processing the functions can do. Several events can be watched via utrace. For example, you can observe events such as system call entry and exit, `fork()`, and signals being sent to the task.

More details about the utrace infrastructure are available at <https://sourceware.org/systemtap/wiki/utrace>.

SystemTap includes support for probing the entry into and return from a function in user space processes, probing predefined markers in user space code, and monitoring user-process events.

To check if the currently running kernel provides the needed utrace support, use the following command:

```
>sudo grep CONFIG_UTRACE /boot/config-`uname -r`
```

For more details about user space probing, refer to https://sourceware.org/systemtap/SystemTap_Beginners_Guide/userspace-probing.html.

4.6. More information

This chapter only provides a short SystemTap overview. Refer to the following links for more information about SystemTap:

<https://sourceware.org/systemtap/>

SystemTap project home page.

<https://sourceware.org/systemtap/wiki/>

A large collection of useful information about SystemTap, ranging from detailed user and developer documentation to reviews and comparisons with other tools, or Frequently Asked Questions and tips. Also contains collections of SystemTap scripts, examples and usage stories and lists recent talks and papers about SystemTap.

<https://sourceware.org/systemtap/documentation.html>

Features a *SystemTap Tutorial*, a *SystemTap Beginner's Guide*, a *Tapset Developer's Guide*, and a *SystemTap Language Reference* in PDF and HTML format. Also lists the relevant man pages.

You can also find the SystemTap language reference and SystemTap tutorial in your installed system under `/usr/share/doc/packages/systemtap`. Example SystemTap scripts are available from the `example` subdirectory.

Chapter 5. Kernel probes

Kernel probes are a set of tools to collect Linux kernel debugging and performance information. Developers and system administrators use them either to debug the kernel, or to find system performance bottlenecks. The reported data can then be used to tune the system for better performance.

You can insert these probes into any kernel routine, and specify a handler to be invoked after a particular break-point is hit. The main advantage of kernel probes is that you no longer need to rebuild the kernel and reboot the system after you make changes in a probe.

To use kernel probes, you typically need to write or obtain a specific kernel module. Such modules include both the *init* and the *exit* function. The *init* function (such as `register_kprobe()`) registers one or more probes, while the *exit* function unregisters them. The registration function defines *where* the probe is inserted and *which handler* is called after the probe is hit. To register or unregister a group of probes at one time, you can use relevant `register_<PROBE_TYPE>probes()` or `unregister_<PROBE_TYPE>probes()` functions.

Debugging and status messages are typically reported with the `printk` kernel routine. `printk` is a kernel space equivalent of a user space `printf` routine. For more information on `printk`, see [Logging kernel messages](#). Normally, you can view these messages by inspecting the output of the `systemd` journal (see Chapter 21, **journalctl**: query the `systemd` journal in “[Administration Guide](#)”). For more information on log files, see *Chapter 3, System log files*.

5.1. Supported architectures

Kernel probes are *fully* implemented on the following architectures:

- x86
- AMD64/Intel 64
- Arm
- POWER

Kernel probes are *partially* implemented on the following architectures:

- IA64 (does not support probes on instruction `slot1`)
- sparc64 (return probes not yet implemented)

5.2. Types of kernel probes

There are three types of kernel probes: *Kprobes*, *Jprobes*, and *Kretprobes*. *Kretprobes* are sometimes called *return probes*. You can find source code examples of all three type of probes in the Linux kernel. See the directory `/usr/src/linux/samples/kprobes/` (package `kernel-source`).

5.1. Kprobes

Kprobes can be attached to any instruction in the Linux kernel. When Kprobes is registered, it inserts a break-point at the first byte of the probed instruction. When the processor hits this break-point, the processor registers are saved, and the processing passes to Kprobes. First, a *pre-handler* is executed, then the probed instruction is stepped, and, finally a *post-handler* is executed. The control is then passed to the instruction following the probe point.

5.2. Jprobes

Jprobes is implemented through the Kprobes mechanism. It is inserted on a function's entry point and allows direct access to the arguments of the function which is being probed. Its handler routine must have the same argument list and return value as the probed function. To end it, call the `jprobe_return()` function.

When a jprobe is hit, the processor registers are saved, and the instruction pointer is directed to the jprobe handler routine. The control then passes to the handler with the same register contents as the function being probed. Finally, the handler calls the `jprobe_return()` function, and switches the control back to the control function.

Generally, you can insert multiple probes on one function. Jprobe is, however, limited to only one instance per function.

5.3. Return probe

Return probes are also implemented through Kprobes. When the `register_kretprobe()` function is called, a kprobe is attached to the entry of the probed function. After hitting the probe, the kernel probes mechanism saves the probed function return address and calls a user-defined return handler. The control is then passed back to the probed function.

Before you call `register_kretprobe()`, you need to set a `maxactive` argument, which specifies how many instances of the function can be probed at the same time. If set too low, a certain number of probes is missed.

5.3. Kprobes API

The programming interface of Kprobes consists of functions which are used to register and unregister all used kernel probes, and associated probe handlers. For a more detailed description of these functions and their arguments, see the information sources in *the section called "More information"*.

register_kprobe()

Inserts a break-point on a specified address. When the break-point is hit, the `pre_handler` and `post_handler` are called.

register_jprobe()

Inserts a break-point in the specified address. The address needs to be the address of the first instruction of the probed function. When the break-point is hit, the specified handler is run. The handler should have the same argument list and return type as the probed.

register_kretprobe()

Inserts a return probe for the specified function. When the probed function returns, a specified handler is run. This function returns 0 on success, or a negative error number on failure.

unregister_kprobe(), unregister_jprobe(), unregister_kretprobe()

Removes the specified probe. You can use it any time after the probe has been registered.

register_kprobes(), register_jprobes(), register_kretprobes()

Inserts each of the probes in the specified array.

unregister_kprobes(), unregister_jprobes(), unregister_kretprobes()

Removes each of the probes in the specified array.

disable_kprobe(), disable_jprobe(), disable_kretprobe()

Disables the specified probe temporarily.

enable_kprobe(), enable_jprobe(), enable_kretprobe()

Temporarily enables disabled probes.

5.4. debugfs Interface

In recent Linux kernels, the Kprobes instrumentation uses the kernel's `debugfs` interface. It can list all registered probes and globally switch all probes on or off.

5.1. Listing registered kernel probes

The list of all currently registered probes is in the `/sys/kernel/debug/kprobes/list` file.

```
saturn.example.com:~ # cat /sys/kernel/debug/kprobes/list
c015d71a k vfs_read+0x0 [DISABLED]
c011a316 j do_fork+0x0
c03dedc5 r tcp_v4_rcv+0x0
```

The first column lists the address in the kernel where the probe is inserted. The second column prints the type of the probe: k for kprobe, j for jprobe, and r for return probe. The third column specifies the symbol, offset and optional module name of the probe. The following optional columns include the status information of the probe. If the probe is inserted on a virtual address which is not valid anymore, it is marked with [GONE]. If the probe is temporarily disabled, it is marked with [DISABLED].

5.2. Globally enabling/disabling kernel probes

The `/sys/kernel/debug/kprobes/enabled` file represents a switch with which you can globally and forcibly turn on or off all the registered kernel probes. To turn them off, simply enter

```
#echo "0" > /sys/kernel/debug/kprobes/enabled
```

on the command line as root. To turn them on again, enter

```
#echo "1" > /sys/kernel/debug/kprobes/enabled
```

With such operations, you do not change the status of the probes. If a probe is temporarily disabled, it is not enabled automatically but remains in the [DISABLED] state after entering the latter command.

5.5. More information

To learn more about kernel probes, look at the following sources of information:

- Thorough but more technically oriented information about kernel probes is in `/usr/src/linux/Documentation/trace/kprobes.txt` (package `kernel-source`).
- Examples of all three types of probes (together with related Makefile) are in the `/usr/src/linux/samples/kprobes/` directory (package `kernel-source`).
- In-depth information about Linux kernel modules and `printk` kernel routine can be found at [The Linux Kernel Module Programming Guide](#)

Chapter 6. Hardware-based performance monitoring with Perf

You can store resulting information in a report. This report contains information about, for example, instruction pointers or what code a thread was executing.

Perf consists of two parts:

- Code integrated into the Linux kernel that instructs the hardware.
- The **perf** user space utility that allows you to use the kernel code and helps you analyze gathered data.

6.1. Hardware-based monitoring

Performance monitoring means collecting information related to how an application or system performs. This information can be obtained either through software-based means or from the CPU or chipset. Perf integrates both of these methods.

Many modern processors contain a performance monitoring unit (PMU). The design and functionality of a PMU is CPU-specific. For example, the number of registers, counters and features supported varies by CPU implementation.

Each PMU model consists of a set of registers: the performance monitor configuration (PMC) and the performance monitor data (PMD). Both can be read, but only PMCs are writable. These registers store configuration information and data.

6.2. Sampling and counting

Perf supports several profiling modes:

- **Counting** Count the number of occurrences of an event.
- **Event-based sampling** A less exact way of counting: A sample is recorded whenever a certain threshold number of events has occurred.
- **Time-based sampling** A less exact way of counting: A sample is recorded in a defined frequency.
- **Instruction-based sampling (AMD64 only)** The processor follows instructions appearing in a given interval and samples which events they produce. This allows following up on individual instructions and seeing which of them is critical to performance.

6.3. Installing Perf

The Perf kernel code is already included with the default kernel. To be able to use the user space utility, install the package `perf`.

6.4. Perf subcommands

To gather the required information, the **perf** tool has several subcommands. This section gives an overview of the most often used commands.

To see help in the form of a man page for any of the subcommands, use either **perf help***SUBCOMMAND* or **man perf-SUBCOMMAND**.

perf stat

Start a program and create a statistical overview that is displayed after the program quits. **perf stat** is used to count events.

perf record

Start a program and create a report with performance counter information. The report is stored as `perf.data` in the current directory. **perf record** is used to sample events.

perf report

Display a report that was previously created with **perf record**.

perf annotate

Display a report file and an annotated version of the executed code. If debug symbols are installed, the source code is also displayed.

perf list

List event types that Perf can report with the current kernel and with your CPU. You can filter event types by category. For example, to see hardware events only, use **perf list hw**.

The man page for **perf_event_open** has short descriptions for the most important events. For example, to find a description of the event `branch-misses`, search for `BRANCH_MISSES` (note the spelling differences):

```
>man perf_event_open | grep -A5 BRANCH_MISSES
```

Sometimes, events may be ambiguous. The lowercase hardware event names are not the names of raw hardware events but instead the names of aliases created by Perf. These aliases map to differently named but similarly defined hardware events on each supported processor.

For example, the `cpu-cycles` event is mapped to the hardware event `UNHALTED_CORE_CYCLES` on Intel processors. On AMD processors, however, it is mapped to the hardware event `CPU_CLK_UNHALTED`.

Perf also allows measuring raw events specific to your hardware. To look up their descriptions, see the Architecture Software Developer's Manual of your CPU vendor. The relevant documents for AMD64/Intel 64 processors are linked to in *the section called "More information"*.

perf top

Display system activity as it happens.

perf trace

This command behaves similarly to **strace**. With this subcommand, you can see which system calls are executed by a particular thread or process and which signals it receives.

6.5. Counting particular types of event

To count the number of occurrences of an event, such as those displayed by **perf list**, use:

```
#perf stat -e EVENT -a
```

To count multiple types of events at once, list them separated by commas. For example, to count cpu-cycles and instructions, use:

```
#perf stat -e cpu-cycles,instructions -a
```

To stop the session, press **Ctrl-C**.

You can also count the number of occurrences of an event within a particular time:

```
#perf stat -e EVENT -a -- sleep TIME
```

Replace *TIME* by a value in seconds.

6.6. Recording events specific to particular commands

There are several ways to sample events specific to a particular command:

- To create a report for a newly invoked command, use:

```
#perf record COMMAND
```

Then, use the started process normally. When you quit the process, the Perf session also stops.

- To create a report for the entire system while a newly invoked command is running, use:

```
#perf record -a COMMAND
```

Then, use the started process normally. When you quit the process, the Perf session also stops.

- To create a report for an already running process, use:

```
#perf record -p PID
```

Replace *PID* with a process ID. To stop the session, press **Ctrl-C**.

Now you can view the gathered data (`perf .data`) using:

```
>perf report
```

This opens a pseudo-graphical interface. To receive help, press **H**. To quit, press **Q**.

If you prefer a graphical interface, try the GTK+ interface of Perf:

```
>perf report --gtk
```

However, the GTK+ interface is limited in functionality.

6.7. More information

This chapter only provides a short overview. Refer to the following links for more information:

https://perf.wiki.kernel.org/index.php/Main_Page

The project home page. It also features a tutorial on using **perf**.

<https://www.brendangregg.com/perf.html>

Unofficial page with many one-line examples of how to use **perf**.

https://web.eece.maine.edu/~vweaver/projects/perf_events/

Unofficial page with several resources, primarily relating to the Linux kernel code of Perf and its API. This page includes, for example, a CPU compatibility table and a programming guide.

<https://www-ssl.intel.com/content/dam/www/public/us/en/documents/manuals/64-ia-32-architectures-software-developer-vol-3b-part-2-manual.pdf>

The *Intel Architectures Software Developer's Manual, Volume 3B*.

<https://support.amd.com/TechDocs/24593.pdf>

The *AMD Architecture Programmer's Manual, Volume 2*.

Chapter 7, OProfile—system-wide profiler

Consult this chapter for other performance optimizations.

Chapter 7. OProfile—system-wide profiler

7.1. Conceptual overview

OProfile consists of a kernel driver and a daemon for collecting data. It uses the hardware performance counters provided on many processors. OProfile is capable of profiling all code including the kernel, kernel modules, kernel interrupt handlers, system shared libraries, and other applications.

Modern processors support profiling through the hardware by performance counters. Depending on the processor, there can be many counters and each of these can be programmed with an event to count. Each counter has a value which determines how often a sample is taken. The lower the value, the more often it is used.

During the post-processing step, all information is collected and instruction addresses are mapped to a function name.

7.2. Installation and requirements

To use OProfile, install the `oprofile` package. OProfile works on AMD64/Intel 64, IBM Z, and POWER processors.

It is useful to install the `*-debuginfo` package for the respective application you want to profile. To profile the kernel, you need the `debuginfo` package as well.

7.3. Available OProfile utilities

OProfile contains several utilities to handle the profiling process and its profiled data. The following list is a short summary of programs used in this chapter:

opannotate

Outputs annotated source or assembly listings mixed with profile information. An annotated report can be used in combination with **addr2line** to identify the source file and line where hotspots potentially exist. See **man addr2line** for more information.

operf

Profiler tool. After profiling stops, the data that is by default stored in `CUR_DIR/oprofile_data/samples/current` can be processed by **opreport**, for example.

ophelp

Lists available events with short descriptions.

opimport

Converts sample database files from a foreign binary format to the format specific to the platform.

opreport

Generates reports from profiled data.

7.4. Using OProfile

With OProfile, you can profile both the kernel and applications. When profiling the kernel, tell OProfile where to find the `vmlinux*` file. Use the `--vmlinux` option and point it to `vmlinux*` (generally available in `/boot`). If you need to profile kernel modules, OProfile does this by default. However, make sure you read <https://oprofile.sourceforge.net/doc/kernel-profiling.html>.

Most applications do not need to profile the kernel, therefore you should use the `--no-vmlinux` option to reduce the amount of information.

7.1. Creating a report

Starting the daemon, collecting data, stopping the daemon, and creating a report for the application *COMMAND*.

1. Open a shell and log in as root.
2. Decide whether to profile with or without the Linux kernel:
 1. **Profile with the Linux kernel** Execute the following commands, because **operf** can only work with uncompressed images:

```
>cp /boot/vmlinux-`uname -r`.gz /tmp
>gunzip /tmp/vmlinux*.gz
>operf --vmlinux=/tmp/vmlinux* COMMAND
```

2. **Profile without the Linux kernel** Use the following command:

```
#operf --no-vmlinux COMMAND
```

To see which functions call other functions in the output, additionally use the `--callgraph` option and set a maximum *DEPTH*:

```
#operf --no-vmlinux --callgraph
DEPTHCOMMAND
```

3. **operf** writes its data to *CUR_DIR*/oprofile_data/samples/current. After the **operf** command is finished (or is aborted by `Ctrl-C`), the data can be analyzed with **oreport**:

```
#opreport
Overflow stats not available
CPU: CPU with timer interrupt, speed 0 MHz (estimated)
Profiling through timer interrupt
      TIMER:0|
samples|      %|
-----|-----
      84877 98.3226 no-vmlinux
...
```

7.2. Getting event configurations

The general procedure for event configuration is as follows:

1. Use first the events `CPU_CLK_UNHALTED` and `INST_RETIRED` to find optimization opportunities.
2. Use specific events to find bottlenecks. To list them, use the command **perf list**.

If you need to profile certain events, first check the available events supported by your processor with the **ophelp** command (example output generated from Intel Core i5 CPU):

```
#ophelp
oprofile: available events for CPU type "Intel Architectural Perfmon"

See Intel 64 and IA-32 Architectures Software Developer's Manual
Volume 3B (Document 253669) Chapter 18 for architectural perfmon events
This is a limited set of fallback events because oprofile does not know your CPU
CPU_CLK_UNHALTED: (counter: all)
    Clock cycles when not halted (min count: 6000)
INST_RETIRED: (counter: all)
    number of instructions retired (min count: 6000)
LLC_MISSES: (counter: all)
    Last level cache demand requests from this core that missed the LLC (min
count: 6000)
    Unit masks (default 0x41)
    -----
    0x41: No unit mask
LLC_REFS: (counter: all)
    Last level cache demand requests from this core (min count: 6000)
    Unit masks (default 0x4f)
    -----
    0x4f: No unit mask
BR_MISS_PRED_RETIRED: (counter: all)
    number of mispredicted branches retired (precise) (min count: 500)
```

Specify the performance counter events with the option `--event`. Multiple options are possible. This option needs an event name (from **ophelp**) and a sample rate, for example:

```
#operf --events CPU_CLK_UNHALTED:100000
```



Setting sampling rates with `CPU_CLK_UNHALTED`

Setting low sampling rates can seriously impair the system performance while high sample rates can disrupt the system to such a high degree that the data is useless. It is recommended to tune the performance metric for being monitored with and without OProfile and to experimentally determine the minimum sample rate that disrupts the performance the least.

7.5. Generating reports

Before generating a report, make sure the **operf** has stopped. Unless you have provided an output directory with `--session-dir`, **operf** has written its data to `CUR_DIR/oprofile_data/samples/current`, and the reporting tools **opreport** and **opannotate** look there by default.

Calling **opreport** without any options gives a complete summary. With an executable as an argument, retrieve profile data only from this executable. If you analyze applications written in C++, use the `--demangle smart` option.

The **opannotate** generates output with annotations from source code. Run it with the following options:

```
#opannotate --source \  
--base-dirs=BASEDIR \  
--search-dirs=SEARCHDIR \  
--output-dir=annotated/ \  
/lib/libfoo.so
```

The option `--base-dir` contains a comma-separated list of paths which is stripped from debug source files. These paths are searched before looking in `--search-dirs`. The `--search-dirs` option is also a comma-separated list of directories to search for source files.



Inaccuracies in annotated source

Because of compiler optimization, code can disappear and appear in a different place. Use the information in <https://oprofile.sourceforge.net/doc/debug-info.html> to fully understand its implications.

7.6. More information

This chapter only provides a short overview. Refer to the following links for more information:

<https://oprofile.sourceforge.net>

The project home page.

Manpages

Details descriptions about the options of the different tools.

`/usr/share/doc/packages/oprofile/oprofile.html`

Contains the OProfile manual.

<https://developer.intel.com/>

Architecture reference for Intel processors.

https://www.ibm.com/support/knowledgecenter/ssw_aix_71/assembler/idalangref_arch_overview.html

Architecture reference for PowerPC64 processors in IBM iSeries, pSeries, and Blade server systems.

Chapter 8. Dynamic debug—kernel debugging messages

You can use dynamic debugging in several situations, such as:

- Troubleshooting kernel issues
- Developing drivers for new hardware
- Tracing and auditing security events

8.1. Benefits of dynamic debugging

Certain benefits of dynamic debugging are listed below:

Real-time debugging

Dynamic debugging enables debugging messages without requiring a system reboot. This real-time capability is crucial for diagnosing issues in production environments.

Selective debugging

You can enable debugging messages for specific parts of the kernel or even individual modules, allowing you to focus on relevant information.

Performance tuning

Use dynamic debugging to monitor and optimize kernel performance by selectively enabling or disabling debugging messages based on the current analysis requirements.

8.2. Checking the status of dynamic debug

For supported kernel versions that are installed by default, dynamic debug is already built in. To check the status of dynamic debug, run the following command as the root user:

```
#zcat /proc/config.gz | grep CONFIG_DYNAMIC_DEBUG
```

If dynamic debug is compiled into the kernel, you should see an output similar to the following:

```
CONFIG_DYNAMIC_DEBUG=y  
CONFIG_DYNAMIC_DEBUG_CORE=y
```

8.3. Using dynamic debug

To enable specific debug messages or logs within the running kernel, you can use the **echo** command and write to the `/sys/kernel/debug/dynamic_debug/control` file.

The following examples illustrate certain simple uses of dynamic debug:

**Note**

Dynamic debug relies on specific debugging macros, such as `pr_debug`, embedded in the kernel code. These macros are used by kernel developers to insert debugging messages into the code.

The examples in this section assume that the `pr_debug` macro works correctly because dynamic debug is allowed for the running kernel.

Enabling debug messages for a specific kernel source code file

To enable the debug messages for a specific kernel source code file, use the following example:

```
#echo "file FILE_NAME.c +p" > /sys/kernel/debug/dynamic_debug/control
```

Enabling debug messages for a specific kernel module

To enable debug messages for a specific kernel module, use the following example:

```
#echo "module MODULE_NAME +p" > /sys/kernel/debug/dynamic_debug/control
```

Disabling debug messages

To disable previously enabled debugging messages for a specific kernel source code file or a kernel module, run the **echo** command with the **-p** option. For example:

```
#echo "file FILE_NAME.c -p" > /sys/kernel/debug/dynamic_debug/control
```

```
#echo "module MODULE_NAME -p" > /sys/kernel/debug/dynamic_debug/control
```

For detailed information about dynamic debug and its use cases, refer to its [official documentation](#).

8.4. Viewing the dynamic debug messages

You can view the dynamic debug messages that were generated based on the configurations you enabled, by running **dmesg** and filtering the output with **grep**. For example:

```
#dmesg | grep -i "FILE_NAME.c"
```

Optionally, to continuously monitor the system messages as they are generated, you can use the **tail** command with the **-f** option:

```
#tail -f /var/log/messages
```

Part IV. Resource management

- 9 General system resource management **91**
- 10 Kernel control groups **95**
- 11 Automatic Non-Uniform Memory Access (NUMA) balancing **102**
- 12 Power management **106**
- 13 Adaptive and dynamic tuning using TuneD **116**

Chapter 9. General system resource management

9.1. Planning the installation

A carefully planned installation ensures that the system is set up exactly as you need it for the given purpose. It also saves considerable time when fine tuning the system. All changes suggested in this section can be made in the *Installation Settings* step during the installation. See the section called “Installation settings” in “[Deployment Guide](#)” for details.

9.1. Partitioning

Depending on the server's range of applications and the hardware layout, the partitioning scheme can influence the machine's performance (although to a lesser extent only). It is beyond the scope of this manual to suggest different partitioning schemes for particular workloads. However, the following rules positively affect performance. They do not apply when using an external storage system.

- Make sure there is always a certain amount of free space available on the disk, since a full disk delivers inferior performance.
- Disperse simultaneous read and write access onto different disks by, for example:
 - using separate disks for the operating system, data, and log files
 - placing a mail server's spool directory on a separate disk
 - distributing the user directories of a home server between different disks

9.2. Installation scope

The installation scope has no direct influence on the machine's performance, but a carefully chosen scope of packages has advantages. It is recommended to install the minimum of packages needed to run the server. A system with a minimum set of packages is easier to maintain and has fewer potential security issues. Furthermore, a tailor made installation scope also ensures that no unnecessary services are started by default.

SUSE Linux Enterprise Server lets you customize the installation scope on the Installation Summary screen. By default, you can select or remove preconfigured patterns for specific tasks, but it is also possible to start the YaST Software Manager for a fine-grained package-based selection.

One or more of the following default patterns may not be needed in all cases:

GNOME desktop environment

Servers rarely need a full desktop environment. In case a graphical environment is needed, a more economical solution such as IceWM can be sufficient.

X Window System

When solely administrating the server and its applications via command line, consider not installing this pattern. However, keep in mind that it is needed to run GUI applications from a remote machine. If your application is managed by a GUI or if you prefer the GUI version of YaST, keep this pattern.

Print server

This pattern is only needed to print from the machine.

9.3. Default target

A running X Window System consumes many resources and is rarely needed on a server. It is strongly recommended to start the system in target `multi-user.target`. You can still remotely start graphical applications.

9.2. Disabling unnecessary services

The default installation starts several services (the number varies with the installation scope). Since each service consumes resources, it is recommended to disable the ones not needed. Run *YaST > System > Services Manager* to start the services management module.

If you are using the graphical version of YaST, you can click the column headlines to sort the list of services. Use this to get an overview of which services are currently running. Use the *Start/Stop* button to disable the service for the running session. To permanently disable it, use the *Enable/Disable* button.

The following list shows services that are started by default after the installation of SUSE Linux Enterprise Server. Check which of the components you need, and disable the others:

alsasound

Loads the Advanced Linux Sound System.

auditd

A daemon for the Audit system (see Part VII, “The Linux Audit Framework” in “[Security and Hardening Guide](#)” for details). Disable this if you do not use Audit.

bluez-coldplug

Handles cold plugging of Bluetooth dongles.

cups

A printer daemon.

java.binfmt_misc

Enables the execution of *.class or *.jar Java programs.

nfs

Services needed to mount NFS.

smbfs

Services needed to mount SMB/CIFS file systems from a Windows* server.

splash / splash_early

Shows the splash screen on start-up.

9.3. File systems and disk access

Hard disks are the slowest components in a computer system and therefore often the cause for a bottleneck. Using the file system that best suits your workload helps to improve performance. Using special mount options or prioritizing a process's I/O priority are further means to speed up the system.

9.1. File systems

SUSE Linux Enterprise Server ships with several file systems, including Btrfs, Ext4, Ext3, Ext2, and XFS. Each file system has its own advantages and disadvantages. Refer to Chapter 1, Overview of file systems in Linux in “[Storage Administration Guide](#)” for detailed information.

9.1. NFS

NFS (Version 3) tuning is covered in detail in the NFS Howto at <https://nfs.sourceforge.net/nfs-howto/>. When mounting NFS shares, start with the experiment of increasing the size of the read-write blocks to 32768, by using the mount options `wsiz` and `rsiz`.

9.2. Time stamp update policy

Each file and directory in a file system has three time stamps associated with it: a time when the file was last read called *access time*, a time when the file data was last modified called *modification time*, and a time when the file metadata was last modified called *change time*. Keeping access time always up to date has significant performance overhead since every read-only access incurs a write operation. By default, every file system updates access time only if current file access time is older than a day, or older than file modification or change time. This feature is called *relative access time* and the corresponding mount option is `relatime`. Updates of access time can be disabled using the `noatime` mount option. However, you need to verify your applications do not use it. This can be true for file and Web servers or for network storage. If the default relative

access time update policy is not suitable for your applications, use the `strictatime` mount option.

Some file systems (for example Ext4) also support lazy time stamp updates. When this feature is enabled using the `lazytime` mount option, updates of all time stamps happen in memory but they are not written to disk. That happens only in response to `fsync` or `sync` system calls, when the file information is written due to another reason such as file size update, when time stamps are older than 24 hours, or when cached file information needs to be evicted from memory.

To update mount options used for a file system, either edit `/etc/fstab` directly, or use the *Fstab Options* dialog when editing or adding a partition with the YaST Partitioner.

9.3. Prioritizing disk access with **ionice**

The **ionice** command lets you prioritize disk access for single processes. This enables you to give less I/O priority to background processes with heavy disk access that are not time-critical, such as backup jobs. **ionice** also lets you raise the I/O priority for a specific process to make sure this process always has immediate access to the disk. The caveat of this feature is that standard writes are cached in the page cache and are written back to persistent storage only later by an independent kernel process. Thus the I/O priority setting generally does not apply for these writes. Also be aware that I/O class and priority setting are obeyed only by *BFQ* I/O scheduler for `blk-mq` I/O path (refer to *the section called “Available I/O elevators with blk-mq I/O path”*). You can set the following three scheduling classes:

Idle

A process from the idle scheduling class is only granted disk access when no other process has asked for disk I/O.

Best effort

The default scheduling class used for any process that has not asked for a specific I/O priority. Priority within this class can be adjusted to a level from 0 to 7 (with 0 being the highest priority). Programs running at the same best-effort priority are served in a round-robin fashion. Some kernel versions treat priority within the best-effort class differently—for details, refer to the `ionice(1)` man page.

Real-time

Processes in this class are always granted disk access first. Fine-tune the priority level from 0 to 7 (with 0 being the highest priority). Use with care, since it can starve other processes.

For more details and the exact command syntax refer to the `ionice(1)` man page. If you need more reliable control over bandwidth available to each application, use Kernel Control Groups as described in *Chapter 10, Kernel control groups*.

Chapter 10. Kernel control groups

10.1. Overview

Every process is assigned exactly one administrative cgroup. cgroups are ordered in a hierarchical tree structure. You can set resource limitations, such as CPU, memory, disk I/O or network bandwidth usage, for single processes or for whole branches of the hierarchy tree.

On SUSE Linux Enterprise Server, systemd uses cgroups to organize all processes in groups, which systemd calls slices. systemd also provides an interface for setting cgroup properties.

The command **systemd-cgls** displays the hierarchy tree.

The kernel cgroup API comes in two variants—v1 and v2. Additionally, there can be multiple cgroup hierarchies exposing different APIs. From many possible combinations, there are two practical choices:

- unified: v2 hierarchy with controllers
- hybrid: v2 hierarchy without controllers, and the controllers are on v1 hierarchies (deprecated)

The default mode is unified. There is a hybrid mode that provides backward compatibility for applications that need it.

You may set only one mode.

10.1. Hybrid cgroup hierarchy



Deprecation notice

cgroup v1 has been deprecated, and might be removed in a future release.

To enable the hybrid control group hierarchy, append `systemd.unified_cgroup_hierarchy=0` as a kernel command-line parameter to the GRUB 2 boot loader. For more details about configuring GRUB 2, refer to Chapter 18, The boot loader GRUB 2 in [“Administration Guide”](#).

10.2. Resource accounting

Organizing processes into different cgroups can be used to obtain per-cgroup resource consumption data.

The accounting has comparatively small but non-zero overhead, whose impact depends on the workload. Activating accounting for one unit also implicitly activates it for all units in the same slice, and for all its parent slices, and the units contained in them.

The accounting can be set on a per-unit basis with directives such as `MemoryAccounting=` or globally for all units in `/etc/systemd/system.conf` with the directive `DefaultMemoryAccounting=`. Refer to **man systemd.resource-control** for the exhaustive list of possible directives.

10.3. Setting resource limits



Implicit resource consumption

Be aware that resource consumption implicitly depends on the environment where your workload executes (for example, size of data structures in libraries/kernel, forking behavior of utilities, computational efficiency). Hence it is recommended to (re)calibrate your limits should the environment change.

Limitations to cgroups can be set with the **systemctl set-property** command. The syntax is:

```
#systemctl set-property [--runtime] NAMEPROPERTY1=VALUE [PROPERTY2=VALUE]
```

The configured value is applied immediately. Optionally, use the `--runtime` option, so that the new values do not persist after reboot.

Replace *NAME* with a systemd service, scope, or slice name.

For a complete list of properties and more details, see **man systemd.resource-control**.

10.4. Preventing fork bombs with TasksMax

systemd supports configuring task count limits both for each individual leaf unit, or aggregated on slices. Upstream systemd ships with defaults that limit the number of tasks in each unit (15% of the kernel global limit, run `/usr/sbin/sysctl kernel.pid_max` to see the total limit). Each user's slice is limited to 33% of the kernel limit. However, this is different for SUSE Linux Enterprise Server.

10.1. Finding the current default TasksMax values

It became apparent, in practice, that there is not a single default that applies to all use cases. SUSE Linux Enterprise Server ships with two custom configurations that override the upstream defaults for system units and for user slices, and sets them both to infinity. `/usr/lib/systemd/system.conf.d/__25-defaults-SLE.conf` contains these lines:

```
[Manager]
DefaultTasksMax=infinity
```

/usr/lib/systemd/system/user-.slice.d/25-defaults-SLE.conf contains these lines:

```
[Slice]
TasksMax=infinity
```

Use **systemctl** to verify the DefaultTasksMax value:

```
>systemctl show --property DefaultTasksMax
DefaultTasksMax=infinity
```

infinity means having no limit. It is not a requirement to change the default, but setting certain limits may help to prevent system crashes from runaway processes.

10.2. Overriding the DefaultTasksMax value

Change the global DefaultTasksMax value by creating a new override file, /etc/systemd/system.conf.d/90-system-tasksmax.conf, and write the following lines to set a new default limit of 256 tasks per system unit:

```
[Manager]
DefaultTasksMax=256
```

Load the new setting, then verify that it changed:

```
>sudo systemctl daemon-reload>systemctl show --property DefaultTasksMax
DefaultTasksMax=256
```

Adjust this default value to suit your needs. You can set different limits on individual services as needed. This example is for MariaDB. First check the current active value:

```
>systemctl status mariadb.service
● mariadb.service - MariaDB database server
  Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; vendor preset>
  Active: active (running) since Tue 2020-05-26 14:15:03 PDT; 27min ago
    Docs: man:mysql(8)
          https://mariadb.com/kb/en/library/systemd/
  Main PID: 11845 (mysqld)
    Status: "Taking your SQL requests now..."
     Tasks: 30 (limit: 256)
    CGroup: /system.slice/mariadb.service
            └─11845 /usr/sbin/mysqld --defaults-file=/etc/my.cnf --user=mysql
```

The Tasks line shows that MariaDB currently has 30 tasks running, and has an upper limit of the default 256, which is inadequate for a database. The following example demonstrates how to raise MariaDB's limit to 8192.

```
>sudo systemctl set-property mariadb.service TasksMax=8192>systemctl status
mariadb.service
● mariadb.service - MariaDB database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; vendor
   preset: disabled)
   Drop-In: /etc/systemd/system/mariadb.service.d
            └─50-TasksMax.conf
   Active: active (running) since Tue 2020-06-02 17:57:48 PDT; 7min ago
     Docs: man:mysqld(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 3446 ExecStartPre=/usr/lib/mysql/mysql-systemd-helper upgrade
   (code=exited, status=0/SUCCESS)
   Process: 3440 ExecStartPre=/usr/lib/mysql/mysql-systemd-helper install
   (code=exited, status=0/SUCCESS)
   Main PID: 3452 (mysqld)
    Status: "Taking your SQL requests now..."
     Tasks: 30 (limit: 8192)
   CGroup: /system.slice/mariadb.service
           └─3452 /usr/sbin/mysqld --defaults-file=/etc/my.cnf --user=mysql
```

systemctl set-property applies the new limit and creates a drop-in file for persistence, `/etc/systemd/system/mariadb.service.d/50-TasksMax.conf`, that contains only the changes you want to apply to the existing unit file. The value does not have to be 8192, but should be whatever limit is appropriate for your workloads.

10.3. Default TasksMax limit on users

The default limit on users should be high, because user sessions need more resources. Set your own default for any user by creating a new file, for example `/etc/systemd/system/user-.slice.d/40-user-taskmask.conf`. The following example sets a default of 16284:

```
[Slice]
TasksMax=16284
```



Numeric prefixes reference

See the section called “Creating drop-in files manually” in “[Administration Guide](#)” to learn what numeric prefixes are expected for drop-in files.

Then reload systemd to load the new value, and verify the change:

```
>sudo systemctl daemon-reload>systemctl show --property TasksMax user-1000.slice
TasksMax=16284
```

How do you know what values to use? This varies according to your workloads, system resources, and other resource configurations. When your TasksMax value is too low, you may see error messages such as *Failed to fork (Resources temporarily unavailable)*, *Can't create thread to handle new connection*, and *Error: Function call 'fork' failed with error code 11, 'Resource temporarily unavailable'*.

For more information on configuring system resources in systemd, see `systemd.resource-control` (5).

10.5. I/O control with cgroups

This section introduces using the Linux kernel's block I/O controller to prioritize or throttle I/O operations. This leverages the means provided by systemd to configure cgroups, and discusses probable pitfalls when dealing with proportional I/O control.

10.1. Prerequisites

The following subsections describe steps that you must take in advance when you design and configure your system, since those aspects cannot be changed during runtime.

10.1. File system

You should use a cgroup-writeback-aware file system (otherwise writeback charging is not possible). The recommended SUSE Linux Enterprise Server file systems added support in the following upstream releases:

- Btrfs (v4.3)
- Ext4 (v4.3)
- XFS (v5.3)

As of SUSE Linux Enterprise Server 15 SP3, any of the named file systems can be used.

10.2. Block I/O scheduler

The throttling policy is implemented higher in the stack, therefore it does not require any additional adjustments. The proportional I/O control policies have two different implementations: the BFQ controller, and the cost-based model. We describe the BFQ controller here. To exert its proportional implementation for a particular device, we must make sure that BFQ is the chosen scheduler. Check the current scheduler:

```
>cat /sys/class/block/sda/queue/scheduler
mq-deadline kyber bfq [none]
```

Switch the scheduler to BFQ:

```
#echo bfq > /sys/class/block/sda/queue/scheduler
```

You must specify the disk device (not a partition). The optimal way to set this attribute is a udev rule specific to the device. SUSE Linux Enterprise Server ships udev rules that already enable BFQ for rotational disk drives.

10.3. Cgroup hierarchy layout

Normally, all tasks reside in the root cgroup and they compete against each other. When the tasks are distributed into the cgroup tree the competition occurs between sibling cgroups only. This

applies to the proportional I/O control; the throttling hierarchically aggregates throughput of all descendants (see the following diagram).

```
r
├─ a      IOWeight=100
│ └─ [c]  IOWeight=300
│   └─ d  IOWeight=100
└─ [b]    IOWeight=200
```

I/O is originating only from cgroups c and b. Even though c has a higher weight, it is treated with lower priority because it is level-competing with b.

10.2. Configuring control quantities

You can apply the values to (long running) services permanently.

```
>sudo systemctl set-property fast.service IOWeight=400>sudo systemctl set-
property slow.service IOWeight=50>sudo systemctl set-property throttled.service
IOReadBandwidthMax="/dev/sda 1M"
```

Alternatively, you can apply I/O control to individual commands, for example:

```
>sudo systemd-run --scope -p IOWeight=400high_prioritized_comman
d>sudo systemd-run --scope -p IOWeight=50low_prioritized_command>sudo systemd-run
--scope -p IOReadBandwidthMax="/dev/sda 1M" dd if=/dev/sda of=/dev/null bs=1M
count=10
```

10.3. I/O control behavior and setting expectations

The following list items describe I/O control behavior, and what you should expect under different conditions.

- I/O control works best for direct I/O operations (bypassing page cache), the situations where the actual I/O is decoupled from the caller (typically writeback via page cache) may manifest variously. For example, delayed I/O control or even no observed I/O control (consider little bursts or competing workloads that happen to never “meet,” submitting I/O at the same time, and saturating the bandwidth). For these reasons, the resulting ratio of I/O throughputs does not strictly follow the ratio of configured weights.
- systemd performs scaling of configured weights (to adjust for narrower BFQ weight range), hence the resulting throughput ratios also differ.
- The writeback activity depends on the amount of dirty pages, besides the global sysctl knobs (vm.dirty_background_ratio and vm.dirty_ratio)). Memory limits of individual cgroups come into play when the dirty limits are distributed among cgroups, and this in turn may affect I/O intensity of affected cgroups.
- Not all storages are equal. The I/O control happens at the I/O scheduler layer, which has ramifications for setups with devices stacked on these that do no actual scheduling. Consider device mapper logical volumes spanning multiple physical devices, MD RAID, or even Btrfs RAID. I/O control over such setups may be challenging.

- There is no separate setting for proportional I/O control of reads and writes.
- Proportional I/O control is only one of the policies that can interact with each other (but responsible resource design perhaps avoids that).
- The I/O device bandwidth is not the only shared resource on the I/O path. Global file system structures are involved, which is relevant when I/O control is meant to guarantee certain bandwidth; it does not, and it may even lead to priority inversion (prioritized cgroup waiting for a transaction of slower cgroup).
- So far, we have been discussing only explicit I/O of file system data, but swap-in and swap-out can also be controlled. Although if such a need arises, it points out to improperly provisioned memory (or memory limits).

10.4. Resource control in user sessions

In order to apply cgroup resource control within user sessions, controllers must be delegated user instances of systemd. SUSE Linux Enterprise Server ships systemd default configuration that delegates *no* controllers.

You can use drop-in files to change the set of delegated controllers. For instance, `/etc/systemd/system/user@.service.d/60-delegate.conf` adds controllers to all users, while `/etc/systemd/system/user@uid.service.d/60-delegate.conf` adds controllers only to a particular user. The content of the file should be like the following:

```
[Service]
Delegate=pids memory
```

Both the systemd instance and the affected user instance must be notified to reload the new configuration.

```
>sudo systemctl daemon-reload>systemctl --user daemon-reexec
```

Alternatively, the affected user may log out and log in instead of applying the second line to restart their user instance.

10.6. More information

- Kernel documentation (package `kernel-source`): files in `/usr/src/linux/Documentation/admin-guide/cgroup-v1` and file `/usr/src/linux/Documentation/admin-guide/cgroup-v2.rst`.
- **man `systemd.resource-control`**
- <https://lwn.net/Articles/604609/>—Brown, Neil: Control Groups Series (2014, 7 parts).
- <https://lwn.net/Articles/243795/>—Corbet, Jonathan: Controlling memory use in containers (2007).
- <https://lwn.net/Articles/236038/>—Corbet, Jonathan: Process containers (2007).

Chapter 11. Automatic Non-Uniform Memory Access (NUMA) balancing

11.1. Implementation

Automatic NUMA balancing happens in three basic steps:

1. A task scanner periodically scans a portion of a task's address space and marks the memory to force a page fault when the data is next accessed.
2. The next access to the data results in a NUMA Hinting Fault. Based on this fault, the data can be migrated to a memory node associated with the task accessing the memory.
3. To keep a task, the CPU it is using and the memory it is accessing together, the scheduler groups tasks that share data.

The unmapping of data and page fault handling incurs overhead. However, commonly the overhead is offset by threads accessing data associated with the CPU.

11.2. Configuration

Static configuration has been the recommended way of tuning workloads on NUMA hardware. To do this, memory policies can be set with **numactl**, **taskset** or **cpusets**. NUMA-aware applications can use special APIs. In cases where the static policies have already been created, automatic NUMA balancing should be disabled as the data access should already be local.

numactl - -hardware shows the memory configuration of the machine and whether it supports NUMA or not. This is example output from a 4-node machine.

```
>numactl --hardware
available: 4 nodes (0-3)
node 0 cpus: 0 4 8 12 16 20 24 28 32 36 40 44
node 0 size: 16068 MB
node 0 free: 15909 MB
node 1 cpus: 1 5 9 13 17 21 25 29 33 37 41 45
node 1 size: 16157 MB
node 1 free: 15948 MB
node 2 cpus: 2 6 10 14 18 22 26 30 34 38 42 46
node 2 size: 16157 MB
node 2 free: 15981 MB
node 3 cpus: 3 7 11 15 19 23 27 31 35 39 43 47
node 3 size: 16157 MB
node 3 free: 16028 MB
node distances:
node  0  1  2  3
 0:  10  20  20  20
 1:  20  10  20  20
 2:  20  20  10  20
 3:  20  20  20  10
```

Automatic NUMA balancing can be enabled or disabled for the current session by writing 1 or 0 to `/proc/sys/kernel/numa_balancing` which enables or disables the feature respectively. To

permanently enable or disable it, use the kernel command line option `numa_balancing=[enable|disable]`.

If Automatic NUMA Balancing is enabled, the task scanner behavior can be configured. The task scanner balances the overhead of Automatic NUMA Balancing with the amount of time it takes to identify the best placement of data.

numa_balancing_scan_delay_ms

The amount of CPU time a thread must consume before its data is scanned. This prevents creating overhead because of short-lived processes.

numa_balancing_scan_period_min_ms and numa_balancing_scan_period_max_ms

Controls how frequently a task's data is scanned. Depending on the locality of the faults, the scan rate increases or decreases. These settings control the min and max scan rates.

numa_balancing_scan_size_mb

Controls how much address space is scanned when the task scanner is active.

11.3. Monitoring

The most important task is to assign metrics to your workload and measure the performance with Automatic NUMA Balancing enabled and disabled to measure the impact. Profiling tools can be used to monitor local and remote memory accesses if the CPU supports such monitoring. Automatic NUMA Balancing activity can be monitored via the following parameters in `/proc/vmstat`:

numa_pte_updates

The amount of base pages that were marked for NUMA hinting faults.

numa_huge_pte_updates

The amount of transparent huge pages that were marked for NUMA hinting faults. In combination with `numa_pte_updates` the total address space that was marked can be calculated.

numa_hint_faults

Records how many NUMA hinting faults were trapped.

numa_hint_faults_local

Shows how many of the hinting faults were to local nodes. In combination with `numa_hint_faults`, the percentage of local versus remote faults can be calculated. A high percentage of local hinting faults indicates that the workload is closer to being converged.

numa_pages_migrated

Records how many pages were migrated because they were misplaced. As migration is a copying operation, it contributes the largest part of the overhead created by NUMA balancing.

11.4. Impact

The following illustrates a simple test case of a 4-node NUMA machine running the SpecJBB 2005 using a single instance of the JVM with no static tuning around memory policies. However, the impact for each workload varies and this example is based on a pre-release version of SUSE Linux Enterprise Server 12.

	Balancing disabled	Balancing enabled
TPut 1	26629.00 (0.00%)	26507.00 (-0.46%)
TPut 2	55841.00 (0.00%)	53592.00 (-4.03%)
TPut 3	86078.00 (0.00%)	86443.00 (0.42%)
TPut 4	116764.00 (0.00%)	113272.00 (-2.99%)
TPut 5	143916.00 (0.00%)	141581.00 (-1.62%)
TPut 6	166854.00 (0.00%)	166706.00 (-0.09%)
TPut 7	195992.00 (0.00%)	192481.00 (-1.79%)
TPut 8	222045.00 (0.00%)	227143.00 (2.30%)
TPut 9	248872.00 (0.00%)	250123.00 (0.50%)
TPut 10	270934.00 (0.00%)	279314.00 (3.09%)
TPut 11	297217.00 (0.00%)	301878.00 (1.57%)
TPut 12	311021.00 (0.00%)	326048.00 (4.83%)
TPut 13	324145.00 (0.00%)	346855.00 (7.01%)
TPut 14	345973.00 (0.00%)	378741.00 (9.47%)
TPut 15	354199.00 (0.00%)	394268.00 (11.31%)
TPut 16	378016.00 (0.00%)	426782.00 (12.90%)
TPut 17	392553.00 (0.00%)	437772.00 (11.52%)
TPut 18	396630.00 (0.00%)	456715.00 (15.15%)
TPut 19	399114.00 (0.00%)	484020.00 (21.27%)
TPut 20	413907.00 (0.00%)	493618.00 (19.26%)
TPut 21	413173.00 (0.00%)	510386.00 (23.53%)
TPut 22	420256.00 (0.00%)	521016.00 (23.98%)
TPut 23	425581.00 (0.00%)	536214.00 (26.00%)
TPut 24	429052.00 (0.00%)	532469.00 (24.10%)
TPut 25	426127.00 (0.00%)	526548.00 (23.57%)
TPut 26	422428.00 (0.00%)	531994.00 (25.94%)
TPut 27	424378.00 (0.00%)	488340.00 (15.07%)
TPut 28	419338.00 (0.00%)	543016.00 (29.49%)
TPut 29	403347.00 (0.00%)	529178.00 (31.20%)
TPut 30	408681.00 (0.00%)	510621.00 (24.94%)
TPut 31	406496.00 (0.00%)	499781.00 (22.95%)
TPut 32	404931.00 (0.00%)	502313.00 (24.05%)
TPut 33	397353.00 (0.00%)	522418.00 (31.47%)
TPut 34	382271.00 (0.00%)	491989.00 (28.70%)
TPut 35	388965.00 (0.00%)	493012.00 (26.75%)
TPut 36	374702.00 (0.00%)	502677.00 (34.15%)
TPut 37	367578.00 (0.00%)	500588.00 (36.19%)
TPut 38	367121.00 (0.00%)	496977.00 (35.37%)
TPut 39	355956.00 (0.00%)	489430.00 (37.50%)
TPut 40	350855.00 (0.00%)	487802.00 (39.03%)
TPut 41	345001.00 (0.00%)	468021.00 (35.66%)
TPut 42	336177.00 (0.00%)	462260.00 (37.50%)
TPut 43	329169.00 (0.00%)	467906.00 (42.15%)
TPut 44	329475.00 (0.00%)	470784.00 (42.89%)
TPut 45	323845.00 (0.00%)	450739.00 (39.18%)
TPut 46	323878.00 (0.00%)	435457.00 (34.45%)
TPut 47	310524.00 (0.00%)	403914.00 (30.07%)
TPut 48	311843.00 (0.00%)	459017.00 (47.19%)
	Balancing Disabled	Balancing Enabled
Expctd Warehouse	48.00 (0.00%)	48.00 (0.00%)
Expctd Peak Bops	310524.00 (0.00%)	403914.00 (30.07%)
Actual Warehouse	25.00 (0.00%)	29.00 (16.00%)
Actual Peak Bops	429052.00 (0.00%)	543016.00 (26.56%)
SpecJBB Bops	6364.00 (0.00%)	9368.00 (47.20%)
SpecJBB Bops/JVM	6364.00 (0.00%)	9368.00 (47.20%)

Automatic NUMA Balancing simplifies tuning workloads for high performance on NUMA machines. Where possible, it is still recommended to statically tune the workload to partition it within each node. However, in all other cases, automatic NUMA balancing should boost performance.

Chapter 12. Power management

12.1. Power management at CPU Level

At the CPU level, you can control power usage in several ways. For example, by using idling power states (C-states), changing CPU frequency (P-states), and throttling the CPU (T-states). The following sections give a short introduction to each approach and its significance for power savings. Detailed specifications can be found at https://uefi.org/sites/default/files/resources/ACPI_Spec_6_4_Jan22.pdf.

12.1. C-states (processor operating states)

Modern processors have several power saving modes called C-states. They reflect the capability of an idle processor to turn off unused components to save power.

When a processor is in the C0 state, it is executing instructions. A processor running in any other C-state is idle. The higher the C number, the deeper the CPU sleep mode: more components are shut down to save power. Deeper sleep states can save large amounts of energy. Their downside is that they introduce latency. This means, it takes more time for the CPU to go back to C0. Depending on workload (threads waking up, triggering CPU usage and then going back to sleep again for a short period of time) and hardware (for example, interrupt activity of a network device), disabling the deepest sleep states can increase overall performance. For details on how to do so, refer to *the section called “Viewing kernel idle statistics with **cpupower**”*.

Some states also have submodes with different power saving latency levels. Which C-states and submodes are supported depends on the respective processor. However, C1 is always available.

Table 12.1, “C-states” gives an overview of the most common C-states.

Table 12.1. C-states

Mode	Definition
C0	Operational state. CPU fully turned on.
C1	First idle state. Stops CPU main internal clocks via software. Bus interface unit and APIC are kept running at full speed.
C2	Stops CPU main internal clocks via hardware. State in which the processor maintains all software-visible states, but may take longer to wake up through interrupts.
C3	Stops all CPU internal clocks. The processor does not need to keep its cache coherent, but maintains other states. Some processors have variations of the C3 state that differ in how long it takes to wake the processor through interrupts.

To avoid needless power consumption, it is recommended to test your workloads with deep sleep states enabled versus deep sleep states disabled. For more information, refer to *the section called “Viewing kernel idle statistics with **cpupower**”* or the **cpupower-idle-set(1)** man page.

12.2. P-states (processor performance states)

While a processor operates (in C0 state), it can be in one of several CPU performance states (P-states). Whereas C-states are idle states (all but C0), P-states are operational states that relate to CPU frequency and voltage.

The higher the P-state, the lower the frequency and voltage at which the processor runs. The number of P-states is processor-specific and the implementation differs across the different types. However, P0 is always the highest-performance state (except for *the section called “Turbo features”*). Higher P-state numbers represent slower processor speeds and lower power consumption. For example, a processor in P3 state runs more slowly and uses less power than a processor running in the P1 state. To operate at any P-state, the processor must be in the C0 state, which means that it is working and not idling. The CPU P-states are also defined in the ACPI specification, see https://uefi.org/sites/default/files/resources/ACPI_Spec_6_5_Aug29.pdf.

C-states and P-states can vary independently of one another.

12.3. Turbo features

Turbo features allow to dynamically overtick active CPU cores while other cores are in deep sleep states. This increases the performance of active threads while still complying with Thermal Design Power (TDP) limits.

However, the conditions under which a CPU core can use turbo frequencies are architecture-specific. Learn how to evaluate the efficiency of those new features in *the section called “The cpupower tools”*.

12.2. In-kernel governors

The in-kernel governors belong to the Linux kernel CPUfreq infrastructure and can be used to dynamically scale processor frequencies at runtime. You can think of the governors as a sort of preconfigured power scheme for the CPU. The CPUfreq governors use P-states to change frequencies and lower power consumption. The dynamic governors can switch between CPU frequencies, based on CPU usage, to allow for power savings while not sacrificing performance.

The following governors are available with the CPUfreq subsystem:

Performance governor

The CPU frequency is statically set to the highest possible for maximum performance. Consequently, saving power is not the focus of this governor.

See also the section called “*Tuning options for P-states*”.

Powersave governor

The CPU frequency is statically set to the lowest possible. This can have severe impact on the performance, as the system never rises above this frequency no matter how busy the processors are. An important exception is the `intel_pstate` which defaults to the powersave mode. This is due to a hardware-specific decision but functionally it operates similarly to the on-demand governor.

However, using this governor often does not lead to the expected power savings as the highest savings can be achieved at idle through entering C-states. With the powersave governor, processes run at the lowest frequency and thus take longer to finish. This means it takes longer until the system can go into an idle C-state.

Tuning options: The range of minimum frequencies available to the governor can be adjusted (for example, with the **cpupower** command line tool).

On-demand governor

The kernel implementation of a dynamic CPU frequency policy: The governor monitors the processor usage. When it exceeds a certain threshold, the governor sets the frequency to the highest available. If the usage is less than the threshold, the next lowest frequency is used. If the system continues to be underemployed, the frequency is again reduced until the lowest available frequency is set.

Drivers and in-kernel governors



Not all drivers use the in-kernel governors to dynamically scale power frequency at runtime. For example, the `intel_pstate` driver adjusts power frequency itself. Use the **cpupower frequency-info** command to find out which driver your system uses.

12.3. The **cpupower** tools

The **cpupower** tools are designed to give an overview of *all* CPU power-related parameters that are supported on a given machine, including turbo (or boost) states. Use the toolset to view and modify settings of the kernel-related CPUfreq and cpuidle systems and other settings not related to frequency scaling or idle states. The integrated monitoring framework can access both kernel-related parameters and hardware statistics. Therefore, it is ideally suited for performance benchmarks. It also helps you to identify the dependencies between turbo and idle states.

After installing the `cpupower` package, view the available **cpupower** subcommands with **cpupower --help**. Access the general man page with **man cpupower**, and the man pages of the subcommands with **man cpupower-SUBCOMMAND**.

12.1. Viewing current settings with **cpupower**

The **cpupower frequency-info** command shows the statistics of the `cpufreq` driver used in the kernel. Additionally, it shows if turbo (boost) states are supported and enabled in the BIOS. Run without any options, it shows an output similar to the following:

Example 12.1. Example output of **cpupower frequency-info**

```
#cpupower frequency-info
analyzing CPU 0:
  driver: intel_pstate
  CPUs which run at the same hardware frequency: 0
  CPUs which need to have their frequency coordinated by software: 0
  maximum transition latency: 0.97 ms.
  hardware limits: 1.20 GHz - 3.80 GHz
  available cpufreq governors: performance, powersave
  current policy: frequency should be within 1.20 GHz and 3.80 GHz.
                  The governor "powersave" may decide which speed to use
                  within this range.
  current CPU frequency is 3.40 GHz (asserted by call to hardware).
  boost state support:
    Supported: yes
    Active: yes
    3500 MHz max turbo 4 active cores
    3600 MHz max turbo 3 active cores
    3600 MHz max turbo 2 active cores
    3800 MHz max turbo 1 active cores
```

To get the current values for all CPUs, use **cpupower -c all frequency-info**.

12.2. Viewing kernel idle statistics with **cpupower**

The **idle-info** subcommand shows the statistics of the `cpuidle` driver used in the kernel. It works on all architectures that use the `cpuidle` kernel framework.

Example 12.2. Example output of `cpupower idle-info`

```
#cpupower idle-info
CPUidle driver: intel_idle
CPUidle governor: menu

Analyzing CPU 0:
Number of idle states: 6
Available idle states: POLL C1-SNB C1E-SNB C3-SNB C6-SNB C7-SNB
POLL:
Flags/Description: CPUIDLE CORE POLL IDLE
Latency: 0
Usage: 163128
Duration: 17585669
C1-SNB:
Flags/Description: MWAIT 0x00
Latency: 2
Usage: 16170005
Duration: 697658910
C1E-SNB:
Flags/Description: MWAIT 0x01
Latency: 10
Usage: 4421617
Duration: 757797385
C3-SNB:
Flags/Description: MWAIT 0x10
Latency: 80
Usage: 2135929
Duration: 735042875
C6-SNB:
Flags/Description: MWAIT 0x20
Latency: 104
Usage: 53268
Duration: 229366052
C7-SNB:
Flags/Description: MWAIT 0x30
Latency: 109
Usage: 62593595
Duration: 324631233978
```

After finding out which processor idle states are supported with `cpupower idle-info`, individual states can be disabled using the `cpupower idle-set` command. Typically one wants to disable the deepest sleep state, for example:

```
#cpupower idle-set -d 5
```

Or, for disabling all CPUs with latencies equal to or higher than 80:

```
#cpupower idle-set -D 80
```

12.3. Monitoring kernel and hardware statistics with `cpupower`

Use the `monitor` subcommand to report processor topology, and monitor frequency and idle power state statistics over a certain period of time. The default interval is 1 second, but it can be changed with the `-i`. Independent processor sleep states and frequency counters are implemented in the tool—some retrieved from kernel statistics, others reading out hardware registers. The available monitors depend on the underlying hardware and the system. List them with `cpupower monitor -l`. For a description of the individual monitors, refer to the `cpupower-monitor` man page.

The **monitor** subcommand allows you to execute performance benchmarks. To compare kernel statistics with hardware statistics for specific workloads, concatenate the respective command, for example:

```
cpupowermonitor db_test.sh
```

Example 12.3. Example **cpupower monitor** output

```
#cpupower monitor
|Mperf || Idle_Stats
①      ②
CPU | C0 | Cx | Freq | POLL | C1 | C2 | C3
0 | 3.71 | 96.29 | 2833 | 0.00 | 0.00 | 0.02 | 96.32
1 | 100.0 | -0.00 | 2833 | 0.00 | 0.00 | 0.00 | 0.00
2 | 9.06 | 90.94 | 1983 | 0.00 | 7.69 | 6.98 | 76.45
3 | 7.43 | 92.57 | 2039 | 0.00 | 2.60 | 12.62 | 77.52
```

- ❶ Mperf shows the average frequency of a CPU, including boost frequencies, over time. Additionally, it shows the percentage of time the CPU has been active (C0) or in any sleep state (Cx). As the turbo states are managed by the BIOS, it is impossible to get the frequency values at a given instant. On modern processors with turbo features the Mperf monitor is the only way to find out about the frequency a certain CPU has been running in.
- ❷ Idle_Stats shows the statistics of the cpuidle kernel subsystem. The kernel updates these values every time an idle state is entered or left. Therefore, there can be a few inaccuracies when cores are in an idle state for some time when the measure starts or ends.

Apart from the (general) monitors in the example above, other architecture-specific monitors are available. For detailed information, refer to the **cpupower-monitor** man page.

By comparing the values of the individual monitors, you can find correlations and dependencies and evaluate how well the power saving mechanism works for a certain workload. In *Example 12.3* you can see that CPU 0 is idle (the value of Cx is near 100%), but runs at a high frequency. This is because the CPUs 0 and 1 have the same frequency values which means that there is a dependency between them.

12.4. Modifying current settings with **cpupower**

You can use **cpupower frequency-set** command as root to modify current settings. It allows you to set values for the minimum or maximum CPU frequency the governor may select or to create a new governor. With the **-c** option, you can also specify for which of the processors the settings should be modified. That makes it easy to use a consistent policy across all processors without adjusting the settings for each processor individually. For more details and the available options, see the man page **cpupower-frequency-set** or run **cpupower frequency-set --help**.

12.4. Special tuning options

The following sections highlight important settings.

12.1. Tuning options for P-states

The CPUfreq subsystem offers several tuning options for P-states: You can switch between the different governors, influence minimum or maximum CPU frequency to be used or change individual governor parameters.

To switch to another governor at runtime, use **cpupower frequency-set** with the **-g** option. For example, running the following command (as root) will activate the performance governor:

```
#cpupower frequency-set -g performance
```

To set values for the minimum or maximum CPU frequency the governor may select, use the **-d** or **-u** option, respectively.

12.5. Troubleshooting

BIOS options enabled?

To use C-states or P-states, check your BIOS options:

- To use C-states, make sure to enable CPU C State or similar options to benefit from power savings at idle.
- To use P-states and the CPUfreq governors, make sure to enable Processor Performance States options or similar.
- Even if P-states and C-states are available, it is possible that the platform firmware is managing CPU frequencies which may be sub-optimal. For example, if pcc-cpufreq is loaded then the OS is only giving hints to the firmware, which is free to ignore the hints. This can be addressed by selecting "OS Management" or similar for CPU frequency managed in the BIOS. After reboot, an alternative driver will be used but the performance impact should be carefully measured.

In case of a CPU upgrade, make sure to upgrade your BIOS, too. The BIOS needs to know the new CPU and its frequency stepping to pass this information on to the operating system.

Log file information?

Check the systemd journal (see Chapter 21, **journalctl**: query the systemd journal in "[Administration Guide](#)") for any output regarding the CPUfreq subsystem. Only severe errors are reported there.

If you suspect problems with the CPUfreq subsystem on your machine, you can also enable additional debug output. To do so, either use **cpufreq.debug=7** as boot parameter or execute the following command as root:

```
#echo 7 > /sys/module/cpufreq/parameters/debug
```

This will cause CPUfreq to log more information to **dmesg** on state transitions, which is useful for diagnosis. But as this additional output of kernel messages can be rather comprehensive, use it only if you are sure that a problem exists.

12.6. More information

Platforms with a Baseboard Management Controller (BMC) may have additional power management configuration options accessible via the service processor. These configurations are vendor specific and therefore not subject of this guide. For more information, refer to the manuals provided by your vendor.

12.7. Monitoring power consumption with powerTOP

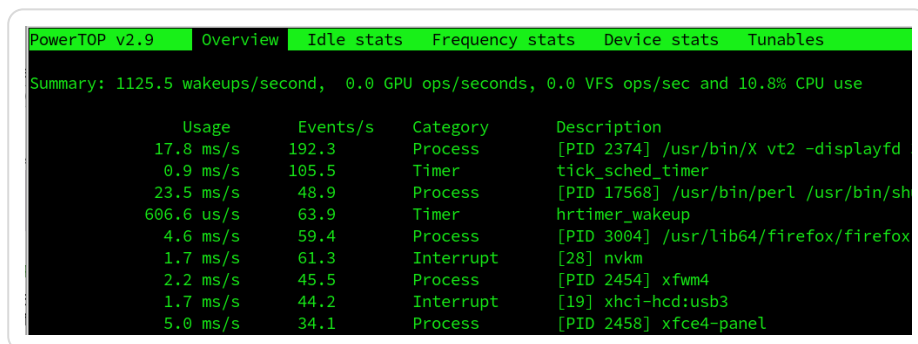
powerTOP helps to identify the causes of unnecessary high power consumption. This is especially useful for laptops, where minimizing power consumption is more important. It supports both Intel and AMD processors. Install it in the usual way:

```
>sudozypper in powertop
```

powerTOP combines several sources of information (analysis of programs, device drivers, kernel options, number and sources of interrupts waking up processors from sleep states) and provides several ways of viewing them. You can launch it in interactive mode, which runs in an ncurses session (see *Figure 12.1, “powerTOP in interactive mode”*):

```
>sudopowertop
```

Figure 12.1. powerTOP in interactive mode



powerTOP supports exporting reports to HTML and CSV. The following example generates a single report of a 240-second run:

```
>sudopowertop --iteration=1 --time=240 --html=POWERREPORT.HTML
```

It can be useful to run separate reports over time. The following example runs powerTOP 10 times for 20 seconds each time, and creates a separate HTML report for each run:

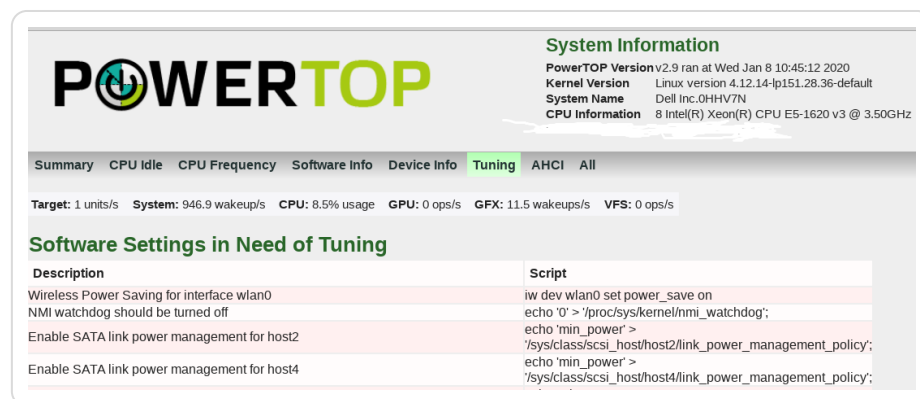
```
>sudo powertop --iteration=10 --time=20 --html=POWERREPORT.HTML
```

This creates 10 time-stamped reports:

```
powerreport-20200108-104512.html  
powerreport-20200108-104451.html  
powerreport-20200108-104431.html  
[...]
```

An HTML report looks like *Figure 12.2, “HTML powerTOP report”*:

Figure 12.2. HTML powerTOP report



The Tuning tab of the HTML reports, and the Tunables tab in the interactive mode, both provide commands for testing the various power settings. The HTML report prints the commands, which you can copy to a root command line for testing, for example **echo '0' > '/proc/sys/kernel/nmi_watchdog'**. The ncurses mode provides a simple toggle between Good and Bad. Good runs a command to enable power saving, and Bad turns off power saving. Enable all powerTOP settings with one command:

```
>sudo powertop --auto-tune
```

None of these changes survive a reboot. To make any changes permanent, use **sysctl**, **udev**, or **systemd** to run your selected commands at boot. powerTOP includes a systemd service file, /usr/lib/systemd/system/powertop.service. This starts powerTOP with the **--auto-tune** option:

```
ExecStart=/usr/sbin/powertop --auto-tune
```

Test this carefully before launching the systemd service, to see if it gives the results that you want. You should not use USB keyboards, and mice should not enter power save mode to avoid constantly waking them up and disturbing other devices. For easier testing and configuration editing, extract the commands from an HTML report with **awk**:

```
>awk -F '</?td ?>' '/tune/ { print $4 }' POWERREPORT.HTML
```

In calibrate mode, powerTOP sets up several runs that use different idle settings for backlight, CPU, Wi-Fi, USB devices, and disks, and helps to identify optimal brightness settings on battery power:

```
> sudo powertop --calibrate
```

You may call a file that creates a workload for more accurate calibration:

```
> sudo powertop --calibrate --workload=FILENAME --html=POWERREPORT.HTML
```

For more information, see:

- The powerTOP project page at <https://01.org/powertop>
- *the section called “System control parameters: /proc/sys/ ”*
- Chapter 19, The systemd daemon in “[Administration Guide](#)”
- Chapter 29, Dynamic kernel device management with udev in “[Administration Guide](#)”

Chapter 13. Adaptive and dynamic tuning using TuneD

13.1. Introduction to TuneD

The TuneD application monitors and adjusts system settings on certain SUSE Linux Enterprise Server workloads based on usage of different system resources, such as the CPU and the disk. The primary goal of TuneD is to deliver energy efficiency, optimized performance and efficient resource utilization without requiring manual configuration of low-level system settings, which can be complex and error-prone.

System administrators can apply predefined or customized *TuneD profiles* based on the current usage scenario or workload on the system. These profiles contain settings for certain system components such as the CPU, disk I/O schedulers, virtual memory, and power management. The system components are managed by *plug-ins*, which tune the system based on the activated profile settings.

13.2. Components of TuneD

The TuneD application consists of the following key components:

tuned daemon

The core daemon process that runs in the background, monitoring system usage and applying tuning settings. It handles profile switching and coordination of the other components. The following items are associated with the daemon:

- The **tuned** command that you can use after enabling TuneD using **systemctl**. By default, when invoked from the terminal manually, **tuned** runs as a normal process. However, you can use the `--daemon` option to run it as a background process.
- A **tuned systemd** service that manages the daemon process, such as starting the daemon at boot time, stopping or restarting it, and ensuring it runs with the necessary permissions and environment.

For more information, read the man page for **tuned**.

```
>man tuned
```

tuned-adm

A command-line utility to manage or administer the **tuned** daemon. To understand the basics of managing the **tuned** daemon using **tuned-adm**, see *the section called “Using **tuned-adm**”*.

Profiles

A *TuneD profile* is a collection of settings for system components that you can apply to tune the system. System administrators can either use the supported profiles that are installed at `/usr/lib/tuned/` as part of the tuned package, or apply customized profiles by creating them at `/etc/tuned/`. If the filenames for a custom and a supported profile matches, the custom profile takes precedence when applied. For more information, see *the section called “TuneD profiles”*.

Plug-ins

A *TuneD plug-in* is an implementation of tuning logic for different subsystems. A plug-in is invoked when a profile contains a setting that is relevant to the subsystem it controls. The plug-in tunes the subsystem based on the value of the relevant setting. A few key and most frequently used plug-ins manage subsystems, such as the CPU, disk, network, virtual machines, video and audio. In addition, there are optional monitoring plug-ins that monitor certain subsystems and pass on relevant information to the daemon for dynamic tuning. For more information, see *the section called “TuneD plug-ins”*.

Extension scripts

System administrators can extend the ecosystem of TuneD profiles and plug-ins using custom scripts that can be executed before or after a profile is applied. To use an extension script, specify the path and the type of the script in the profile configuration. Such extensibility allows the implementation of custom tuning logic and offers more control over the entire process of applying profiles. For more information, see *the section called “Using profile hooks”*.

13.3. Managing TuneD

This section covers the essential tasks for managing the tuned service lifecycle on SUSE Linux Enterprise Server. It guides you through installing the tuned package from official repositories, enabling and disabling the associated systemd service for automatic startup, and running the tuned daemon either as a background process managed by systemd or as a normal foreground process. Proper management of the tuned service ensures that the dynamic tuning capabilities are available when required and can be controlled effectively, allowing you to optimize system performance and resource utilization based on your needs.

13.1. Installing TuneD

The recommended way to install a supported version of TuneD is to install the tuned package using the **zypper** package manager.



tuned is available only for SUSE Linux Enterprise Server

The tuned package is available only in the official SUSE Linux Enterprise Server repositories. There is no equivalent package available in the official SUSE Linux Enterprise Desktop repositories.

1. To install the tuned daemon, command-line utilities, profiles and plug-ins, run the following command:

```
>sudozypper install -y tuned
```

2. Verify the installation by running the following commands:

```
>tuned --help
```

```
>tuned-adm --help
```

3. To know more, read the *man pages* for tuned, tuned-adm, tuned-profiles and tuned.conf.

13.2. Enabling tuned

To enable tuned, perform the following procedure:

1. Check the status of the tuned service:

```
>sudo systemctl status tuned
```

By default, the status of the service is *disabled*.

2. To enable a systemd service for future boots so that the tuned daemon starts automatically at the boot time, run the following command:

```
>sudo systemctl enable tuned
```

However, this command does not start the tuned process. If you check the status of the systemd service now, it should be *enabled* and *inactive*.

13.3. Starting tuned

After enabling the associated systemd service for tuned, for the current session you can start tuned either as a background daemon or as a normal process connected to the TTY.

Alternatively, you can enable the associated systemd service and start the daemon simultaneously.

1. Depending on how you want to start tuned across sessions, you have the following options:

- To start tuned only for the current session, perform one of the following steps:

1. Start tuned as a background process and let systemd manage the states of the daemon:

```
>sudo systemctl start tuned
```



Profile run by tuned

Once activated, tuned starts tuning the system based on the default profile that is appropriate for the system, or the currently active profile. To change it, use the **tuned-adm** command. For more information, see *the section called “TuneD profiles”* and *the section called “TuneD plug-ins”*.

2. Start tuned as a normal process connected to the TTY only for the current session:

```
>sudo tuned
```



Running the tuned command

You can use this command to activate a profile, or even run it as a daemon. For more information, run the **tuned --help** command.

- To enable a systemd service for future boots and simultaneously start the tuned daemon in the current session, run the following command:

```
>sudo systemctl enable --now tuned
```

13.4. Disabling tuned

As a best practice, perform the following steps to disable tuned:

1. Check the status of the tuned systemd service:

```
>sudo systemctl status tuned
```

2. Turn off or disable all tuning settings applied earlier:

```
>sudo tuned-adm off
```

3. If the status of the tuned daemon is *active*, stop the daemon:

```
>sudo systemctl stop tuned
```


4. To stop the `tuned` daemon from being automatically activated during the next boot, you have two options:

- Disable only the associated `systemd` service:

```
>sudo systemctl disable tuned
```

- Simultaneously disable the `tuned` `systemd` service and stop the `tuned` daemon immediately:

```
>sudo systemctl disable --now tuned
```

13.4. TuneD profiles

TuneD optimizes SUSE Linux Enterprise Server systems using predefined profiles with settings tailored for different use cases. These profiles adjust several system settings, including CPU governor policies, disk I/O scheduling, network parameters, and kernel parameters, to enhance performance, energy efficiency, or other system characteristics. Each profile is designed for specific scenarios, such as high throughput, low latency, power saving, or virtualized environments. System administrators can switch between profiles using the `tuned-adm` command and customize or combine profiles to meet unique requirements. This section covers the basics of managing, creating and customizing TuneD profiles.

13.1. Supported TuneD profiles

SUSE Linux Enterprise Server supports the following profiles:

balanced

The `balanced` profile provides a general-purpose optimization of the system, offering a good compromise between performance and energy consumption. It dynamically adjusts CPU frequency and power states, and balances I/O and network performance with power saving, making it suitable for most desktop and server environments.

cpu-partitioning

The `cpu-partitioning` profile is designed for systems where CPU resources need to be partitioned for specific tasks or applications. It configures CPU isolation and affinity settings and adjusts scheduler parameters to ensure predictable performance and reduce interference between processes. The configuration variables for this profile are defined in `/etc/tuned/cpu-partitioning-variables.conf`.

desktop

The `desktop` profile optimizes the system for desktop environments, enhancing performance for graphical interfaces and desktop applications. It optimizes CPU and I/O

performance, settings for low-latency audio and video playback, and reduces power-saving measures to ensure a smooth and responsive user experience.

latency-performance

The `latency-performance` profile prioritizes low latency and deterministic performance over power savings. It sets the CPU governor to performance mode, adjusts kernel parameters to reduce latency, and disables power-saving features that could introduce delays, making it suitable for real-time applications and high-performance computing.

mssql

The `mssql` profile optimizes the system for running Microsoft SQL Server by tuning CPU and memory settings, optimizing disk I/O for database access patterns, and enhancing network settings for improved database connectivity.

network-latency

The `network-latency` profile optimizes the network performance for applications requiring low latency, such as financial trading platforms and real-time communication systems. It configures network settings to reduce latency, sets the CPU governor to performance mode, and adjusts kernel parameters to prioritize network traffic.

network-throughput

The `network-throughput` profile enhances the system for sustained high data transfer rates, particularly on older CPUs or high-speed networks. It tunes network stack parameters for maximum throughput and optimizes CPU settings to handle high network loads.

powersave

Optimizes the system for energy efficiency, possibly at the cost of performance. While using this configuration as a stand-alone or merged profile, carefully analyze the deployment use case and ensure that it saves more power than the `balanced` profile.

throughput-performance

The `throughput-performance` profile maximizes overall system performance for general-purpose servers handling diverse tasks. It sets the CPU governor to performance mode, tunes I/O and network settings for high throughput, and adjusts kernel parameters to enhance system performance.

virtual-guest

The `virtual-guest` profile optimizes performance and efficiency for virtual machines running as guests. It tunes CPU and memory settings for virtualized environments and

adjusts disk and network settings to ensure optimal resource utilization and VM performance.

virtual-host

The `virtual-host` profile enhances systems running KVM guests by optimizing resource allocation and performance. It configures CPU and memory settings for hosting multiple virtual machines, tunes I/O and network settings, and enhances kernel parameters to support virtualization.

For information on the TuneD profile configuration files, see the following sections:

- the section called “Supported profile configuration”
- the section called “Custom profile configuration”

13.2. Managing TuneD profiles

If you just want to activate a profile, pass the profile name to the **tuned** command:

```
>sudo tuned --profile PROFILE_NAME
```

However, the best practice for managing the lifecycle of TuneD profiles is to use the **tuned-adm** command-line tool.

13.1. Using tuned-adm

You can use the **tuned-adm** command to perform the following tasks:

- List all available profiles.

```
>sudo tuned-adm list
```

- List active profile.

```
>sudo tuned-adm active
```

- Display information about the current profile.

```
>sudo tuned-adm profile_info
```

- Display information about a specified profile.

```
>sudo tuned-adm profile_info PROFILE_NAME
```

- Recommend a profile based on the current system usage.

```
>sudo tuned-adm recommend
```

By default, TuneD in SUSE Linux Enterprise Server recommends a profile based on the configuration mentioned in `/usr/lib/tuned/recommend.d/50-tuned.conf`. You can

also define custom recommendation rules by creating the file `/etc/tuned/recommend.conf`, which takes precedence over the default rules.

- Display the current profile selection mode.

```
>sudo tuned-adm profile_mode
```

- Select a profile automatically, and switch to the recommended profile.

```
>sudo tuned-adm auto_profile
```

- Manually switch to a specified profile.

```
>sudo tuned-adm profile PROFILE_NAME
```

- Verify that a profile has been successfully applied, and the system state matches the profile's configurations.

```
>sudo tuned-adm verify
```

- Turn off all tunings.

```
>sudo tuned-adm off
```

For more information on **tuned-adm**, see its help information or man page.

```
>sudo tuned-adm --help
```

```
>man tuned-adm
```

13.2. Using profile hooks

Profile hooks are specific scripts that are executed at different stages of a profile's lifecycle. They allow for more granular control and customization of what happens when a profile is applied. Profile hooks are generally placed in the profile directory under `/etc/tuned/PROFILE_NAME/` for custom profiles, and `/usr/lib/tuned/PROFILE_NAME/` for the default or supported profiles.

Profile hooks can be used to execute custom commands or scripts before or after certain events, such as:

- Starting a profile
- Stopping a profile
- Verifying a profile
- Fully rolling back a profile

Example 13.1. Creating and applying a profile hook

For example, in a custom profile that mixes the `balanced` and `powersave` profiles, we can include a profile hook that displays certain relevant information before applying the profile.

1. Create a script named `start.sh` in the custom profile directory `/etc/tuned/CUSTOM_PROFILE_NAME/` with the following content:

Example 13.2. Custom hook script to handle specific tasks before applying the custom profile

```
#!/bin/bash

# Example task: Log the current CPU governor and energy performance bias
echo "Current CPU governor: $(cat /sys/devices/system/cpu/cpu*/cpufreq/
scaling_governor)" >>
/var/log/tuned/CUSTOM_PROFILE_NAME.log
echo "Current energy performance bias: $(cat /sys/devices/system/cpu/cpu*/
cpufreq/energy_performance_bias)" >> /var/log/tuned/CUSTOM_PROFILE_NAME.log

# Example task: Check for specific conditions before applying settings
if [ ! -d "/sys/devices/system/cpu/cpufreq/policy0" ]; then
    echo "CPU frequency scaling not available, aborting profile
application." >> /var/log/tuned/CUSTOM_PROFILE_NAME.log
    exit 1
fi

# Example task: Notify about the upcoming profile application
echo "Preparing to apply custom profile settings." >> /var/log/tuned/CUSTOM
_PROFILE_NAME.log
```

2. Ensure the script is executable:

```
>sudochmod +x /etc/tuned/CUSTOM_PROFILE_NAME/start.sh
```

3. Include the path to the custom script in `/etc/tuned/CUSTOM_PROFILE_NAME/tuned.conf` by using the `[script]` section:

Example 13.3. TuneD profile with custom script path

```
[main]
include=balanced

[cpu]
governor=powersave
energy_perf_bias=powersave

[disk]
# Inherit settings from balanced profile

[sysctl]
vm.swappiness=10

[script]
script=/etc/tuned/CUSTOM_PROFILE_NAME/start.sh
```

4. Apply the custom profile to test the custom script:

```
>sudo tuned-adm profile CUSTOM_PROFILE_NAME
```

13.5. TuneD configuration

Configuration files form the foundation of TuneD and its profiles. In SUSE Linux Enterprise Server, the `/etc/tuned/` file is the main configuration file for the `tuned` daemon. This file contains global settings that affect the overall behavior of TuneD.

Also, each TuneD profile has its `tuned.conf` configuration file that defines the specific tuning parameters and plug-in settings for a particular profile. The profile-specific configuration files are divided into sections, each corresponding to a specific TuneD plug-in or a group of related settings. System administrators can specify system tuning parameters across different subsystems, such as CPU, disk and network. For more information on TuneD plug-ins, see *the section called “TuneD plug-ins”*.

13.1. Global TuneD configuration

When `tuned` starts running (as a daemon, by default), the settings in the `/etc/tuned/tuned-main.conf` global configuration file are applied on the system. This action prepares the system to apply additional profile-specific tuning in the subsequent stages.

The default `/etc/tuned/tuned-main.conf` file contains the following global settings:

Example 13.4. Default global TuneD configuration

```
daemon = 1 ❶
dynamic_tuning = 1 ❷
sleep_interval = 1 ❸
update_interval = 10 ❹
recommend_command = 1 ❺
reapply_sysctl = 1 ❻
default_instance_priority = 0 ❼
udev_buffer_size = 1MB ❽
log_file_count = 2 ❾
log_file_max_size = 1MB ❿
```

- ❶ Specifies whether to use the `tuned` daemon. When set to 1, the daemon is used, enabling features such as D-Bus integration, settings rollback, hotplug support, and dynamic tuning. Disabling the daemon by setting it to 0 is *not recommended*, as it limits TuneD's functionality to static tuning only.
- ❷ Enables or disables dynamic tuning of devices. When set to 1, TuneD dynamically adjusts settings based on system activity. When disabled by setting it to 0, only static tuning is applied. Since dynamic tuning relies on the daemon to function, setting `dynamic_tuning=1` becomes irrelevant when the daemon is disabled by setting it to `daemon=0`.
- ❸ Defines the interval in seconds that the `tuned` daemon sleeps before checking for events. A higher value reduces overhead but increases response time to changes.

- ❹ Sets the interval in seconds for updating dynamic tunings. This value must be a multiple of `sleep_interval`.
- ❺ Controls the availability of the **tuned-adm recommend** command. When enabled by setting it to 1, tuned parses the custom recommendation configuration at `/etc/tuned/recommend.conf` or the default recommendation configuration `/usr/lib/tuned/recommend.d/50-tuned.conf` and provides profile recommendations. When disabled by setting it to 0, the daemon returns a single hard-coded profile, typically balanced.
- ❻ Determines whether system `sysctl` settings from files such as `/etc/sysctl.conf` and `/etc/sysctl.d` should be reapplied after TuneD `sysctl` settings are applied. When enabled by setting to 1, system `sysctl` settings override TuneD `sysctl` settings.
- ❼ Specifies the default priority assigned to the TuneD instances. Higher values indicate a higher priority.
- ❽ Defines the buffer size for `udev` events. This setting helps manage the amount of data processed from `udev`.
- ❾ Sets the number of log files to keep in the directory `/var/log/tuned/`. This helps in log rotation, where older logs are archived and new logs are created.
- ❿ Specifies the maximum size of each log file before it is rotated. This prevents log files from growing indefinitely. For example, when the default log file `/var/log/tuned/tuned.log` reaches the size set by this parameter, a new file is started.



Warning

Modify the values of the global settings in `/etc/tuned/` only when you are sure of its effects on the behavior of TuneD.

For a detailed overview of the main TuneD configuration file, read its man page:

```
>man 5 tuned-main.conf
```

13.2. Supported profile configuration

Every supported TuneD profile has its own directory under `/usr/lib/tuned/`, and each such profile directory contains the `/usr/lib/tuned/PROFILE_NAME/tuned.conf` configuration file. The configuration file contains a default set of plug-ins, parameters and options that are appropriate for the profile.

13.1. General syntax of TuneD profiles

The general syntax for a profile configuration file is as follows:

```
[main]
summary=SUMMARY_TEXT_STRING
include=ANOTHER_PROFILE_NAME_AS_BASE

[PLUG-IN]
PARAMETER=OPTIONPARAMETER=OPTION
...

[PLUG-IN]
# COMMENT_STRINGPARAMETER=OPTION
...
```

For an overview of the Tuned configuration files, read its man page:

```
>man 5 tuned.conf
```

13.3. Custom profile configuration

Also, you can define **custom** profiles. For each custom profile, create a *PROFILE_NAME* directory in */etc/tuned/*, and place its configuration file within its own directory. For example, the configuration file for a custom profile can be at the path */etc/tuned/PROFILE_NAME/tuned.conf*. However, if there is a match between *PROFILE_NAME* in the default path */usr/lib/tuned/PROFILE_NAME* path and the custom path */etc/tuned/PROFILE_NAME*, the custom profile configuration is prioritized and applied. As a best practice for customizing a supported profile, copy it from */usr/lib/tuned/* to */etc/tuned/* and then modify it. This preserves the original profile and separates custom configurations.



Path to Tuned profile configuration files

Although *not recommended*, you can specify paths in the */etc/tuned/* other than the standard paths for supported and custom profiles:

```
profile_dirs=/etc/tuned/,/usr/lib/tuned/,/PATH/TO/CUSTOM/DIRECTORY/
```

The directories are searched for profiles according to the order of their listing. In the above example, */etc/tuned/* is searched first, followed by */usr/lib/tuned/* and */PATH/TO/CUSTOM/DIRECTORY/*.



Warning

SUSE does not support any profile that is not part of the supported profiles supplied with the tuned from official SUSE Linux Enterprise Server repositories. Create and apply a custom profile on a production system only if you are sure of its effect.

13.1. Creating a custom profile by merging configurations

As a best practice, always merge two or more profiles by manually creating a custom profile configuration. This approach makes you aware of the exact mix of parameters and plug-ins in your custom merged profile. It is a more involved process compared to applying two or more profiles simultaneously without checking the result of the merge that gets generated on-the-fly.

As an example, consider a simple merge of the `balanced` and `powersave` profiles.

Example 13.5. Merging `balanced` and `powersave` Tuned profiles

1. Create the path for the custom profile:

```
>sudo mkdir /etc/tuned/CUSTOM_PROFILE_NAME
```

2. Create a profile configuration file in the following directory:

```
>sudo vi /etc/tuned/CUSTOM_PROFILE_NAME/tuned.conf
```

3. Add the following configuration to the file:

Example 13.6. Merged configuration for `balanced` and `powersave` Tuned profiles

```
[main]
include=balanced ❶

[cpu] ❷
governor=powersave
energy_perf_bias=powersave

[disk] ❸
# Inherit settings from balanced profile

[sysctl]
vm.swappiness=10 ❹
```

- ❶ Include the `balanced` profile as a base, which is suitable for general purpose use.
 - ❷ Overrides CPU parameters such as `governor` and `energy-perf-bias` to `powersave`.
 - ❸ Inherit disk settings from the `balanced` profile.
 - ❹ Add a `sysctl` parameter to reduce swappiness, which can be beneficial for both performance and power consumption in certain scenarios.
4. Save the configuration file and exit the text editor.
 5. Apply the new custom profile after merging certain parameters from `balanced` and `powersave` profiles.

```
>sudo tuned-adm profile CUSTOM_PROFILE_NAME
```

13.6. TuneD plug-ins

TuneD plug-ins are modular components that extend the functionality of the TuneD system daemon. These plug-ins allow system administrators to create custom optimization profiles tailored to specific workloads or hardware configurations. By leveraging several system metrics and user-defined parameters, TuneD plug-ins can dynamically adjust kernel settings, CPU frequencies, disk I/O schedulers, and other low-level system parameters to achieve optimal performance and energy efficiency. While the core tuned package includes several preconfigured profiles, you can extend it through custom plug-ins for fine-grained control over the system.

The general syntax for including plug-ins in the TuneD profile configuration files is as follows:

```
...
[PLUGIN-NAME]
PARAMETER_1=VALUE_1PARAMETER_2=VALUE_2
...
```

The supported TuneD profiles for SUSE Linux Enterprise Server have multiple plug-ins and their parameters included within them. To see a list of all such plug-ins, run the following command:

```
>sudo grep -r '\[.*\]' /usr/lib/tuned/*/tuned.conf | sort -u
```

13.1. Supported TuneD plug-ins

On SUSE Linux Enterprise Server, the following TuneD plug-ins are supported:

main

`main` is not a plug-in, but a special section in the TuneD profile configuration files that provides overall settings and metadata for the profile. The most commonly used parameters for the `[main]` section are as follows:

- `summary`: A summary of the key functionality of the profile.
- `include`: Provision to include other profiles as a parent, so that its configuration can be inherited as a base. In case of conflict, the configuration of the child profile takes precedence over the configuration of the included profile.



Impact of updated TuneD profile

When the configuration for a parent profile is updated, the child profiles that inherit the parent profile also get impacted.

modules

The `modules` plug-in manages kernel modules within the profiles. It can load or unload specific modules when a profile is activated, allowing administrators to optimize system

behavior for different workloads by controlling which kernel modules are active. When this plug-in is used, Tuned writes information about the kernel modules in the `/etc/modprobe.d/tuned.conf` file.

If you need to add a kernel module parameter that should be handled by tuned, include it in the profile configuration using the following syntax:

```
[modules]
MODULE_NAME=MODULE_PARAMETERS❶
```

❶ You can mention multiple module parameters separated by comma.

If you want the module to be reloaded automatically, use the `+r` option:

```
[modules]
MODULE_NAME=+r,MODULE_PARAMETERS
```

For example:

Example 13.7. Automatic reload of kernel modules by Tuned

```
[modules]
cpufreq_conservative=+r,down_threshold=20,up_threshold=80,sampling_rate=20000❶
```

- ❶
 - `cpufreq_conservative` is a kernel module that adjusts the CPU clock speed based on the system load. It is more conservative in its frequency scaling compared to other governors like `ondemand`.
 - The `=+r` syntax indicates that the module should be loaded if it is not already, and Tuned keeps track of how many profiles are using this module. When a module with `+r` is loaded, Tuned increments the reference count for this module. If the profile is later deactivated, the reference count is decremented. The module is unloaded only when the reference count reaches zero, which implies that no other active profile is using it.
 - The parameters mean the following:
 - `down_threshold=20` indicate that when the CPU usage drops below 20%, the governor lowers the CPU frequency to conserve power.
 - `up_threshold=80` indicate that when the CPU usage exceeds 80%, the governor raises the CPU frequency to improve performance.
 - `sampling_rate=20000` indicate that the governor samples the CPU usage every 20,000 microseconds (or 20 milliseconds).

For more information on kernel modules, refer to Chapter 28, Managing kernel modules in “[Administration Guide](#)”. Additionally, you can check the parameters of the installed modules on your system by looking at `/sys/module/MODULE_NAME/parameters/`.

cpu

The `cpu` plug-in manages and optimizes CPU-related settings to enhance performance, power savings, or a balance between the two. This plug-in allows fine-grained control over how the CPU operates, including setting the frequency governor, energy performance bias, and other parameters that directly influence the CPU's behavior and performance characteristics.

Certain commonly used parameters for the `cpu` plug-in are as follows:

- **priority**: Sets the priority of CPU tuning operations. Higher values give the tuning operation a higher priority.
- **governor**: Sets the CPU frequency scaling governor. Common options for this parameter include the following:
 - **performance**: Forces the CPU to run at the maximum frequency, providing the best performance at the cost of higher power consumption.
 - **ondemand**: Dynamically adjusts the CPU frequency based on system load, providing a balance between performance and power saving.
 - **conservative**: Similar to the `ondemand` governor, but increases or decreases the CPU frequency more gradually. It is suitable for systems where power saving is important but performance should still be maintained.
- **energy_perf_bias**: Hints to the CPU how to balance between power consumption and performance. Common options for this parameter include the following:
 - **performance**: Bias towards maximum performance, possibly increasing power consumption.
 - **powersave**: Bias towards power saving, possibly reducing performance.
 - **power**: A more aggressive power-saving setting compared to `powersave`, but potentially less demanding compared to `performance`.
 - **normal**: A balanced setting between performance and power saving.
- **force_latency**: Controls the forced latency setting for the CPU. Lower values mean the system aggressively applies power-saving measures, possibly at the cost of increased latency. Higher values reduce the aggressiveness of power-saving measures, possibly improving responsiveness.
- **min_perf_pct**: Specifies the minimum performance level as a percentage of the CPU's maximum performance. For example, a value of 100 means the CPU always

runs at its maximum performance level. Lower values allow the CPU to run at lower performance levels when full performance is not needed, which can save power.

For example, a profile created for CPU-intensive workload, such as high-performance computing, can configure the plug-in for maximum CPU usage:

Example 13.8. TuneD plug-in configuration for CPU-intensive workload

```
[cpu]
priority=1
governor=performance
energy_perf_bias=performance
force_latency=10
min_perf_pct=100
```

For more information on CPU performance scaling, refer to <https://www.kernel.org/doc/Documentation/admin-guide/pm/cpufreq.rst>.

audio

The audio plug-in tunes the system to optimize audio performance. This can involve adjusting CPU scheduling policies, setting CPU affinity, and configuring other system parameters to minimize audio latency, ensure smooth audio playback, and record real-time audio activities. For example:

Example 13.9. TuneD plug-in configuration for optimal audio performance

```
[audio]
timeout=10 ❶
reset_controller=False ❷
```

- ❶ The `timeout` parameter specifies the time period (in seconds) for which the system should remain in the optimized state after the last audio activity is detected. This is intended to prevent the system from frequently switching back and forth between optimized and non-optimized states, which can cause instability or glitches in audio performance.
- ❷ The `reset_controller` parameter manages the behavior of the audio module's power-saving controller. It allows you to enable or disable the resetting of the controller, which can influence how the system handles power management for audio devices. The parameter interacts with a system file located at `/sys/module/AUDIO_MODULE_NAME/parameters/power_save_controller` (for example, `snd_hda_intel`). By default, it is set to `False`.

video

The video plug-in provides options for tuning power-saving settings for certain graphics cards, particularly AMD Radeon cards. For example:

[video]
`radeon_powersave=dpm-balanced`

Common parameters used with this plug-in are as follows:

- `radeon_powersave`: This parameter controls the power-saving mode of AMD Radeon graphics cards. It determines how the GPU handles power management, impacting performance and energy consumption. Configuration options used with this parameter include the following:
 - `default`: The default power-saving profile, which is the default setting provided by the graphics driver.
 - `auto`: Automatically adjusts power-saving settings based on current system usage.
 - `low`: A low-power profile that reduces performance to save energy.
 - `mid`: A medium-power profile that balances performance and power savings.
 - `high`: A high-power profile that maximizes performance at the cost of higher power consumption.
 - `dynpm`: Dynamic Power Management (DPM)—a mode that allows for dynamic adjustments based on the workload.
 - `dpm-battery`: A DPM profile optimized for battery usage.
 - `dpm-balanced`: A DPM profile that balances performance and power efficiency.
 - `dpm-performance`: A DPM profile optimized for maximum performance.

disk

The disk plug-in tunes and optimizes the performance and power consumption of disk drives.

Common parameters and their possible values are as follows:

- `devices`: Specifies on which disk devices the tuning parameters should be applied. By default, all devices are included. You can specify devices separated by comma.
- `dynamic`: Boolean values that enable or disable dynamic tuning based on disk load. By default, it is set to `True`.
- `elevator`: Configures the I/O scheduler (elevator) for the disk. Depending on the system's available schedulers, the possible options include `noop`, `deadline` or `cfq`.
- `apm`: An integer (1–254) that sets the Advanced Power Management (APM) level of the disk. Higher values typically mean better performance, but higher power consumption.

- **spindown:** The integer value for the `spindown` parameter is a special encoding used by the `hdparm` utility to set the spin-down timeout for a disk. For an explanation of the acceptable spin-down values, read about the `-S` option in the man page of `hdparm`.
- **readahead:** An integer that sets the `readahead` value for the disk in kilobytes. You can change this to sectors by adding the appropriate suffix. For example, `readahead = 8192 s`. Ensure there is at least one space between the number and the suffix.
- **readahead_multiply:** A float value that indicates the factor by which the value of the `readahead` parameter is multiplied.
- **scheduler_quantum:** An integer that sets the number of I/O requests the scheduler handles simultaneously. Setting a higher quantum can improve throughput for sequential I/O operations by allowing more requests to be processed before switching. Conversely, a lower quantum can improve responsiveness for random I/O operations by reducing latency.

As an example, consider the following disk plug-in configuration in a TuneD profile:

Example 13.10. disk plug-in configuration in a TuneD profile

```
[disk]
# Comma separated list of devices, all devices if commented out.
devices=sda,sdb

dynamic=True
elevator=deadline
apm=128
spindown=60

# Readahead adjusted to sectors by specifying relevant suffix.
# There must be at least one space between the number and suffix.
readahead=8192 s

readahead_multiply=1.5
scheduler_quantum=64
```

scsi_host

The `scsi_host` plug-in is used to manage power-saving settings for SCSI (Small Computer System Interface) host adapters on Linux systems. Commonly used parameters for this plug-in include the following:

- **alpm:** The Aggressive Link Power Management (ALPM) setting controls the power management policy for Serial ATA (SATA) links. It can take different values to balance between power savings and performance. The possible options for this parameter include the following:
 - **min_power:** This setting is more aggressive in terms of power savings. It puts the system into a lower power state when the SATA links are idle, which can save more power but may increase latency when accessing the disks.

- `medium_power`: This setting provides a balance between power savings and performance. It aims to reduce power consumption without impacting system performance too much.
- `max_performance`: This setting keeps SCSI hosts and devices in their highest power state. It prioritizes speed and responsiveness over energy efficiency, disabling power-saving features to minimize latency and maximize throughput.

For example, consider the following configuration that is appropriate for battery-powered devices, low-usage servers, and systems with long idle periods:

```
[scsi_host]
alpm=min_power
```



Read/write of link power management policy by Tuned

The possible values are typically read from or written to the `/sys/class/scsi_host/SCSI_HOST_NAME/link_power_management_policy` file. The exact values may depend on what the kernel and the hardware support.

variables

The `variables` plug-in allows for the definition of variables that you can use across different Tuned profiles. This is particularly useful for setting parameters that may change based on the system environment or user preferences. By centralizing variable definitions, you can simplify profile management and enhance the flexibility of your tuning configurations.

Within the `[variables]` section of your profile configuration, you can specify variables using the following methods:

```
[variables]
include=PATH/TO/VARIABLE/CONFIGURATION/FILE❶VAR_NAME=value❷VAR_ANOTHER=${VAR_NAME_suffix}❸VAR_DYNAMIC=$(uname -r)❹
```

- ❶ Include a configuration file that already contains variable definitions for a certain context. For example, you can include the `/etc/tuned/cpu-partitioning-variables.conf` file to make variables such as `isolated_cores` and its values available for the ongoing profile configuration.
- ❷ Define a variable directly by assigning a value.
- ❸ Define a variable using another variable.
- ❹ Define a variable using a command output.

For example, consider the following configuration for variables and its subsequent use in configuring other plug-ins for a profile configuration:

Example 13.11. Using dynamic variables for plug-in configurations in a Tuned profile

```
[variables]
include=/etc/tuned/cpu-partitioning-variables.conf

SWAPPINESS_VALUE=10
NETWORK_INTERFACE=$(ip -o -4 route show to default | awk '{print $5}')

[sysctl]
vm.swappiness=${SWAPPINESS_VALUE}

[net]
interface=${NETWORK_INTERFACE}
disable_offload=yes
```

For more technically involved variable definitions, inspect the `[variables]` plug-in configurations available in the supported profile configuration files. To get a list of such profile configuration paths where the `[variables]` plug-in is included, run the following command:

```
>sudo grep -r '\[.*\]' /usr/lib/tuned/*/tuned.conf | sort -u | grep
variables
```

sysctl

The `sysctl` plug-in offers a powerful way to customize several kernel parameters at runtime. This allows system administrators to optimize system performance, enhance security, and adjust network behavior without directly editing system files or running manual `sysctl` commands.

The `sysctl` plug-in accepts a wide range of parameters. Certain common categories include:

- Kernel parameters (`kernel.*`)
- Virtual Memory parameters (`vm.*`)
- Network parameters (`net.*`)
- File system parameters (`fs.*`)

where `*` is the wild card expression denoting permissible parameters for that category. When applying these parameters, Tuned might use the **`sysctl`** under the hood. For example:

```
>sysctl -q -w CATEGORY.PARAMETER=VALUE
```

Common parameters and their possible values for the `sysctl` plug-in are summarized below:

Understand the impact before modifying `sysctl` parameters

Modifying these parameters can considerably impact system behavior. It is crucial to understand their implications before making changes. Consult the official kernel documentation for detailed descriptions.

- `kernel.hung_task_timeout_secs`: Time in seconds before a task is considered hung (default: 120).
- `kernel.numa_balancing`: Enable (1) or disable (0) automatic NUMA balancing.
- `kernel.nmi_watchdog`: Enable (1) or disable (0) the NMI watchdog.
- `kernel.timer_migration`: Enable (1) or disable (0) timer migration.
- `kernel.sched_autogroup_enabled`: Enable (1) or disable (0) scheduler autogrouping.
- `kernel.sched_min_granularity_ns`: Minimum preemption granularity (in nanoseconds) for CPU-bound tasks.
- `kernel.sched_migration_cost_ns`: The time (in nanoseconds) the scheduler considers a migrated process “cache hot”.
- `vm.stat_interval`: Interval between updates of virtual machine statistics (default: 1).
- `vm.dirty_ratio`: The percentage of system memory that can be filled with dirty pages before the processes must write them to disk.
- `vm.dirty_background_ratio`: The percentage of system memory that can be filled with dirty pages before the background writeback process starts writing them to disk.
- `vm.swappiness`: The tendency of the kernel to swap out idle processes. A swappiness of 0 instructs the kernel to keep the processes in the main memory for as long as possible.
- `vm.max_map_count`: Maximum number of memory map areas a process may have.
- `net.core.busy_read`: Number of busy loops for `recvmsg` (default: 50).
- `net.core.busy_poll`: Number of busy loops for `poll` (default: 50).
- `net.ipv4.tcp_fastopen`: Enable (1) or disable (0) TCP Fast Open.
- `net.ipv4.tcp_rmem`: TCP read buffer size (minimum, default and maximum values separated by space).
- `net.ipv4.tcp_wmem`: TCP write buffer size (minimum, default and maximum values separated by space).
- `net.ipv4.udp_mem`: UDP buffer size (minimum, default and maximum values separated by space).

- `fs.aio-max-nr`: Maximum number of allowed concurrent asynchronous I/O requests.
- `fs.file-max`: Maximum number of file handles that the Linux kernel allocates.
- `fs.inotify.max_user_instances`: Maximum number of inotify instances per user.
- `fs.inotify.max_user_watches`: Maximum number of inotify watches per user.
- `fs.nr_open`: Maximum number of file descriptors a process can have.
- `fs.suid_dumpable`: Controls whether core dumps are produced for set-user-ID or set-group-ID programs.

As an example, consider the following configuration for the `sysctl` plug-in:

Example 13.12. Sample configuration for the `sysctl` plug-in in a TuneD profile

```
[sysctl]

# Kernel Parameters
kernel.hung_task_timeout_secs = 600
kernel.nmi_watchdog = 0
kernel.timer_migration = 1
kernel.sched_autogroup_enabled = 1
kernel.sched_min_granularity_ns = 100000000
kernel.sched_migration_cost_ns = 50000000
kernel.sched_latency_ns = 600000000
kernel.sched_wakeup_granularity_ns = 20000000
kernel.numa_balancing = 0

# Virtual Machine Parameters
vm.stat_interval = 10
vm.dirty_ratio = 10
vm.dirty_background_ratio = 3
vm.swappiness = 10
vm.max_map_count = 800000

# Network Parameters
net.core.busy_read = 50
net.core.busy_poll = 50
net.ipv4.tcp_fastopen = 3
net.ipv4.tcp_rmem = "4096 87380 16777216"
net.ipv4.tcp_wmem = "4096 16384 16777216"
net.ipv4.udp_mem = "3145728 4194304 16777216"

# Filesystem Parameters
fs.aio-max-nr = 1048576
fs.file-max = 2097152
fs.inotify.max_user_instances = 1024
fs.inotify.max_user_watches = 524288
fs.nr_open = 1048576
fs.suid_dumpable = 1
```



Best practices for changing `sysctl` settings

Optimal `sysctl` settings can vary greatly depending on the specific use case, hardware and workload of your system. It is recommended to test changes thoroughly and monitor system performance after applying new `sysctl` configurations.

sysfs

The `sysfs` plug-in accepts any valid file path under the system file system, or the `/sys/` directory. The plug-in checks whether a file path is valid and can read and write the values in the files that are part of the configuration. For example, consider the following configuration of the `sysfs` plug-in:

Example 13.13. `sysfs` plug-in configuration in a Tuned profile

```
[sysfs]
/sys/bus/workqueue/devices/writeback/cpumask = 2,6 ❶
/sys/devices/virtual/workqueue/cpumask = 3-5 ❷
/sys/devices/system/machinecheck/machinecheck*/ignore_ce = 1 ❸
```

- ❶ Single CPUs separated by comma. Only those CPUs that are not isolated using the `isolcpus` kernel boot parameter are allowed. To check the CPUs that are already isolated, run the **`grep isolcpus /proc/cmdline`** command. To check the total number of CPUs in your system, run the **`lscpu | grep "^CPU(s):"`** command.
- ❷ Range of CPUs. Only those CPUs that are not isolated using the `isolcpus` kernel boot parameter are allowed. To check the CPUs that are already isolated, run the command **`grep isolcpus /proc/cmdline`**. To check the total number of CPUs in your system, run the **`lscpu | grep "^CPU(s):"`** command.
- ❸ Corrected Errors (CEs) are hardware errors that the system has detected and corrected automatically. The `ignore_ce` attribute is a `sysfs` setting that controls whether the system must report these corrected errors. When `ignore_ce` is set to 1, the system ignores corrected errors, meaning they are not logged or reported to the operating system's error handling mechanisms. When `ignore_ce` is set to 0 (the default value), corrected errors are reported and logged by the system, allowing administrators to monitor them.

systemd

The `systemd` plug-in focuses on configuring the parameters of `systemd`, which is the system and service manager for SUSE Linux Enterprise Server. For example, consider the plug-in configuration for CPU affinity of `systemd` processes:

Example 13.14. systemd plug-in configuration in a Tuned profile

```
[systemd]
cpu_affinity=0,1,2 ❶
```

- ❶ Binds the systemd services to multiple specified CPU cores. By default, it is None. In this example, the services are configured to run on CPU cores 0, 1, and 2. Alternate ways of passing the values include passing a range of CPU cores, such as 0-2. However, you *cannot* pass on the CPU cores as values which you have declared as `isolated_cores` in the file `/etc/tuned/cpu-partitioning-variables.conf`. Configuring the `cpu_affinity` parameter in the plug-in is equivalent to configuring the `CPUAffinity` in the systemd configuration file `/etc/systemd/system.conf`.

script

The `script` plug-in allows users to execute custom scripts that Tuned runs at different stages. System administrators can use it to automate several system configurations and optimizations by specifying paths to such scripts. Common parameters and their possible values include the following:

- `priority`: An integer value that sets the priority level for the script. Tuned executes scripts based on their priority, with lower numbers indicating higher priority. So, a priority of 5 means this script has a low priority compared to scripts with a priority number lower than 5.
- `script`: This specifies the path to the script that Tuned executes. You can either provide the absolute path to the script as its value, or use the `PROFILE_DIR` variable to pass the relative path to the script.

For example, consider the following use of a relative script path:

Example 13.15. Using variables for including a relative script path in a Tuned profile

```
[script]
priority=5
script=${i:PROFILE_DIR}/script.sh
```



Resource for definition of variables

For a list of useful variables, inspect the constants defined in the source file <https://github.com/redhat-performance/tuned/blob/v2.10.0/tuned/consts.py>.

scheduler

The `scheduler` plug-in is used to manage the isolation of CPU cores and the selection of processes for management based on adding to an approved list or a blocklist. This plug-in is

primarily concerned with isolating specific CPU cores and managing which processes should or should not be managed by the scheduler. Commonly used parameters and their possible values include the following:

- `isolated_cores`: Isolates specific CPU cores from the general scheduler. Processes can be pinned to these isolated cores to ensure they run without interference from other system processes. To pass values to this parameter, use comma-separated integers representing the cores, or the bitmask of the integers in the hexadecimal format. For example, `isolated_cores=0,1` can also be represented by `0x3` in a more compact manner.
- `ps_whitelist`: A list of allowed process names that the scheduler plug-in is empowered to manage. Only processes matching this pattern are considered for scheduling adjustments. Pass regular expressions of process names as values to this parameter. For example, `.*` matches all processes, and `^bash$` matches only the bash process.
- `ps_blacklist`: A list of process names that the plug-in excludes from management. Processes matching this pattern are not adjusted by the plug-in. Pass regular expressions of process names as values to this parameter. For example, `.*` matches all processes, and `^idle$` matches only the idle process.

For example, consider the following configuration for the scheduler plug-in:

Example 13.16. scheduler plug-in configuration in a Tuned profile

```
isolated_cores=0x3
ps_whitelist=.*
ps_blacklist=^idle$;.*pmd.*;.*PMD.*;^DPDK;.*qemu-kvm.*
```

bootloader

The `bootloader` plug-in is used to apply performance profiles that require changes to the boot parameters of the system. These changes are made in the boot loader configuration file (typically GRUB 2 configuration) and are applied when the system boots up. This plug-in is particularly useful for optimizing system performance by setting specific kernel parameters, such as those controlling CPU isolation or power management, as well as other boot-time settings that can affect the overall system behavior and performance.

Common parameters and their values are summarized below:

- `grub2_cfg_file`: Specifies the GRUB 2 configuration file to modify. If not provided, the plug-in uses the default GRUB2 configuration files located in the system.
- `initrd_dst_img`: Sets the destination path for the initrd image. If specified, the initrd image is copied to this location. The path must be absolute, and if it is not provided, the default boot directory is used.

- `initrd_add_img`: Adds the path to an initrd image in the GRUB 2 configuration, so that custom initrd images can be used during the boot process.
- `initrd_add_dir`: Adds a directory to be included in the initrd image. This parameter specifies the path to a directory whose contents are packaged into an initrd image and included in the boot configuration. This is useful for adding custom modules or scripts to the initrd.
- `initrd_remove_dir`: A boolean option to specify whether to remove the directory specified in `initrd_add_dir` after it has been packaged into the initrd image. If set to `True`, the source directory is deleted after the initrd image is created.
- `cmdline`: Specifies additional kernel command-line parameters to be added to the GRUB 2 configuration. This allows the addition of custom kernel parameters that can optimize system performance or behavior based on the specific needs of the workload.

As an example, consider the following configuration for the `bootloader` plug-in:

Example 13.17. `bootloader` plug-in configuration in a Tuned profile

```
[bootloader]
cmdline="isolcpus=1-3 nohz_full=1-3 intel_pstate=disable"
initrd_add_img="/path/to/custom-initrd.img"
initrd_add_dir="/path/to/custom-initrd-dir"
initrd_remove_dir=True
initrd_dst_img="/boot/initrd-custom.img"
grub2_cfg_file="/etc/grub2.cfg"
```

net

The `net` plug-in is used to configure several parameters for optimizing network performance. Common parameters and their values used with this plug-in are summarized below:

- `devices`: Specifies a comma-separated list of the network device names. For example, `devices=eth0,eth1`. By default, the settings are applied on all network devices.
- `dynamic`: Boolean values (`true/false`) that enable or disable dynamic tuning based on network load.
- `wake_on_lan`: Configure Wake-on-LAN (WoL) settings using any combination of the following characters:
 - `p`: Wake on PHY activity.
 - `u`: Wake on unicast messages.
 - `m`: Wake on multicast messages.
 - `b`: Wake on broadcast messages.
 - `a`: Wake on ARP.

- g: Wake on MagicPacket™.
- s: Enable SecureOn™ for MagicPacket™.
- d: Disable WoL.
- `nf_conntrack_hashsize`: Integer value setting the size of the connection-tracking hash table.
- `features`: A dictionary of network device feature names and their desired states. For example, `tx-checksum: off`, `sg: on`.
- `coalesce`: A dictionary of packet coalescing parameters.
 - `adaptive-rx` and `adaptive-tx` have the possible values `on` or `off`.
 - The following ones have integer values: `rx-usecs`, `rx-frames`, `rx-usecs-irq`, `rx-frames-irq`, `tx-usecs`, `tx-frames`, `tx-usecs-irq`, `tx-frames-irq`, `stats-block-usecs`, `pkt-rate-low`, `rx-usecs-low`, `rx-frames-low`, `tx-usecs-low`, `tx-frames-low`, `pkt-rate-high`, `rx-usecs-high`, `rx-frames-high`, `tx-usecs-high`, `tx-frames-high`, `sample-interval`
- `pause`: A dictionary of pause frame parameters. The following pause frame parameters have `on` or `off` values: `autoneg`, `rx` and `tx`.
- `ring`: A dictionary of ring buffer sizes. Each of the following ring buffer size parameter has integer values: `rx`, `rx-mini`, `rx-jumbo` and `tx`.

As an example of configuring network device parameters, consider the following:

Example 13.18. `net` plug-in configuration in a Tuned profile

```
[net]
devices=eth0,eth1
dynamic=true
wake_on_lan=g
nf_conntrack_hashsize=16384
features=tx-checksum:off, sg:on
coalesce=adaptive-rx:on, rx-usecs:50, rx-frames:32
pause=autoneg:on, rx:on, tx:off
ring=rx:2048, tx:1024
```

vm

The `vm` plug-in optimizes memory management in virtualized environments. It provides mechanisms for tuning memory-related parameters, specifically focusing on the management of Transparent Hugepages (THP). Commonly used parameters and their values for this plug-in include the following:

- `transparent_hugepage` or `transparent_hugepages`: Often used interchangeably, these parameters are aliases for each other and perform the function

of controlling the behavior of Transparent Hugepages. The common properties and values associated with this parameter are as follows:

- `enabled`: Controls the overall enabling of transparent hugepages. Possible values are `always`, `madvise` and `never`.
- `defrag`: Determines when, if defragmentation should occur. Possible values are `always`, `defer`, `madvise`, `defer+madvise` and `never`.
- `khugepaged`: Configures settings related to the khugepaged daemon, including options such as `scan_sleep_millisecs` and `alloc_sleep_millisecs`.

For example, consider the following configuration of the `vm` plug-in in a TunedD profile:

Example 13.19. `vm` plug-in configuration in a TunedD profile

```
[vm]
transparent_hugepages.enabled=always
transparent_hugepage.defrag=always
transparent_hugepages.khugepaged.scan_sleep_millisecs=10000
```

`eeepc_she`

The `eeepc_she` plug-in enhances power management on ASUS EeePC netbooks, specifically those equipped with the Super Hybrid Engine (SHE) technology. It optimizes system performance by dynamically adjusting CPU frequencies and power states to achieve a balance between energy efficiency and processing power. The commonly used parameters and their values are as follows:

- `load_threshold_normal`: Threshold for load to switch to normal mode. Default value is `0.6`.
- `load_threshold_powersave`: Threshold for load to switch to powersave mode. Default value is `0.4`.
- `she_powersave`: Value to set SHE to power saving mode. Default value is `2`.
- `she_normal`: Value to set SHE to normal mode. Default value is `1`.

13.7. More information

For more information, refer to the following resources:

- The official Web site of the TunedD project: <https://tuned-project.org/>
- Source code repository on GitHub: <https://github.com/redhat-performance/tuned>. You may find the following resources to be particularly helpful:
 - Configuration files for several profiles are available at <https://github.com/redhat-performance/tuned/tree/v2.10.0/profiles>. These configuration examples can provide a good starting point for creating a custom profile that is not available as part of the tuned package for SUSE Linux Enterprise Server.

- If you are interested in implementing the plug-ins that monitor and dynamically tune the system based on values provided in the profiles, the source code is available at <https://github.com/redhat-performance/tuned/tree/v2.10.0/tuned/plugins>.



Refer to appropriate version of Tuned source code

Check for the version of `tuned` in your SUSE Linux Enterprise Server system by running the **`tuned --version`** command. On the GitHub repository, select the corresponding branch or tag for inspecting the source code of the specific `tuned` version installed on your system.

Part V. Kernel tuning

- 14 Tuning I/O performance **147**
- 15 Tuning the task scheduler **152**
- 16 Tuning the memory management subsystem **163**
- 17 Tuning the network **174**
- 18 Tuning SUSE Linux Enterprise for SAP **178**

Chapter 14. Tuning I/O performance

I/O scheduling controls how input/output operations are submitted to storage. SUSE Linux Enterprise Server offers several I/O algorithms—called elevators—suited for different workloads. Elevators can help to reduce seek operations and can prioritize I/O requests.

Choosing the best suited I/O elevator not only depends on the workload but on the hardware too. For example, single ATA disk systems, SSDs, RAID arrays, or network storage systems each require different tuning strategies.

14.1. Switching I/O scheduling

SUSE Linux Enterprise Server picks a default I/O scheduler at boot-time, which can be changed on the fly per block device. This makes it possible to set different algorithms, for example, for the device hosting the system partition and the device hosting a database.

The default I/O scheduler is chosen for each device based on whether the device reports to be rotational disk or not. For rotational disks, the BFQ I/O scheduler is picked. Other devices default to MQ-DEADLINE or NONE.

To change the elevator for a specific device in the running system, run the following command:

```
>sudo echo SCHEDULER > /sys/block/DEVICE/queue/scheduler
```

Here, *SCHEDULER* is one of bfq, none, kyber, or mq-deadline. *DEVICE* is the block device (sda for example). This change does not persist during reboot. For permanent I/O scheduler change for a particular device, copy `/usr/lib/udev/rules.d/60-io-scheduler.rules` to `/etc/udev/rules.d/60-io-scheduler.rules`, and edit the latter file to suit your needs.



Default scheduler on IBM Z

On IBM Z, the default I/O scheduler for a storage device is set by the device driver.



elevator boot parameter removed

The elevator boot parameter has been removed. The blk-mq I/O path replaces cfq and does not include the elevator boot parameter.

14.2. Available I/O elevators with blk-mq I/O path

Below is a list of elevators available on SUSE Linux Enterprise Server for devices that use the blk-mq I/O path. If an elevator has tunable parameters, they can be set with the command:

```
echo VALUE > /sys/block/DEVICE/queue/iosched/TUNABLE
```

In the command above, *VALUE* is the desired value for the *TUNABLE* and *DEVICE* is the block device.

To find out what elevators are available for a device (sda for example), run the following command (the currently selected scheduler is listed in brackets):

```
>cat /sys/block/sda/queue/scheduler  
[mq-deadline] kyber bfq none
```



Scheduler options when switching from Legacy Block to blk-mq I/O path

When switching from legacy block to blk-mq I/O path for a device, the none option is roughly comparable to noop, mq-deadline is comparable to deadline, and bfq is comparable to cfq.

14.1. MQ-DEADLINE

MQ-DEADLINE is a latency-oriented I/O scheduler. MQ-DEADLINE has the following tunable parameters:

Table 14.1. MQ-DEADLINE tunable parameters

File	Description
writes_starved	Controls how many times reads are preferred over writes. A value of 3 means that three read operations can be done before writes and reads are dispatched on the same selection criteria. Default is 3.
read_expire	Sets the deadline (current time plus the read_expire value) for read operations in milliseconds. Default is 500.
write_expire	Sets the deadline (current time plus the write_expire value) for write operations in milliseconds. Default is 5000.
front_merges	Enables (1) or disables (0) attempts to front merge requests. Default is 1.

File	Description
<code>fifo_batch</code>	Sets the maximum number of requests per batch (deadline expiration is only checked for batches). This parameter allows to balance between latency and throughput. When set to 1 (that is, one request per batch), it results in “first come, first served” behavior and lowest latency, with higher values increasing the throughput. Default is 16.

14.2. NONE

When **NONE** is selected as I/O elevator option for blk-mq, no I/O scheduler is used, and I/O requests are passed down to the device without further I/O scheduling interaction.

NONE is the default for NVM Express devices. With no overhead compared to other I/O elevator options, it is considered the fastest way of passing down I/O requests on multiple queues to such devices.

There are no tunable parameters for **NONE**.

14.3. BFQ (Budget Fair Queueing)

BFQ is a fairness-oriented scheduler. It is described as “a proportional-share storage-I/O scheduling algorithm based on the slice-by-slice service scheme of CFQ. But BFQ assigns budgets, measured in number of sectors, to processes instead of time slices.” (Source: [linux-4.12/block/bfq-iosched.c](https://www.kernel.org/doc/Documentation/block/bfq-iosched.c))

BFQ allows to assign I/O priorities to tasks which are taken into account during scheduling decisions (see the section called “Prioritizing disk access with **ionice**”).

BFQ scheduler has the following tunable parameters:

Table 14.2. BFQ tunable parameters

File	Description
<code>slice_idle</code>	Value in milliseconds specifies how long to idle, waiting for next request on an empty queue. Default is 8.
<code>slice_idle_us</code>	Same as <code>slice_idle</code> but in microseconds. Default is 8000.

File	Description
<code>low_latency</code>	Enables (1) or disables (0) BFQ's low latency mode. This mode prioritizes certain applications (for example, if interactive) such that they observe lower latency. Default is 1.
<code>back_seek_max</code>	Maximum value (in Kbytes) for backward seeking. Default is 16384.
<code>back_seek_penalty</code>	Used to compute the cost of backward seeking. Default is 2.
<code>fifo_expire_async</code>	Value (in milliseconds) is used to set the timeout of asynchronous requests. Default is 250.
<code>fifo_expire_sync</code>	Value in milliseconds specifies the timeout of synchronous requests. Default is 125.
<code>timeout_sync</code>	Maximum time in milliseconds that a task (queue) is serviced after it has been selected. Default is 124.
<code>max_budget</code>	Limit for number of sectors that are served at maximum within <code>timeout_sync</code> . If set to 0 BFQ internally calculates a value based on <code>timeout_sync</code> and an estimated peak rate. Default is 0 (set to auto-tuning).
<code>strict_guarantees</code>	Enables (1) or disables (0) BFQ specific queue handling required to give stricter bandwidth sharing guarantees under certain conditions. Default is 0.

14.4. KYBER

KYBER is a latency-oriented I/O scheduler. It makes it possible to set target latencies for reads and synchronous writes and throttles I/O requests in order to try to meet these target latencies.

Table 14.3. KYBER tunable parameters

File	Description
<code>read_lat_nsec</code>	Sets the target latency for read operations in nanoseconds. Default is 2000000.
<code>write_lat_nsec</code>	Sets the target latency for write operations in nanoseconds. Default is 10000000.

14.3. I/O barrier tuning

Certain file systems (for example, Ext3 or Ext4) send write barriers to disk after `fsync` or during transaction commits. Write barriers enforce proper ordering of writes, making volatile disk write caches safe to use (with performance penalty). If your disks are battery-backed in one way or another, disabling barriers can safely improve performance.

nobarrier is deprecated in XFS



The `nobarrier` option is deprecated for XFS and is not a valid mount option in SUSE Linux Enterprise 15 SP2 and upward. Any XFS mount command that explicitly specifies the flag may fail to mount the file system. To prevent this from happening, make sure that no scripts or `fstab` entries contain the `nobarrier` option.

Sending write barriers can be disabled using the `nobarrier` mount option.

Disabling barriers can lead to data loss



Disabling barriers when disks cannot guarantee caches are properly written in case of power failure can lead to severe file system corruption and data loss.

Chapter 15. Tuning the task scheduler

Modern operating systems, such as SUSE® Linux Enterprise Server, normally run many tasks at the same time. For example, you can be searching in a text file while receiving an e-mail and copying a big file to an external hard disk. These simple tasks require many additional processes to be run by the system. To provide each task with its required system resources, the Linux kernel needs a tool to distribute available system resources to individual tasks. And this is exactly what the *task scheduler* does.

The following sections explain the most important terms related to a process scheduling. They also introduce information about the task scheduler policy, scheduling algorithm, description of the task scheduler used by SUSE Linux Enterprise Server, and references to other sources of relevant information.

15.1. Introduction

The Linux kernel controls the way that tasks (or processes) are managed on the system. The task scheduler, sometimes called *process scheduler*, is the part of the kernel that decides which task to run next. It is responsible for the optimal use of system resources to guarantee that multiple tasks are being executed simultaneously. This makes it a core component of any multitasking operating system.

15.1. Preemption

The theory behind task scheduling is simple. If there are runnable processes in a system, at least one process must always be running. If there are more runnable processes than processors in a system, not all the processes can be running all the time.

Therefore, certain processes need to be stopped temporarily, or *suspended*, so that others can be running again. The scheduler decides what process in the queue runs next.

As already mentioned, Linux, like all other Unix variants, is a *multitasking* operating system. That means that several tasks can be running at the same time. Linux provides a so called *preemptive* multitasking, where the scheduler decides when a process is suspended. This forced suspension is called *preemption*. All Unix flavors have been providing preemptive multitasking since the beginning.

15.2. Timeslice

The time period for which a process runs before it is *preempted* is defined in advance. It is called a *timeslice* of a process and represents the amount of processor time that is provided to each process. By assigning timeslices, the scheduler makes global decisions for the running system, and prevents individual processes from taking control of the processor resources.

15.3. Process priority

The scheduler evaluates processes based on their priority. To calculate the current priority of a process, the task scheduler uses complex algorithms. As a result, each process is given a value according to which it is “allowed” to run on a processor.

15.2. Process classification

Processes are classified according to their purpose and behavior. Although the borderline is not always distinct, generally two criteria are used to sort them. These criteria are independent and do not exclude each other.

One approach is to classify a process either *I/O-bound* or *processor-bound*.

I/O-bound

I/O stands for Input/Output devices, such as keyboards, mice, or optical and hard disks. *I/O-bound processes* spend most of the time submitting and waiting for requests. They are run frequently, but at short time intervals, not to block other processes waiting for I/O requests.

processor-bound

Processor-bound tasks use their time to execute a code, and run until they are preempted by the scheduler. They do not block processes waiting for I/O requests, and, therefore, can be run less frequently but for longer time intervals.

Another approach is to divide processes by type into *interactive*, *batch*, and *real-time* processes.

- *Interactive* processes spend a lot of time waiting for I/O requests, such as keyboard or mouse operations. The scheduler must wake up such processes quickly on user request, or the user finds the environment unresponsive. The typical delay is approximately 100 ms. Office applications, text editors or image manipulation programs represent typical interactive processes.
- *Batch* processes often run in the background and do not need to be responsive. They receive lower priority from the scheduler. Multimedia converters, database search engines, or log files analyzers are typical examples of batch processes.
- *Real-time* processes must never be blocked by low-priority processes, and the scheduler guarantees a short response time to them. Applications for editing multimedia content are a good example here.

15.3. Completely Fair Scheduler

Since the Linux kernel version 2.6.23, a new approach has been taken to the scheduling of runnable processes. Completely Fair Scheduler (CFS) became the default Linux kernel scheduler.

Since then, important changes and improvements have been made. The information in this chapter applies to SUSE Linux Enterprise Server with kernel version 2.6.32 and higher (including 3.x kernels). The scheduler environment was divided into several parts, and three main new features were introduced:

Modular scheduler core

The core of the scheduler was enhanced with *scheduling classes*. These classes are modular and represent scheduling policies.

Completely Fair Scheduler

Introduced in kernel 2.6.23 and extended in 2.6.24, CFS tries to assure that each process obtains its “fair” share of the processor time.

Group scheduling

For example, if you split processes into groups according to which user is running them, CFS tries to provide each of these groups with the same amount of processor time.

As a result, CFS brings optimized scheduling for both servers and desktops.

15.1. How CFS works

CFS tries to guarantee a fair approach to each runnable task. To find the most balanced way of task scheduling, it uses the concept of *red-black tree*. A red-black tree is a type of self-balancing data search tree which provides inserting and removing entries in a reasonable way so that it remains well balanced.

When CFS schedules a task it accumulates “virtual runtime” or *vruntime*. The next task picked to run is always the task with the minimum accumulated vruntime so far. By balancing the red-black tree when tasks are inserted into the *run queue* (a planned time line of processes to be executed next), the task with the minimum vruntime is always the first entry in the red-black tree.

The amount of vruntime a task accrues is related to its priority. High priority tasks gain vruntime at a slower rate than low priority tasks, which results in high priority tasks being picked to run on the processor more often.

15.2. Grouping processes

Since the Linux kernel version 2.6.24, CFS can be tuned to be fair to groups rather than to tasks only. Runnable tasks are then grouped to form entities, and CFS tries to be fair to these entities instead of individual runnable tasks. The scheduler also tries to be fair to individual tasks within these entities.

The kernel scheduler lets you group runnable tasks using control groups. For more information, see *Chapter 10, Kernel control groups*.

15.3. Kernel configuration options

Basic aspects of the task scheduler behavior can be set through the kernel configuration options. Setting these options is part of the kernel compilation process. Because kernel compilation process is a complex task and out of this document's scope, refer to relevant source of information.



Kernel compilation

If you run SUSE Linux Enterprise Server on a kernel that was not shipped with it, for example on a self-compiled kernel, you lose the entire support entitlement.

15.4. Terminology

Documents regarding task scheduling policy often use several technical terms which you need to know to understand the information correctly. A few of them are as follows:

Latency

Delay between the time a process is scheduled to run and the actual process execution.

Granularity

The relation between granularity and latency can be expressed by the following equation:

$$gran = (lat / rtasks) - (lat / rtasks / rtasks)$$

where *gran* stands for granularity, *lat* stand for latency, and *rtasks* is the number of running tasks.

15.1. Scheduling policies

The Linux kernel supports the following scheduling policies:

SCHED_FIFO

Scheduling policy designed for special time-critical applications. It uses the First In-First Out scheduling algorithm.

SCHED_BATCH

Scheduling policy designed for CPU-intensive tasks.

SCHED_IDLE

Scheduling policy intended for *very* low-priority tasks.

SCHED_OTHER

Default Linux time-sharing scheduling policy used by most of the processes.

SCHED_RR

Similar to SCHED_FIFO, but uses the Round Robin scheduling algorithm.

15.5. Changing real-time attributes of processes with **chrt**

The **chrt** command sets or retrieves the real-time scheduling attributes of a running process, or runs a command with the specified attributes. You can get or retrieve both the scheduling policy and priority of a process.

In the following examples, a process whose PID is 16244 is used.

To *retrieve* the real-time attributes of an existing task:

```
#chrt -p 16244
pid 16244's current scheduling policy: SCHED_OTHER
pid 16244's current scheduling priority: 0
```

Before setting a new scheduling policy on the process, you need to find out the minimum and maximum valid priorities for each scheduling algorithm:

```
#chrt -m
SCHED_SCHED_OTHER min/max priority : 0/0
SCHED_SCHED_FIFO min/max priority : 1/99
SCHED_SCHED_RR min/max priority : 1/99
SCHED_SCHED_BATCH min/max priority : 0/0
SCHED_SCHED_IDLE min/max priority : 0/0
```

In the above example, SCHED_OTHER, SCHED_BATCH, SCHED_IDLE policies only allow for priority 0, while that of SCHED_FIFO and SCHED_RR can range from 1 to 99.

To set SCHED_BATCH scheduling policy:

```
#chrt -b -p 0 16244
pid 16244's current scheduling policy: SCHED_BATCH
pid 16244's current scheduling priority: 0
```

For more information on **chrt**, see its man page (**man 1 chrt**).

15.6. Runtime tuning with **sysctl**

The **sysctl** interface for examining and changing kernel parameters at runtime introduces important variables, using which you can change the default behavior of the task scheduler. The

syntax of the **sysctl** is simple, and all the following commands must be entered on the command line as root.

To read a value from a kernel variable, enter

```
#sysctl VARIABLE
```

To assign a value, enter

```
#sysctl VARIABLE=VALUE
```

To get a list of all scheduler-related variables, run the **sysctl** command, and use **grep** to filter the output:

```
#sysctl -A | grep "sched" | grep -v "domain"
kernel.sched_cfs_bandwidth_slice_us = 5000
kernel.sched_child_runs_first = 0
kernel.sched_compat_yield = 0
kernel.sched_latency_ns = 24000000
kernel.sched_migration_cost_ns = 500000
kernel.sched_min_granularity_ns = 8000000
kernel.sched_nr_migrate = 32
kernel.sched_rr_timeslice_ms = 25
kernel.sched_rt_period_us = 1000000
kernel.sched_rt_runtime_us = 950000
kernel.sched_schedstats = 0
kernel.sched_shares_window_ns = 10000000
kernel.sched_time_avg_ms = 1000
kernel.sched_tunable_scaling = 1
kernel.sched_wakeup_granularity_ns = 10000000
```

Variables ending with “_ns” and “_us” accept values in nanoseconds and microseconds, respectively.

A list of the most important task scheduler **sysctl** tuning variables (located at `/proc/sys/kernel/`) with a short description follows:

sched_cfs_bandwidth_slice_us

When CFS bandwidth control is in use, this parameter controls the amount of runtime (bandwidth) transferred to a run queue from the task’s control group bandwidth pool. Small values allow the global bandwidth to be shared in a fine-grained manner among tasks, larger values reduce transfer overhead. See <https://www.kernel.org/doc/Documentation/scheduler/sched-bwc.txt>.

sched_child_runs_first

A freshly forked child runs before the parent continues execution. Setting this parameter to 1 is beneficial for an application in which the child performs an execution after fork.

sched_compat_yield

Enables the aggressive CPU yielding behavior of the old $O(1)$ scheduler by moving the relinquishing task to the end of the runnable queue (right-most position in the red-black tree). Applications that depend on the `sched_yield(2)` syscall behavior may see performance improvements by giving other processes a chance to run when there are highly contended resources (such as locks). Given that this call occurs in context switching, misusing the call can hurt the workload. Use it only when you see a drop in performance. The default value is 0.

sched_migration_cost_ns

Amount of time after the last execution that a task is considered to be “cache hot” in migration decisions. A “hot” task has fewer chances of migration to another CPU, so increasing this variable reduces task migrations. The default value is 500000 (ns).

If the CPU idle time is higher than expected when there are runnable processes, try reducing this value. If tasks bounce between CPUs or nodes too often, try increasing it.

sched_latency_ns

Targeted preemption latency for CPU bound tasks. Increasing this variable increases a CPU bound task's timeslice. A task's timeslice is its weighted fair share of the scheduling period:

$$\text{timeslice} = \text{scheduling period} * (\text{task's weight} / \text{total weight of tasks in the run queue})$$

The task's weight depends on the task's nice level and the scheduling policy. Minimum task weight for a SCHED_OTHER task is 15, corresponding to nice 19. The maximum task weight is 88761, corresponding to nice -20.

Timeslices become smaller as the load increases. When the number of runnable tasks exceeds `sched_latency_ns/sched_min_granularity_ns`, the slice becomes `number_of_running_tasks * sched_min_granularity_ns`. Before that, the slice is equal to `sched_latency_ns`.

This value also specifies the maximum amount of time during which a sleeping task is considered to be running for entitlement calculations. Increasing this variable increases the amount of time a waking task may consume before being preempted, thus increasing scheduler latency for CPU bound tasks. The default value is 6000000 (ns).

sched_min_granularity_ns

Minimal preemption granularity for CPU bound tasks. See `sched_latency_ns` for details. The default value is 4000000 (ns).

sched_wakeup_granularity_ns

The wake-up preemption granularity. Increasing this variable reduces wake-up preemption, reducing disturbance of compute bound tasks. Lowering it improves wake-up latency and throughput for latency critical tasks, particularly when a short duty cycle load component must compete with CPU bound components. The default value is 2500000 (ns).

**Setting the right wake-up granularity value**

Settings larger than half of `sched_latency_ns` result in no wake-up preemption. Short duty cycle tasks cannot compete with CPU hogs effectively.

sched_rr_timeslice_ms

Quantum that SCHED_RR tasks are allowed to run before they are preempted and put to the end of the task list.

sched_rt_period_us

Period over which real-time task bandwidth enforcement is measured. The default value is 1000000 (μ s).

sched_rt_runtime_us

Quantum allocated to real-time tasks during `sched_rt_period_us`. Setting to -1 disables RT bandwidth enforcement. By default, RT tasks may consume 95%CPU/sec, thus leaving 5%CPU/sec or 0.05 s to be used by SCHED_OTHER tasks. The default value is 950000 (μ s).

sched_nr_migrate

Controls how many tasks can be migrated across processors for load-balancing purposes. Because balancing iterates the runqueue with interrupts disabled (softirq), it can incur in irq-latency penalties for real-time tasks. Therefore, increasing this value may give a performance boost to large SCHED_OTHER threads at the expense of increased irq-latencies for real-time tasks. The default value is 32.

sched_time_avg_ms

This parameter sets the period over which the time spent running real-time tasks is averaged. That average assists CFS in making load-balancing decisions and gives an indication of how busy a CPU is with high-priority real-time tasks.

The optimal setting for this parameter is highly workload dependent and depends, among other things, on how frequently real-time tasks are running and for how long.



A few scheduler parameters have been moved to debugfs

If the default Linux kernel version of your operating system is 5.13 or later (which can be checked using the command `rpm -q kernel-default`), you might notice messages in the kernel logs that are similar to the following example:

```
[ 20.485624] The sched.sched_min_granularity_ns sysctl was moved to
debugfs in kernel 5.13 for CPU scheduler debugging only. This sysctl
will be removed in a future SLE release.
[ 20.485632] The sched.sched_wakeup_granularity_ns sysctl was moved
to debugfs in kernel 5.13 for CPU scheduler debugging only. This
sysctl will be removed in a future SLE release.
```

This happens because six scheduler parameters have been moved from `/proc/sys/kernel/sched_*` to `/sys/kernel/debug/sched/*`. The affected scheduler parameters are as follows:

- `sched_latency_ns`
- `sched_migration_cost_ns`
- `sched_min_granularity_ns`
- `sched_nr_migrate`
- `sched_tunable_scaling`
- `sched_wakeup_granularity_n`

For temporary convenience, the `sysctl` for these scheduler parameters still exists in SUSE Linux Enterprise Server. However, due to planned changes in the CPU scheduler implementation, there is no guarantee that either the `sysctl` or the `debugfs` options will exist in a future release of SUSE Linux Enterprise Server.

If any of your current system tuning configurations depend on these six scheduler parameters, we strongly recommend that you find an alternate method of achieving your objective and stop depending on them for production workloads.

15.7. Debugging interface and scheduler statistics

CFS comes with a new improved debugging interface, and provides runtime statistics information. Relevant files were added to the `/proc` file system, which can be examined simply with the **cat** or **less** command. A list of the related `/proc` files follows with their short description:

`/proc/sched_debug`

Contains the current values of all tunable variables (see *the section called “Runtime tuning with **sysctl**”*) that affect the task scheduler behavior, CFS statistics, and information about the run queues (CFS, RT and deadline) on all available processors. A summary of the task

running on each processor is also shown, with the task name and PID, along with scheduler specific statistics. The first being the `tree-key` column, it indicates the task's virtual runtime, and its name comes from the kernel sorting all runnable tasks by this key in a red-black tree. The `switches` column indicates the total number of switches (involuntary or not), and the `prio` refers to the process priority. The `wait-time` value indicates the amount of time the task waited to be scheduled. Finally both `sum-exec` and `sum-sleep` account for the total amount of time (in nanoseconds) the task was running on the processor or asleep, respectively.

```
#cat /proc/sched_debug
Sched Debug Version: v0.11, 6.4.0-150700.38-default #1
ktime                : 23533900.395978
sched_clk             : 23543587.726648
cpu_clk              : 23533900.396165
jiffies              : 4300775771
sched_clock_stable   : 0

sysctl_sched
.sysctl_sched_latency      : 6.000000
.sysctl_sched_min_granularity : 2.000000
.sysctl_sched_wakeup_granularity : 2.500000
.sysctl_sched_child_runs_first : 0
.sysctl_sched_features     : 154871
.sysctl_sched_tunable_scaling : 1 (logarithmic)

cpu#0, 2666.762 MHz
.nr_running            : 1
.load                 : 1024
.nr_switches           : 1918946
[...]

cfs_rq[0]:/
.exec_clock            : 170176.383770
.MIN_vruntime          : 0.000001
.min_vruntime          : 347375.854324
.max_vruntime          : 0.000001
[...]

rt_rq[0]:/
.rt_nr_running         : 0
.rt_throttled          : 0
.rt_time               : 0.000000
.rt_runtime            : 950.000000

dl_rq[0]:
.dl_nr_running         : 0

  task  PID      tree-key  switches  prio    wait-time    [...]
-----
R  cc1 63477    98876.717832    197    120     0.000000     ...
```

/proc/schedstat

Displays statistics relevant to the current run queue. Also domain-specific statistics for SMP systems are displayed for all connected processors. Because the output format is not user-friendly, read the contents of `/usr/src/linux/Documentation/scheduler/sched-stats.txt` for more information.

/proc/*PID*/sched

Displays scheduling information on the process with id *PID*.

```
#cat /proc/$(pidof gdm)/sched
gdm (744, #threads: 3)
-----
se.exec_start          :          8888.758381
se.vruntime            :          6062.853815
se.sum_exec_runtime    :           7.836043
se.statistics.wait_start :          0.000000
se.statistics.sleep_start :          8888.758381
se.statistics.block_start :          0.000000
se.statistics.sleep_max :          1965.987638
[...]
se.avg.decay_count     :           8477
policy                 :              0
prio                   :             120
clock-delta            :             128
mm->numa_scan_seq      :              0
numa_migrations, 0
numa_faults_memory, 0, 0, 1, 0, -1
numa_faults_memory, 1, 0, 0, 0, -1
```

15.4. More information

To get a compact knowledge about Linux kernel task scheduling, you need to explore several information sources. Here are some:

- For task scheduler System Calls description, see the relevant manual page (for example **man 2 sched_setaffinity**).
- A useful lecture on Linux scheduler policy and algorithm is available in <https://www.inf.fu-berlin.de/lehre/SS01/OS/Lectures/Lecture08.pdf>.
- A good overview of Linux process scheduling is given in *Linux Kernel Development* by Robert Love (ISBN-10: 0-672-32512-8). See <https://www.informit.com/articles/article.aspx?p=101760>.
- A comprehensive overview of the Linux kernel internals is given in *Understanding the Linux Kernel* by Daniel P. Bovet and Marco Cesati (ISBN 978-0-596-00565-8).
- Technical information about task scheduler is covered in files under `/usr/src/linux/Documentation/scheduler`.

Chapter 16. Tuning the memory management subsystem

To understand and tune the memory management behavior of the kernel, it is important to first have an overview of how it works and cooperates with other subsystems.

The memory management subsystem, also called the virtual memory manager, will subsequently be called “VM”. The role of the VM is to manage the allocation of physical memory (RAM) for the entire kernel and user programs. It is also responsible for providing a virtual memory environment for user processes (managed via POSIX APIs with Linux extensions). Finally, the VM frees up RAM when there is a shortage, either by trimming caches or swapping out “anonymous” memory.

The most important thing to understand when examining and tuning VM is how its caches are managed. The basic goal of the VM's caches is to minimize the cost of I/O as generated by swapping and file system operations (including network file systems). This is achieved by avoiding I/O or by submitting I/O in better patterns.

Free memory is used and filled up by these caches as required. The more memory is available for caches and anonymous memory, the more effectively the caches and swapping operate. However, if a memory shortage is encountered, the caches are trimmed or the memory is swapped out.

For a particular workload, the first thing that can be done to improve performance is to increase memory and reduce the frequency that memory must be trimmed or swapped. The second thing is to change the way caches are managed by changing kernel parameters.

Finally, the workload itself should be examined and tuned as well. If an application is allowed to run more processes or threads, effectiveness of VM caches can be reduced, if each process is operating in its own area of the file system. Memory overheads are also increased. If applications allocate their own buffers or caches, larger caches mean that less memory is available for VM caches. However, more processes and threads can mean more opportunity to overlap and pipeline I/O, and may take better advantage of multiple cores. Experimentation is required for the best results.

16.1. Memory usage

Memory allocations can be characterized as “pinned” (also known as “unreclaimable”), “reclaimable” or “swappable”.

16.1. Anonymous memory

Anonymous memory tends to be program heap and stack memory (for example, `>malloc()`). It is reclaimable, except in special cases such as `mlock` or if there is no available swap space. Anonymous memory must be written to swap before it can be reclaimed. Swap I/O (both swapping in and swapping out pages) tends to be less efficient than pagecache I/O, because of allocation and access patterns.

16.2. Pagecache

A cache of file data. When a file is read from disk or network, the contents are stored in pagecache. No disk or network access is required, if the contents are up-to-date in pagecache. tmpfs and shared memory segments count toward pagecache.

When a file is written to, the new data is stored in pagecache before being written back to a disk or the network (making it a write-back cache). When a page has new data not written back yet, it is called “dirty”. Pages not classified as dirty are “clean”. Clean pagecache pages can be reclaimed if there is a memory shortage by simply freeing them. Dirty pages must first be made clean before being reclaimed.

16.3. Buffercache

This is a type of pagecache for block devices (for example, /dev/sda). A file system typically uses the buffercache when accessing its on-disk metadata structures such as inode tables, allocation bitmaps, and so forth. Buffercache can be reclaimed similarly to pagecache.

16.4. Buffer heads

Buffer heads are small auxiliary structures that tend to be allocated upon pagecache access. They can generally be reclaimed easily when the pagecache or buffercache pages are clean.

16.5. Writeback

As applications write to files, the pagecache becomes dirty and the buffercache may become dirty. When the amount of dirty memory reaches a specified number of pages in bytes (*vm.dirty_background_bytes*), or when the amount of dirty memory reaches a specific ratio to total memory (*vm.dirty_background_ratio*), or when the pages have been dirty for longer than a specified amount of time (*vm.dirty_expire_centisecs*), the kernel begins writeback of pages starting with files that had the pages dirtied first. The background bytes and ratios are mutually exclusive and setting one will overwrite the other. Flusher threads perform writeback in the background and allow applications to continue running. If the I/O cannot keep up with applications dirtying pagecache, and dirty data reaches a critical setting (*vm.dirty_bytes* or *vm.dirty_ratio*), then applications begin to be throttled to prevent dirty data exceeding this threshold.

16.6. Readahead

The VM monitors file access patterns and may attempt to perform readahead. Readahead reads pages into the pagecache from the file system that have not been requested yet. It is done to allow fewer, larger I/O requests to be submitted (more efficient). And for I/O to be pipelined (I/O performed at the same time as the application is running).

16.7. VFS caches

16.1. Inode cache

This is an in-memory cache of the inode structures for each file system. These contain attributes such as the file size, permissions and ownership, and pointers to the file data.

16.2. Directory entry cache

This is an in-memory cache of the directory entries in the system. These contain a name (the name of a file), the inode which it refers to, and children entries. This cache is used when traversing the directory structure and accessing a file by name.

16.2. Reducing memory usage

16.1. Reducing malloc (anonymous) usage

Applications running on SUSE Linux Enterprise Server15 SP7 can allocate more memory compared to older releases. This is because of glibc changing its default behavior while allocating user space memory. See https://www.gnu.org/s/libc/manual/html_node/Malloc-Tunable-Parameters.html for explanation of these parameters.

To restore behavior similar to older releases, `M_MMAP_THRESHOLD` should be set to `128*1024`. This can be done with `mallopt()` call from the application, or via setting `MALLOC_MMAP_THRESHOLD_` environment variable before running the application.

16.2. Reducing kernel memory overheads

Kernel memory that is reclaimable (caches, described above) is trimmed automatically during memory shortages. Most other kernel memory cannot be easily reduced but is a property of the workload given to the kernel.

Reducing the requirements of the user space workload reduces the kernel memory usage (fewer processes, fewer open files and sockets, etc.).

16.3. Memory controller (memory cgroups)

If the memory cgroups feature is not needed, it can be switched off by passing `cgroup_disable=memory` on the kernel command line, reducing memory consumption of the kernel a bit. There is also a slight performance benefit as there is a small amount of accounting overhead when memory cgroups are available even if none are configured.

16.3. Virtual memory manager (VM) tunable parameters

When tuning the VM, it should be understood that certain changes take time to affect the workload and take full effect. If the workload changes throughout the day, it may behave differently at different times. A change that increases throughput under certain conditions may decrease it under other conditions.

16.1. Reclaim ratios

/proc/sys/vm/swappiness

This control is used to define how aggressively the kernel swaps out anonymous memory relative to pagecache and other caches. Increasing the value increases the amount of swapping. The default value is 60.

Swap I/O tends to be much less efficient than other I/O. However, certain pagecache pages are accessed much more frequently than less used anonymous memory. The right balance should be found here.

If swap activity is observed during slowdowns, it may be worth reducing this parameter. If there is a lot of I/O activity and the amount of pagecache in the system is rather small, or if there are large dormant applications running, increasing this value can improve performance.

The more data is swapped out, the longer the system takes to swap data back in when it is needed.

/proc/sys/vm/vfs_cache_pressure

This variable controls the tendency of the kernel to reclaim the memory which is used for caching of VFS caches, versus pagecache and swap. Increasing this value increases the rate at which VFS caches are reclaimed.

It is difficult to know when this should be changed, other than by experimentation. The **slabtop** command (part of the package `procps`) shows top memory objects used by the kernel. The vfs caches are the "dentry" and the "*_inode_cache" objects. If these are consuming a large amount of memory in relation to pagecache, it may be worth trying to increase pressure. Could also help to reduce swapping. The default value is 100.

/proc/sys/vm/min_free_kbytes

This controls the amount of memory that is kept free for use by special reserves including "atomic" allocations (those which cannot wait for reclaim). This should not normally be lowered unless the system is being carefully tuned for memory usage (normally useful for embedded rather than server applications). If "page allocation failure" messages and stack

traces are frequently seen in logs, `min_free_kbytes` could be increased until the errors disappear. There is no need for concern if these messages are infrequent. The default value depends on the amount of RAM.

`/proc/sys/vm/watermark_scale_factor`

Broadly speaking, free memory has high, low and min watermarks. When the low watermark is reached then **kswapd** wakes to reclaim memory in the background. It stays awake until free memory reaches the high watermark. Applications will stall and reclaim memory when the min watermark is reached.

The `watermark_scale_factor` defines the amount of memory left in a node/system before kswapd is woken up and how much memory needs to be free before kswapd goes back to sleep. The unit is in fractions of 10,000. The default value of 10 means the distances between watermarks are 0.1% of the available memory in the node/system. The maximum value is 1000, or 10% of memory.

Workloads that frequently stall in direct reclaim, accounted by `allocstall` in `/proc/vmstat`, may benefit from altering this parameter. Similarly, if **kswapd** is sleeping prematurely, as accounted for by `kswapd_low_wmark_hit_quickly`, then it may indicate that the number of pages kept free to avoid stalls is too low.

16.2. Writeback parameters

One important change in writeback behavior since SUSE Linux Enterprise Server 10 is that modification to file-backed `mmap()` memory is accounted immediately as dirty memory (and subject to writeback). Whereas previously it would only be subject to writeback after it was unmapped, upon an `msync()` system call, or under heavy memory pressure.

Some applications do not expect `mmap` modifications to be subject to such writeback behavior, and performance can be reduced. Increasing writeback ratios and times can improve this type of slowdown.

`/proc/sys/vm/dirty_background_ratio`

This is the percentage of the total amount of free and reclaimable memory. When the amount of dirty pagecache exceeds this percentage, writeback threads start writing back dirty memory. The default value is 10 (%).

`/proc/sys/vm/dirty_background_bytes`

This contains the amount of dirty memory at which the background kernel flusher threads start writeback. `dirty_background_bytes` is the counterpart of `dirty_background_ratio`. If one of them is set, the other one will automatically be read as 0.

`/proc/sys/vm/dirty_ratio`

Similar percentage value as for `dirty_background_ratio`. When this is exceeded, applications that want to write to the pagecache are blocked and wait for kernel background flusher threads to reduce the amount of dirty memory. The default value is 20 (%).

`/proc/sys/vm/dirty_bytes`

This file controls the same tunable as `dirty_ratio` however the amount of dirty memory is in bytes as opposed to a percentage of reclaimable memory. Since both `dirty_ratio` and `dirty_bytes` control the same tunable, if one of them is set, the other one is automatically read as 0. The minimum value allowed for `dirty_bytes` is two pages (in bytes); any value lower than this limit is ignored and the old configuration will be retained.

`/proc/sys/vm/dirty_expire_centisecs`

The data which has been dirty in-memory for longer than this interval is written out next time a flusher thread wakes up. Expiration is measured based on the modification time of a file's inode. Therefore, multiple dirtied pages from the same file are all written when the interval is exceeded.

`dirty_background_ratio` and `dirty_ratio` together determine the pagecache writeback behavior. If these values are increased, more dirty memory is kept in the system for a longer time. With more dirty memory allowed in the system, the chance to improve throughput by avoiding writeback I/O and to submitting more optimal I/O patterns increases. However, more dirty memory can either harm latency when memory needs to be reclaimed or at points of data integrity ("synchronization points") when it needs to be written back to disk.

16.3. Timing differences of I/O writes between SUSE Linux Enterprise 12 and SUSE Linux Enterprise 11

The system is required to limit what percentage of the system's memory contains file-backed data that needs writing to disk. This guarantees that the system can always allocate the necessary data structures to complete I/O. The maximum amount of memory that can be dirty and requires writing at any time is controlled by `vm.dirty_ratio` (`/proc/sys/vm/dirty_ratio`). The defaults are:

```
SLE-11-SP3:    vm.dirty_ratio = 40
SLE-12:        vm.dirty_ratio = 20
```

The primary advantage of using the lower ratio in SUSE Linux Enterprise 12 is that page reclamation and allocation in low memory situations completes faster as there is a higher probability that old clean pages are quickly found and discarded. The secondary advantage is that if all data on the system must be synchronized, then the time to complete the operation on SUSE Linux Enterprise 12 is lower than SUSE Linux Enterprise 11 SP3 by default. Most workloads will

not notice this change as data is synchronized with `fsync()` by the application or data is not dirtied quickly enough to hit the limits.

There are exceptions, and if your application is affected by this, it can manifest as an unexpected stall during writes. To prove it is affected by dirty data rate limiting then monitor `/proc/PID_OF_APPLICATION/stack` and it will be observed that the application spends significant time in `balance_dirty_pages_ratelimited`. If this is observed and it is a problem, then increase the value of `vm.dirty_ratio` to 40 to restore the SUSE Linux Enterprise 11 SP3 behavior.

Important



The overall I/O throughput is the same regardless of the setting. The only difference is the timing of when the I/O is queued.

This is an example of using **dd** to asynchronously write 30% of memory to disk which would happen to be affected by the change in `vm.dirty_ratio`:

```
#MEMTOTAL_MBYTES=`free -m | grep Mem: | awk '{print $2}'`
#sysctl vm.dirty_ratio=40
#dd if=/dev/zero of=zerofile ibs=1048576 count=$((MEMTOTAL_MBYTES*30/100))
2507145216 bytes (2.5 GB) copied, 8.00153 s, 313 MB/s
#sysctl vm.dirty_ratio=20
dd if=/dev/zero of=zerofile ibs=1048576 count=$((MEMTOTAL_MBYTES*30/100))
2507145216 bytes (2.5 GB) copied, 10.1593 s, 247 MB/s
```

The parameter affects the time it takes for the command to complete and the apparent write speed of the device. With `dirty_ratio=40`, more of the data is cached and written to disk in the background by the kernel. The speed of I/O is identical in both cases. To demonstrate, this is the result when **dd** synchronizes the data before exiting:

```
#sysctl vm.dirty_ratio=40
#dd if=/dev/zero of=zerofile ibs=1048576 count=$((MEMTOTAL_MBYTES*30/100))
conv=fdatasync
2507145216 bytes (2.5 GB) copied, 21.0663 s, 119 MB/s
#sysctl vm.dirty_ratio=20
#dd if=/dev/zero of=zerofile ibs=1048576 count=$((MEMTOTAL_MBYTES*30/100))
conv=fdatasync
2507145216 bytes (2.5 GB) copied, 21.7286 s, 115 MB/s
```

As observed, `dirty_ratio` had almost no impact here and is within the natural variability of a command. Hence, `dirty_ratio` does not directly impact I/O performance but it may affect the apparent performance of a workload that writes data asynchronously without synchronizing.

16.4. Readahead parameters

`/sys/block/<bdev>/queue/read_ahead_kb`

If one or more processes are sequentially reading a file, the kernel reads certain data in advance (ahead) to reduce the amount of time that processes need to wait for data to be available. The actual amount of data being read in advance is computed dynamically, based

on the extent of *sequentiality* of the I/O. This parameter sets the maximum amount of data that the kernel reads ahead for a single file. If you observe that large sequential reads from a file are not fast enough, you can try increasing this value. Increasing it too far may result in readahead thrashing where pagecache used for readahead is reclaimed before it can be used, or slowdowns because of a large amount of useless I/O. The default value is 512 (KB).

16.5. Transparent HugePage parameters

Transparent HugePages (THP) provide a way to dynamically allocate huge pages either on-demand by the process or deferring the allocation until later via the **khugepaged** kernel thread. This method is distinct from the use of `hugetlbfs` to manually manage their allocation and use. Workloads with contiguous memory access patterns can benefit greatly from THP. A 1000-fold decrease in page faults can be observed when running synthetic workloads with contiguous memory access patterns.

There are cases when THP may be undesirable. Workloads with sparse memory access patterns can perform poorly with THP due to excessive memory usage. For example, 2 MB of memory may be used at fault time instead of 4 KB for each fault and ultimately lead to premature page reclaim. On releases older than SUSE Linux Enterprise 12 SP2, it was possible for an application to stall for long periods of time trying to allocate a THP which frequently led to a recommendation of disabling THP. Such recommendations should be re-evaluated for SUSE Linux Enterprise 12 SP3 and later releases.

The behavior of THP may be configured via the `transparent_hugepage=` kernel parameter or via `sysfs`. For example, it may be disabled by adding the kernel parameter `transparent_hugepage=never`, rebuilding your `grub2` configuration, and rebooting. Verify if THP is disabled with:

```
#cat /sys/kernel/mm/transparent_hugepage/enabled
always madvise [never]
```

If disabled, the value `never` is shown in square brackets like in the example above. A value of `always` mandatorily tries and uses THP at fault time but defers to **khugepaged** if the allocation fails. A value of `madvise` will only allocate THP for address spaces explicitly specified by an application.

`/sys/kernel/mm/transparent_hugepage/defrag`

This parameter controls how much effort an application commits when allocating a THP. A value of `always` is the default for SUSE Linux Enterprise 12 SP1 and earlier releases that supported THP. If a THP is not available, the application tries to defragment memory. It potentially incurs large stalls in an application if the memory is fragmented and a THP is not available.

A value of `madvise` means that THP allocation requests will only defragment if the application explicitly requests it. This is the default for SUSE Linux Enterprise 12 SP2 and later releases.

`defer` is only available on SUSE Linux Enterprise 12 SP2 and later releases. If a THP is not available, the application falls back to using small pages if a THP is not available. It wakes the **kswapd** and **kcompactd** kernel threads to defragment memory in the background and a THP will be allocated later by **khugepaged**.

The final option `never` uses small pages if a THP is unavailable but no other action will take place.

16.6. khugepaged parameters

khugepaged is automatically started when `transparent_hugepage` is set to `always` or `madvise`, and it will be automatically shut down if it is set to `never`. Normally this runs at low frequency but the behavior can be tuned.

/sys/kernel/mm/transparent_hugepage/khugepaged/defrag

A value of 0 will disable **khugepaged** even though THP may still be used at fault time. This may be important for latency-sensitive applications that benefit from THP but cannot tolerate a stall if **khugepaged** tries to update an application memory usage.

/sys/kernel/mm/transparent_hugepage/khugepaged/pages_to_scan

This parameter controls how many pages are scanned by **khugepaged** in a single pass. A scan identifies small pages that can be reallocated as THP. Increasing this value will allocate THP in the background faster at the cost of CPU usage.

/sys/kernel/mm/transparent_hugepage/khugepaged/scan_sleep_millisecs

khugepaged sleeps for a short interval specified by this parameter after each pass to limit how much CPU usage is used. Reducing this value allocates THP in the background faster at the cost of CPU usage. A value of 0 will force continual scanning.

/sys/kernel/mm/transparent_hugepage/khugepaged/alloc_sleep_millisecs

This parameter controls how long **khugepaged** will sleep in the event it fails to allocate a THP in the background waiting for **kswapd** and **kcompactd** to take action.

The remaining parameters for **khugepaged** are rarely useful for performance tuning but are fully documented in `/usr/src/linux/Documentation/vm/transhuge.txt`

16.7. Further VM parameters

For the complete list of the VM tunable parameters, see `/usr/src/linux/Documentation/sysctl/vm.txt` (available after having installed the kernel-source package).

16.4. Monitoring VM behavior

Some simple tools that can help monitor VM behavior:

1. **vmstat**: This tool gives a good overview of what the VM is doing. See *the section called “vmstat”* for details.
2. `/proc/meminfo`: This file gives a detailed breakdown of where memory is being used. See *the section called “Detailed memory usage: /proc/meminfo”* for details.
3. **slabtop**: This tool provides detailed information about kernel slab memory usage. `buffer_head`, `dentry`, `inode_cache`, `ext3_inode_cache`, etc. are the major caches. This command is available with the package `procps`.
4. `/proc/vmstat`: This file gives a detailed breakdown of internal VM behavior. The information contained within is implementation specific and may not always be available. Some information is duplicated in `/proc/meminfo` and other information can be presented in a friendly fashion by utilities. For maximum utility, this file needs to be monitored over time to observe rates of change. The most important pieces of information that are hard to derive from other sources are as follows:

pgscan_kswapd_*, pgsteal_kswapd_*

These report respectively the number of pages scanned and reclaimed by **kswapd** since the system started. The ratio between these values can be interpreted as the reclaim efficiency with a low efficiency implying that the system is struggling to reclaim memory and may be thrashing. Light activity here is generally not something to be concerned with.

pgscan_direct_*, pgsteal_direct_*

These report respectively the number of pages scanned and reclaimed by an application directly. This is correlated with increases in the `allocstall` counter. This is more serious than **kswapd** activity as these events indicate that processes are stalling. Heavy activity here combined with **kswapd** and high rates of `pgpgin`, `pgpout` and/or high rates of `pswapin` or `pswpout` are signs that a system is thrashing heavily.

More detailed information can be obtained using tracepoints.

thp_fault_alloc, thp_fault_fallback

These counters correspond to how many THPs were allocated directly by an application and how many times a THP was not available and small pages were used.

Generally a high fallback rate is harmless unless the application is sensitive to TLB pressure.

thp_collapse_alloc, thp_collapse_alloc_failed

These counters correspond to how many THPs were allocated by **khugepaged** and how many times a THP was not available and small pages were used. A high fallback rate implies that the system is fragmented and THPs are not being used even when the memory usage by applications would allow them. It is only a problem for applications that are sensitive to TLB pressure.

compact*_scanned, compact_stall, compact_fail, compact_success

These counters may increase when THP is enabled and the system is fragmented. `compact_stall` is incremented when an application stalls allocating THP. The remaining counters account for pages scanned, the number of defragmentation events that succeeded or failed.

Chapter 17. Tuning the network

The network subsystem is complex and its tuning highly depends on the system use scenario and on external factors such as software clients or hardware components (switches, routers, or gateways) in your network. The Linux kernel aims more at reliability and low latency than low overhead and high throughput. Other settings can mean less security, but better performance.

17.1. Configurable kernel socket buffers

Most of modern networking is based on the TCP/IP protocol and a socket interface for communication; for more information about TCP/IP, see Chapter 23, Basic networking in “[Administration Guide](#)”. The Linux kernel handles data it receives or sends via the socket interface in socket buffers. These kernel socket buffers are tunable.

TCP autotuning



Since kernel version 2.6.17 full autotuning with 4 MB maximum buffer size exists. This means that manual tuning does not improve networking performance considerably. It is often the best not to touch the following variables, or, at least, to check the outcome of tuning efforts carefully.

If you update from an older kernel, it is recommended to remove manual TCP tunings in favor of the autotuning feature.

The special files in the `/proc` file system can modify the size and behavior of kernel socket buffers; for general information about the `/proc` file system, see *the section called “The /proc file system”*. Find networking related files in:

```
/proc/sys/net/core  
/proc/sys/net/ipv4  
/proc/sys/net/ipv6
```

General net variables are explained in the kernel documentation (`linux/Documentation/sysctl/net.txt`). Special `ipv4` variables are explained in `linux/Documentation/networking/ip-sysctl.txt` and `linux/Documentation/networking/ipv6-sysctl.txt`.

In the `/proc` file system, for example, it is possible to either set the Maximum Socket Receive Buffer and Maximum Socket Send Buffer for all protocols, or both these options for the TCP protocol only (in `ipv4`) and thus overriding the setting for all protocols (in `core`).

`/proc/sys/net/ipv4/tcp_moderate_rcvbuf`

If `/proc/sys/net/ipv4/tcp_moderate_rcvbuf` is set to 1, autotuning is active and buffer size is adjusted dynamically.

/proc/sys/net/ipv4/tcp_rmem

The three values setting the minimum, initial, and maximum size of the Memory Receive Buffer per connection. They define the actual memory usage, not only TCP window size.

/proc/sys/net/ipv4/tcp_wmem

The same as `tcp_rmem`, but for Memory Send Buffer per connection.

/proc/sys/net/core/rmem_max

Set to limit the maximum receive buffer size that applications can request.

/proc/sys/net/core/wmem_max

Set to limit the maximum send buffer size that applications can request.

Via `/proc` it is possible to disable TCP features that you do not need (all TCP features are switched on by default). For example, check the following files:

/proc/sys/net/ipv4/tcp_timestamps

TCP time stamps are defined in RFC1323.

/proc/sys/net/ipv4/tcp_window_scaling

TCP window scaling is also defined in RFC1323.

/proc/sys/net/ipv4/tcp_sack

Select acknowledgments (SACKS).

Use **sysctl** to read or write variables of the `/proc` file system. **sysctl** is preferable to **cat** (for reading) and **echo** (for writing), because it also reads settings from `/etc/sysctl.conf` and, thus, those settings survive reboots reliably. With **sysctl** you can read all variables and their values easily; as root use the following command to list TCP related settings:

```
>sudo sysctl -a | grep tcp
```

**Side effects of tuning network variables**

Tuning network variables can affect other system resources such as CPU or memory use.

17.2. Detecting network bottlenecks and analyzing network traffic

Before starting with network tuning, it is important to isolate network bottlenecks and network traffic patterns. There are certain tools that can help you with detecting those bottlenecks.

The following tools can help analyzing your network traffic: **netstat**, **tcpdump**, and **wireshark**. Wireshark is a network traffic analyzer.

17.3. Netfilter

The Linux firewall and masquerading features are provided by the Netfilter kernel modules. This is a highly configurable rule based framework. If a rule matches a packet, Netfilter accepts or denies it or takes special action (“target”) as defined by rules such as address translation.

There are many properties Netfilter can take into account. Thus, the more rules are defined, the longer packet processing may last. Also advanced connection tracking could be rather expensive and, thus, slowing down overall networking.

When the kernel queue becomes full, all new packets are dropped, causing existing connections to fail. The 'fail-open' feature allows a user to temporarily disable the packet inspection and maintain the connectivity under heavy network traffic. For reference, see https://home.regit.org/netfilter-en/using-nfqueue-and-libnetfilter_queue/.

For more information, see the home page of the Netfilter and iptables project, <https://www.netfilter.org>.

17.4. Improving the network performance with receive packet steering (RPS)

Modern network interface devices can move so many packets that the host can become the limiting factor for achieving maximum performance. To keep up, the system must be able to distribute the work across multiple CPU cores.

Some modern network interfaces can help distribute the work to multiple CPU cores through the implementation of multiple transmission and multiple receive queues in hardware. However, others are only equipped with a single queue and the driver must deal with all incoming packets in a single, serialized stream. To work around this issue, the operating system must “parallelize” the stream to distribute the work across multiple CPUs. On SUSE Linux Enterprise Server this is done via Receive Packet Steering (RPS). RPS can also be used in virtual environments.

RPS creates a unique hash for each data stream using IP addresses and port numbers. The use of this hash ensures that packets for the same data stream are sent to the same CPU, which helps to increase performance.

RPS is configured per network device receive queue and interface. The configuration file names match the following scheme:

```
/sys/class/net/<device>/queues/<rx-queue>/rps_cpus
```

<device> stands for the network device, such as eth0, eth1. *<rx-queue>* stands for the receive queue, such as rx-0, rx-1.

If the network interface hardware only supports a single receive queue, only `rx-0` exists. If it supports multiple receive queues, there is an `rx-N` directory for each receive queue.

These configuration files contain a comma-delimited list of CPU bitmaps. By default, all bits are set to 0. With this setting, RPS is disabled and therefore the CPU that handles the interrupt also processes the packet queue.

To enable RPS and enable specific CPUs to process packets for the receive queue of the interface, set the value of their positions in the bitmap to 1. For example, to enable CPUs 0-3 to process packets for the first receive queue for `eth0`, set the bit positions 0-3 to 1 in binary: 00001111. This representation then needs to be converted to hex—which results in `F` in this case. Set this hex value with the following command:

```
>sudo echo "f" > /sys/class/net/eth0/queues/rx-0/rps_cpus
```

If you wanted to enable CPUs 8-15:

```
1111 1111 0000 0000 (binary)
15    15    0    0 (decimal)
F     F     0    0 (hex)
```

The command to set the hex value of `ff00` would be:

```
>sudo echo "ff00" > /sys/class/net/eth0/queues/rx-0/rps_cpus
```

On NUMA machines, best performance can be achieved by configuring RPS to use the CPUs on the same NUMA node as the interrupt for the interface's receive queue.

On non-NUMA machines, all CPUs can be used. If the interrupt rate is high, excluding the CPU handling the network interface can boost performance. The CPU being used for the network interface can be determined from `/proc/interrupts`. For example:

```
>sudo cat /proc/interrupts
          CPU0      CPU1      CPU2      CPU3
...
51:  113915241          0          0          0   Phys-fasteoi   eth0
...
```

In this case, CPU 0 is the only CPU processing interrupts for `eth0`, since only CPU0 contains a non-zero value.

On x86 and AMD64/Intel 64 platforms, **irqbalance** can be used to distribute hardware interrupts across CPUs. See **man 1 irqbalance** for more details.

Chapter 18. Tuning SUSE Linux Enterprise for SAP

This chapter presents information about preparing and tuning SUSE Linux Enterprise Server to work optimally with SAP applications with `sapconf`. `sapconf` is for SUSE Linux Enterprise systems that install SAP applications. Customers who have SUSE Linux Enterprise Server for SAP applications should use **saptune**.



The **sapconf** command has been removed

In SUSE Linux Enterprise Server and SUSE Linux Enterprise Server 11 and 12, the **sapconf** command was included in the package with the same name.

For SUSE Linux Enterprise Server and SUSE Linux Enterprise Server 15, this has been changed: The command **sapconf** has been removed from the `sapconf` package. The package contains a `systemd` service only. There is no `sapconf` command line tool anymore, no `sapconf/tuned` profiles, and no `tuned`.

18.1. Tuning SLE Systems with `sapconf` 5

The package `sapconf` is available in SUSE Linux Enterprise Server and SUSE Linux Enterprise Server for SAP applications. It sets recommended parameters for the following types of SAP applications: SAP NetWeaver, SAP HANA, and SAP HANA-based applications.

Overview of **sapconf5** in SUSE® Linux Enterprise Server 12

sapconf5 (without `tuned`)

- `sapconf - netweaver` (`sapconf` profile as a replacement for `tuned` profile)
- `sapconf - hana` (`sapconf` profile as a replacement for `tuned` profile)
- `sapconf - bobj` (`sapconf` profile as a replacement for `tuned` profile)
- `sapconf - ase` (`sapconf` profile as a replacement for `tuned` profile)

Overview of **sapconf5** in SUSE® Linux Enterprise Server 15

sapconf5 (without `tuned`)

no profiles anymore

If you previously made changes to the system tuning, those changes may be overwritten by `sapconf`.

`sapconf` 5 ships a `systemd` service which applies the tuning and ensures that related services are running.

To use `sapconf`, make sure that the package `sapconf` is installed on your system.



No profiles in SUSE Linux Enterprise Server and SUSE Linux Enterprise Server 15 SP7

In SUSE Linux Enterprise Server and SUSE Linux Enterprise Server 15, `sapconf` no longer supports profiles.

18.1. Verifying `sapconf` setup

With `sapconf` 5.0.2 and up, the check tool **`sapconf_check`** is available, which verifies the correct setup of `sapconf`. For example:

```
#sapconf_check
This is sapconf_check v1.0.
It verifies if sapconf is set up correctly and will give advice to do so.
Please keep in mind:
{{ - This tool does not check, if the tuning itself works correctly.}}
{{ - Follow the hints from top to down to minimize side effects.}}
Checking sapconf
=====
[ OK ] sapconf package has version 5.0.2
[ OK ] saptune.service is inactive
[ OK ] saptune.service is disabled
[WARN] tuned.service is enabled/active with profile 'virtual-guest -> Sapconf
does not require tuned! Run 'systemctl stop tuned.service', if not needed
otherwise.
[FAIL] sapconf.service is inactive -> Run 'systemctl start sapconf.service' to
activate the tuning now.
[FAIL] sapconf.service is disabled -> Run 'systemctl enable sapconf.service' to
activate sapconf at boot.1 warning(s) have been found.
2 error(s) have been found.
Sapconf will not work properly!
```

If **`sapconf_check`** finds problems, it gives hints on how to resolve the issue. The tool does not verify whether the system has been tuned correctly. It only checks that `sapconf` is set up correctly and has been started.

18.2. Enabling and disabling `sapconf` and viewing its status

After the installation of `sapconf`, the `sapconf` service is enabled.

You can inspect or change the status of `sapconf` as described in the following:

- To see the status of the service `sapconf`:

```
#systemctl status sapconf
```

The service should be displayed as *active (exited)*.

- To start the service `sapconf`:

```
#systemctl start sapconf
```

- Should `sapconf` be disabled, enable and start it with:

```
#systemctl enable --now sapconf
```

- To stop the service `sapconf`:

```
#systemctl stop sapconf
```

This command disables most of the optimizations immediately. The only exceptions from this rule are options that require a system reboot to enable/disable.

- To disable `sapconf`, use:

```
#systemctl disable sapconf
```

If you have not specifically enabled any of the services that `sapconf` depends on, this also disables most tuning parameters and all services used by `sapconf`.



Additional services that `sapconf` relies on

Besides the `sapconf` service, it also relies on the following two services:

- `sysstat` which collects data on system activity.
- `uidd` which generates time-based UUIDs that are guaranteed to be unique even in settings where many processor cores are involved. This is necessary for SAP applications.

18.3. Configuring `sapconf5`

Generally, the default configuration of `sapconf` already uses the parameter values recommended by SAP. However, if you have special needs, you can configure the tool to better suit those.

All parameters of `sapconf` can be found in the file `/etc/sysconfig/sapconf`. The file can be edited directly. All parameters in this file are explained by comments and references to SAP Notes, which can be viewed at <https://launchpad.support.sap.com/>.

When `sapconf` is updated, all customized parameters from this file are preserved as much as possible. However, sometimes parameters cannot be transferred cleanly to the new configuration file. Therefore, after updating it is advisable to check the difference between the previous custom configuration, which during the update is moved to `/etc/sysconfig/sapconf.rpmsave`, and the new version at `/etc/sysconfig/sapconf`.

Log messages related to this file are written to `/var/log/sapconf.log`.

When editing either of these files, you can find that certain values are commented by a `#` character at the beginning of the line. This means that while the parameter is relevant for tuning, there is no suitable default for it.

Conversely, you can add `#` characters to the beginning of the line to comment specific parameters. However, you should avoid this practice, as it can lead to `sapconf` not properly applying the profile.

To apply the edited configuration, restart `sapconf`:

```
#systemctl restart sapconf
```

Confirming that a certain parameter value was applied correctly works differently for different parameters. Hence, the following serves as an example only:

Example 18.1. Checking Parameters

To confirm that the setting for `TCP_SLOW_START` was applied, do the following:

- View the log file of `sapconf` to see whether it applied the value. Within `/var/log/sapconf.log`, check for a line containing this text:

```
Change net.ipv4.tcp_slow_start_after_idle from 1 to 0
```

Alternatively, the parameter may have already been set correctly before `sapconf` was started. In this case, `sapconf` does not change its value:

```
Leaving net.ipv4.tcp_slow_start_after_idle unchanged at 1
```

- The underlying option behind `TCP_SLOW_START` can be manually configured at `/proc/sys/net.ipv4.tcp_slow_start_after_idle`. To check its actual current value, use:

```
#sysctl net.ipv4.tcp_slow_start_after_idle
```

18.4. Removing `sapconf`

To remove `sapconf` from a system, uninstall its package with:

```
#zypper rm sapconf
```

When doing this, dependencies of `sapconf` remain installed. However, the service `sysstat` goes into a disabled state. If it is still relevant to you, make sure to enable it again.

18.5. For more information

The following man pages provide additional information about `sapconf`:

- Detailed description of all tuning parameters set by `sapconf`: **man 5 `sapconf`**
- Information about configuring and customizing the `sapconf` profile: **man 7 `sapconf`**

Also see the blog series detailing the updated version of `sapconf` at <https://www.suse.com/c/a/new-sapconf-is-available/>.

18.6. Using **tuned** together with **sapconf**

With version 5, **sapconf** does not rely on **tuned** anymore. This means both tools can be used independently. **sapconf** prints a warning in its log if the **tuned** service is started.

Using **tuned** and **sapconf** together



If you are going to use **tuned** and **sapconf** simultaneously, ensure that both tools do not configure the same system parameters.

Part VI. Handling system dumps

- 19 Tracing tools **184**
- 20 Kexec and Kdump **193**
- 21 Using **systemd-coredump** to debug application crashes **210**

Chapter 19. Tracing tools

SUSE Linux Enterprise Server comes with several tools that help you obtain useful information about your system. You can use the information for different purposes. For example, to debug and find problems in your program, to discover places causing performance drops, or to trace a running process to find out what system resources it uses. Most of the tools are part of the installation media. In certain cases, they need to be installed from the SUSE Software Development Kit, which is a separate download.



Tracing and impact on performance

While a running process is being monitored for system or library calls, the performance of the process is heavily reduced. You are advised to use tracing tools only for the time you need to collect the data.

19.1. Tracing system calls with strace

The **strace** command traces system calls of a process and signals received by the process. **strace** can either run a new command and trace its system calls, or you can attach **strace** to an already running command. Each line of the command's output contains the system call name, followed by its arguments in parentheses and its return value.

To run a new command and start tracing its system calls, enter the command to be monitored as you normally do, and add **strace** at the beginning of the command line:

```
>strace ls
execve("/bin/ls", ["ls"], [/ * 52 vars */]) = 0
brk(0) = 0x618000
mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) \
= 0x7f9848667000
mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) \
= 0x7f9848666000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT \
(No such file or directory)
open("/etc/ld.so.cache", O_RDONLY) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=200411, ...}) = 0
mmap(NULL, 200411, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f9848635000
close(3) = 0
open("/lib64/librt.so.1", O_RDONLY) = 3
[...]
mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) \
= 0x7fd780f79000
write(1, "Desktop\nDocuments\nbin\ninst-sys\n", 31Desktop
Documents
bin
inst-sys
) = 31
close(1) = 0
munmap(0x7fd780f79000, 4096) = 0
close(2) = 0
exit_group(0) = ?
```

To attach **strace** to an already running process, you need to specify the **-p** with the process ID (PID) of the process that you want to monitor:

```
>strace -p `pidof cron`
Process 1261 attached
restart syscall(<... resuming interrupted call ...>) = 0
stat("/etc/localtime", {st_mode=S_IFREG|0644, st_size=2309, ...}) = 0
select(5, [4], NULL, NULL, {0, 0}) = 0 (Timeout)
socket(PF_LOCAL, SOCK_STREAM|SOCK_CLOEXEC|SOCK_NONBLOCK, 0) = 5
connect(5, {sa_family=AF_LOCAL, sun_path="/var/run/nscd/socket"}, 110) = 0
sendto(5, "\2\0\0\0\0\0\0\0\5\0\0\0root\0", 17, MSG_NOSIGNAL, NULL, 0) = 17
poll([{fd=5, events=POLLIN|POLLERR|POLLHUP}], 1, 5000) = 1 ([{fd=5,
revents=POLLIN|POLLHUP}])
read(5, "\2\0\0\0\1\0\0\0\5\0\0\0\2\0\0\0\0\0\0\0\0\5\0\0\0\6\0\0\0"...
, 36) = 36
read(5, "root\0x\0root\0/root\0/bin/bash\0", 28) = 28
close(5) = 0
rt_sigprocmask(SIG_BLOCK, [CHLD], [], 8) = 0
rt_sigaction(SIGCHLD, NULL, {0x7f772b9ea890, [], SA_RESTORER|SA_RESTART,
0x7f772adf7880}, 8) = 0
rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
nanosleep({60, 0}, 0x7fff87d8c580) = 0
stat("/etc/localtime", {st_mode=S_IFREG|0644, st_size=2309, ...}) = 0
select(5, [4], NULL, NULL, {0, 0}) = 0 (Timeout)
socket(PF_LOCAL, SOCK_STREAM|SOCK_CLOEXEC|SOCK_NONBLOCK, 0) = 5
connect(5, {sa_family=AF_LOCAL, sun_path="/var/run/nscd/socket"}, 110) = 0
sendto(5, "\2\0\0\0\0\0\0\0\5\0\0\0root\0", 17, MSG_NOSIGNAL, NULL, 0) = 17
poll([{fd=5, events=POLLIN|POLLERR|POLLHUP}], 1, 5000) = 1 ([{fd=5,
revents=POLLIN|POLLHUP}])
read(5, "\2\0\0\0\1\0\0\0\5\0\0\0\2\0\0\0\0\0\0\0\0\5\0\0\0\6\0\0\0"...
, 36) = 36
read(5, "root\0x\0root\0/root\0/bin/bash\0", 28) = 28
close(5)
[...]
```

The **-e** option understands several sub-options and arguments. For example, to trace all attempts to open or write to a particular file, use the following:

```
>strace -e trace=open,write ls ~
open("/etc/ld.so.cache", 0_RDONLY) = 3
open("/lib64/librt.so.1", 0_RDONLY) = 3
open("/lib64/libselinux.so.1", 0_RDONLY) = 3
open("/lib64/libacl.so.1", 0_RDONLY) = 3
open("/lib64/libc.so.6", 0_RDONLY) = 3
open("/lib64/libpthread.so.0", 0_RDONLY) = 3
[...]
```

```
open("/usr/lib/locale/cs.CZ.utf8/LC_CTYPE", 0_RDONLY) = 3
open(".", 0_RDONLY|0_NONBLOCK|0_DIRECTORY|0_CLOEXEC) = 3
write(1, "addressbook.db.bak\nbin\ncxoffice\n"... , 311) = 311
```

To trace only network related system calls, use **-e trace=network**:

```
>strace -e trace=network -p 26520
Process 26520 attached - interrupt to quit
socket(PF_NETLINK, SOCK_RAW, 0) = 50
bind(50, {sa_family=AF_NETLINK, pid=0, groups=00000000}, 12) = 0
getsockname(50, {sa_family=AF_NETLINK, pid=26520, groups=00000000}, \
[12]) = 0
sendto(50, "\24\0\0\0\26\0\1\3~p\315K\0\0\0\0\0\0\0", 20, 0,
{sa_family=AF_NETLINK, pid=0, groups=00000000}, 12) = 20
[...]
```

The **-c** calculates the time the kernel spent on each system call:

```
>strace -c find /etc -name xorg.conf
/etc/X11/xorg.conf
% time      seconds    usecs/call     calls     errors syscall
-----
32.38      0.000181         181         1          execve
22.00      0.000123          0        576          getdents64
19.50      0.000109          0        917          31 open
19.14      0.000107          0        888          close
4.11       0.000023          2         10          mprotect
0.00       0.000000          0          1          write
[...]
0.00       0.000000          0          1          getrlimit
0.00       0.000000          0          1          arch_prctl
0.00       0.000000          0          3          1 futex
0.00       0.000000          0          1          set_tid_address
0.00       0.000000          0          4          fadvise64
0.00       0.000000          0          1          set_robust_list
-----
100.00     0.000559                3633         33 total
```

To trace all child processes of a process, use -f:

```
>strace -f systemctl status apache2.service
execve("/usr/bin/systemctl", ["systemctl", "status", "apache2.service"], \
 0x7ffea44a3318 /* 56 vars */) = 0
brk(NULL)                               = 0x5560f664a000
[...]
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7f98c58a5000
mmap(NULL, 4420544, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f98c524a000
mprotect(0x7f98c53f4000, 2097152, PROT_NONE) = 0
[...]
[pid 9130] read(0, "\342\227\217 apache2.service - The Apache"..., 8192) = 165
[pid 9130] read(0, "", 8027) = 0
● apache2.service - The Apache Webserver227\217 apache2.service - Th"..., 193
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; vendor
  preset: disabled)
  Active: inactive (dead)
) = 193
[pid 9130] ioctl(3, SNDCTL_TMR_STOP or TCSETSW, {B38400 opost isig icanon
echo ...}) = 0
[pid 9130] exit_group(0) = ?
[pid 9130] +++ exited with 0 +++
<... waitid resumed>{si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=9130, \
 si_uid=0, si_status=0, si_utime=0, si_stime=0}, WEXITED, NULL) = 0
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=9130, si_uid=0, \
 si_status=0, si_utime=0, si_stime=0} ---
exit_group(3) = ?
+++ exited with 3 +++
```

If you need to analyze the output of **strace** and the output messages are too long to be inspected directly in the console window, use -o. In that case, unnecessary messages, such as information about attaching and detaching processes, are suppressed. You can also suppress these messages (normally printed on the standard output) with -q. To add time stamps at the beginning of each line with a system call, use -t:

```
>strace -t -o strace_sleep.txt sleep 1; more strace_sleep.txt
08:44:06 execve("/bin/sleep", ["sleep", "1"], [/ * 81 vars *]) = 0
08:44:06 brk(0) = 0x606000
08:44:06 mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, \
-1, 0) = 0x7f8e78cc5000
[...]
08:44:06 close(3) = 0
08:44:06 nanosleep({1, 0}, NULL) = 0
08:44:07 close(1) = 0
08:44:07 close(2) = 0
08:44:07 exit_group(0) = ?
```

The behavior and output format of **strace** can be controlled. For more information, see the relevant manual page (man 1 strace).

19.2. Tracing library calls with ltrace

ltrace traces dynamic library calls of a process. It is used in a similar way to **strace**, and most of their parameters have a similar or identical meaning. By default, **ltrace** uses `/etc/ltrace.conf` or `~/.ltrace.conf` configuration files. You can, however, specify an alternative one with the `-F CONFIG_FILE` option.

In addition to library calls, **ltrace** with the `-S` option can trace system calls as well:

```
>ltrace -S -o ltrace_find.txt find /etc -name \
xorg.conf; more ltrace_find.txt
SYS_brk(NULL) = 0x00628000
SYS_mmap(0, 4096, 3, 34, 0xffffffff) = 0x7f1327ea1000
SYS_mmap(0, 4096, 3, 34, 0xffffffff) = 0x7f1327ea0000
[...]
fnmatch("xorg.conf", "xorg.conf", 0) = 0
free(0x0062db80) = <void>
__errno_location() = 0x7f1327e5d698
__ctype_get_mb_cur_max(0x7fff25227af0, 8192, 0x62e020, -1, 0) = 6
__ctype_get_mb_cur_max(0x7fff25227af0, 18, 0x7f1327e5d6f0, 0x7fff25227af0,
0x62e031) = 6
__fprintf_chk(0x7f1327821780, 1, 0x420cf7, 0x7fff25227af0, 0x62e031
<unfinished ...>
SYS_fstat(1, 0x7fff25227230) = 0
SYS_mmap(0, 4096, 3, 34, 0xffffffff) = 0x7f1327e72000
SYS_write(1, "/etc/X11/xorg.conf\n", 19) = 19
[...]
```

You can change the type of traced events with the `-e` option. The following example prints library calls related to `fnmatch` and `strlen` functions:

```
>ltrace -e fnmatch,strlen find /etc -name xorg.conf
[...]
fnmatch("xorg.conf", "xorg.conf", 0) = 0
strlen("Xresources") = 10
strlen("Xresources") = 10
strlen("Xresources") = 10
fnmatch("xorg.conf", "Xresources", 0) = 1
strlen("xorg.conf.install") = 17
[...]
```

To display only the symbols included in a specific library, use `-l /path/to/library`:

```
>ltrace -l /lib64/librt.so.1 sleep 1
clock_gettime(1, 0x7fff4b5c34d0, 0, 0, 0) = 0
clock_gettime(1, 0x7fff4b5c34c0, 0xfffffffff600180, -1, 0) = 0
+++ exited (status 0) +++
```

You can make the output more readable by indenting each nested call by the specified number of space with the `-n NUM_OF_SPACES`.

19.3. Debugging and profiling with Valgrind

Valgrind is a set of tools to debug and profile your programs so that they can run both faster and with fewer errors. Valgrind can detect problems related to memory management and threading, or can also serve as a framework for building new debugging tools. It is well known that this tool can incur high overhead, causing, for example, higher runtimes or changing the normal program behavior under concurrent workloads based on timing.

19.1. Installation

Valgrind is not shipped with standard SUSE Linux Enterprise Server distribution. To install it on your system, you need to obtain SUSE Software Development Kit, and either install it and run

```
zypperinstall VALGRIND
```

or browse through the SUSE Software Development Kit directory tree, locate the Valgrind package and install it with

```
rpm-i valgrind-VERSION_ARCHITECTURE.rpm
```

The SDK is a module for SUSE Linux Enterprise and is available via an online channel from the SUSE Customer Center. Refer to *Modules and Extensions Quick Start* for details.

19.2. Supported architectures

SUSE Linux Enterprise Server supports Valgrind on the following architectures:

- AMD64/Intel 64
- POWER
- IBM Z

19.3. General information

The main advantage of Valgrind is that it works with existing compiled executables. You do not need to recompile or modify your programs to use it. Run Valgrind like this:

```
valgrindVALGRIND_OPTIONS your-prog YOUR-PROGRAM-OPTIONS
```

Valgrind consists of several tools, and each provides specific functionality. Information in this section is general and valid regardless of the used tool. The most important configuration option is `--tool`. This option tells Valgrind which tool to run. If you omit this option, memcheck is selected by default. For example, to run `find --name .bashrc` with Valgrind's memcheck tools, enter the following in the command line:

```
valgrind --tool=memcheck find ~ -name .bashrc
```

A list of standard Valgrind tools with a brief description follows:

memcheck

Detects memory errors. It helps you tune your programs to behave correctly.

cachegrind

Profiles cache prediction. It helps you tune your programs to run faster.

callgrind

Works in a similar way to cachegrind but also gathers additional cache-profiling information.

exp-drd

Detects thread errors. It helps you tune your multi-threaded programs to behave correctly.

helgrind

Another thread error detector. Similar to exp-drd but uses different techniques for problem analysis.

massif

A heap profiler. Heap is an area of memory used for dynamic memory allocation. This tool helps you tune your program to use less memory.

lackey

An example tool showing instrumentation basics.

19.4. Default options

Valgrind can read options at start-up. There are three places which Valgrind checks:

1. The file `.valgrindrc` in the home directory of the user who runs Valgrind.
2. The environment variable `$VALGRIND_OPTS`

3. The file `.valgrindrc` in the current directory where Valgrind is run from.

These resources are parsed exactly in this order, while later given options take precedence over earlier processed options. Options specific to a particular Valgrind tool must be prefixed with the tool name and a colon. For example, if you want cachegrind to always write profile data to the `/tmp/cachegrind_PID.log`, add the following line to the `.valgrindrc` file in your home directory:

```
--cachegrind:cachegrind-out-file=/tmp/cachegrind_%p.log
```

19.5. How Valgrind works

Valgrind takes control of your executable before it starts. It reads debugging information from the executable and related shared libraries. The executable's code is redirected to the selected Valgrind tool, and the tool adds its own code to handle its debugging. Then the code is handed back to the Valgrind core and the execution continues.

For example, memcheck adds its code, which checks every memory access. As a consequence, the program runs much slower than in the native execution environment.

Valgrind simulates every instruction of your program. Therefore, it not only checks the code of your program, but also all related libraries (including the C library), libraries used for graphical environment, and so on. If you try to detect errors with Valgrind, it also detects errors in associated libraries (like C, X11, or Gtk libraries). Because you do not need all these errors frequently, Valgrind can selectively suppress these error messages to suppression files. The `--gen-suppressions=yes` tells Valgrind to report these suppressions which you can copy to a file.

You should supply a real executable (machine code) as a Valgrind argument. If your application is run, for example, from a shell or Perl script, you can by mistake get error reports related to `/bin/sh` (or `/usr/bin/perl`). In such cases, you can use `--trace-children=yes` to work around this issue. However, using the executable itself avoids any confusion over this issue.

19.6. Messages

During its runtime, Valgrind reports messages with detailed errors and important events. The following example explains the messages:

```

>valgrind --tool=memcheck find ~ -name .bashrc
[...]
==6558== Conditional jump or move depends on uninitialised value(s)
==6558==    at 0x400AE79: _dl_relocate_object (in /lib64/ld-2.11.1.so)
==6558==    by 0x4003868: _dl_main (in /lib64/ld-2.11.1.so)
[...]
==6558== Conditional jump or move depends on uninitialised value(s)
==6558==    at 0x400AE82: _dl_relocate_object (in /lib64/ld-2.11.1.so)
==6558==    by 0x4003868: _dl_main (in /lib64/ld-2.11.1.so)
[...]
==6558== ERROR SUMMARY: 2 errors from 2 contexts (suppressed: 0 from 0)
==6558== malloc/free: in use at exit: 2,228 bytes in 8 blocks.
==6558== malloc/free: 235 allocs, 227 frees, 489,675 bytes allocated.
==6558== For counts of detected errors, rerun with: -v
==6558== searching for pointers to 8 not-freed blocks.
==6558== checked 122,584 bytes.
==6558==
==6558== LEAK SUMMARY:
==6558==    definitely lost: 0 bytes in 0 blocks.
==6558==    possibly lost: 0 bytes in 0 blocks.
==6558==    still reachable: 2,228 bytes in 8 blocks.
==6558==    suppressed: 0 bytes in 0 blocks.
==6558== Rerun with --leak-check=full to see details of leaked memory.

```

The `==6558==` introduces Valgrind's messages and contains the process ID number (PID). You can easily distinguish Valgrind's messages from the output of the program itself, and decide which messages belong to a particular process.

To make Valgrind's messages more detailed, use `-v` or even `-v -v`.

You can make Valgrind send its messages to three different places:

1. By default, Valgrind sends its messages to the file descriptor 2, which is the standard error output. You can tell Valgrind to send its messages to any other file descriptor with the `--log-fd=FILE_DESCRIPTOR_NUMBER` option.
2. The second, and a more useful way, is to send Valgrind's messages to a file with `--log-file=FILENAME`. This option accepts several variables, for example, `%p` gets replaced with the PID of the currently profiled process. This way you can send messages to different files based on their PID. `%q{env_var}` is replaced with the value of the related `env_var` environment variable.

The following example checks for possible memory errors during the Apache Web server restart, while following children processes and writing detailed Valgrind's messages to separate files distinguished by the current process PID:

```

>valgrind -v --tool=memcheck --trace-children=yes \
--log-file=valgrind_pid_%p.log systemctl restart apache2.service

```

This process created 52 log files in the testing system, and took 75 seconds instead of the usual 7 seconds needed to run `sudo systemctl restart apache2.service` without Valgrind, which is approximately 10 times more.


```
>ls -l valgrind_pid*.log
valgrind_pid_11780.log
valgrind_pid_11782.log
valgrind_pid_11783.log
[...]
valgrind_pid_11860.log
valgrind_pid_11862.log
valgrind_pid_11863.log
```

3. You may also prefer to send the Valgrind's messages over the network. You need to specify the `aa.bb.cc.dd` IP address and `port_num` port number of the network socket with the `--log-socket=AA.BB.CC.DD:PORT_NUM` option. If you omit the port number, 1500 is used.

It is useless to send Valgrind's messages to a network socket if no application is capable of receiving them on the remote machine. That is why **valgrind-listener**, a simple listener, is shipped together with Valgrind. It accepts connections on the specified port and copies everything it receives to the standard output.

19.7. Error messages

Valgrind remembers all error messages, and if it detects a new error, the error is compared against old error messages. This way Valgrind checks for duplicate error messages. In case of a duplicate error, it is recorded but no message is shown. This mechanism prevents you from being overwhelmed by millions of duplicate errors.

The `-v` option adds a summary of all reports (sorted by their total count) to the end of the Valgrind's execution output. Moreover, Valgrind stops collecting errors if it detects either 1000 different errors, or 10 000 000 errors in total. To suppress this limit and see all error messages, use `--error-limit=no`.

Certain errors cause other errors. Therefore, fix errors in the same order as they appear and re-check the program continuously.

19.4. More information

- For a complete list of options related to the described tracing tools, see the corresponding man page (**man 1 strace**, **man 1 ltrace**, and **man 1 valgrind**).
- To describe advanced usage of Valgrind is beyond the scope of this document. It is well documented, see [Valgrind User Manual](#). These pages are indispensable if you need more advanced information on Valgrind or the usage and purpose of its standard tools.

Chapter 20. Kexec and Kdump

Kexec is a tool to boot to another kernel from the currently running one. You can perform faster system reboots without any hardware initialization. You can also prepare the system to boot to another kernel if the system crashes.

20.1. Introduction

With Kexec, you can replace the running kernel with another one without a hard reboot. The tool is useful for several reasons:

- Faster system rebooting

If you need to reboot the system frequently, Kexec can save you significant time.

- Avoiding unreliable firmware and hardware

Computer hardware is complex and serious problems may occur during the system start-up. You cannot always replace unreliable hardware immediately. Kexec boots the kernel to a controlled environment with the hardware already initialized. The risk of unsuccessful system start is then minimized.

- Saving the dump of a crashed kernel

Kexec preserves the contents of the physical memory. After the *production* kernel fails, the *capture* kernel (an additional kernel running in a reserved memory range) saves the state of the failed kernel. The saved image can help you with the subsequent analysis.

- Booting without GRUB 2 configuration

When the system boots a kernel with Kexec, it skips the boot loader stage. The normal booting procedure can fail because of an error in the boot loader configuration. With Kexec, you do not depend on a working boot loader configuration.

20.2. Required packages

To use Kexec on SUSE® Linux Enterprise Server to speed up reboots or avoid potential hardware problems, make sure that the package `kexec-tools` is installed. It contains a script called **kexec-bootloader**, which reads the boot loader configuration and runs Kexec using the same kernel options as the normal boot loader.

To set up an environment that helps you obtain debug information in case of a kernel crash, make sure that the package `makedumpfile` is installed.

The preferred method of using Kdump in SUSE Linux Enterprise Server is through the YaST Kdump module. To use the YaST module, make sure that the package `yast2-kdump` is installed.

20.3. Kexec internals

The most important component of Kexec is the `/sbin/kexec` command. You can load a kernel with Kexec in two different ways:

- Load the kernel to the address space of a production kernel for a regular reboot:

```
#kexec -l KERNEL_IMAGE
```

You can later boot to this kernel with **kexec -e**.

- Load the kernel to a reserved area of memory:

```
#kexec -p KERNEL_IMAGE
```

This kernel is booted automatically when the system crashes.

To boot another kernel and preserve the data of the production kernel when the system crashes, you need to reserve a dedicated area of the system memory. The production kernel never loads to this area because it must be always available. It is used for the capture kernel so that the memory pages of the production kernel can be preserved.

To reserve the area, append the option `crashkernel` to the boot command line of the production kernel. To determine the necessary values for `crashkernel`, follow the instructions in *the section called "Calculating crashkernel allocation size"*.

This is not a parameter of the capture kernel. The capture kernel does not use Kexec.

The capture kernel is loaded to the reserved area and waits for the kernel to crash. Then, Kdump tries to invoke the capture kernel because the production kernel is no longer reliable at this stage. This means that even Kdump can fail.

To load the capture kernel, you need to include the kernel boot parameters. In most cases, the initial RAM file system is used for booting. You can specify it with `--initrd=FILENAME`. With `--append=CMDLINE`, you append options to the command line of the kernel to boot.

It is required to include the command line of the production kernel. You can simply copy the command line with `--append="$(cat /proc/cmdline)"` or add more options with `--append="$(cat /proc/cmdline) more_options"`.

For example, to load the `/boot/vmlinuz-6.4.0-150700.38-default` kernel image with the command line of the currently running production kernel and the `/boot/initrd` file, run the following command:

```
# kexec -l /boot/vmlinuz-6.4.0-150700.38-default \
--append="$(cat /proc/cmdline)" --initrd=/boot/initrd
```

You can always unload the previously loaded kernel. To unload a kernel that was loaded with the `-l` option, use the **kexec** -u command. To unload a crash kernel loaded with the `-p` option, use **kexec** -p -u command.

20.4. Calculating crashkernel allocation size

To use Kexec with a capture kernel and to use Kdump in any way, RAM needs to be allocated for the capture kernel. The allocation size depends on the expected hardware configuration of the computer, therefore you need to specify it.

The allocation size also depends on the hardware architecture of your computer. Make sure to follow the procedure intended for your system architecture.

Procedure 20.1. Allocation size on AMD64/Intel 64

1. To find out the base value for the computer, run the following command:

```
#kdumptool calibrate
Total: 49074
Low: 72
High: 180
MinLow: 72
MaxLow: 3085
MinHigh: 0
MaxHigh: 45824
```

All values are given in megabytes.

2. Take note of the values of Low and High.



Significance of Low and High values

On AMD64/Intel 64 computers, the High value stands for the memory reservation for all available memory. The Low value stands for the memory reservation in the DMA32 zone, that is, all the memory up to the 4 GB mark.

SIZE_LOW is the amount of memory required by 32-bit-only devices. The kernel allocates 64M for DMA32 bounce buffers. If your server does not have any 32-bit-only devices, everything should work with the default allocation of 72M for SIZE_LOW. A possible exception to this is on NUMA machines, which may make it appear that more Low memory is needed. The Kdump kernel may be booted with `numa=off` to make sure normal kernel allocations do not use Low memory.

3. Adapt the High value from the previous step for the number of LUN kernel paths (paths to storage devices) attached to the computer. A sensible value in megabytes can be calculated using this formula:

```
SIZE_HIGH = RECOMMENDATION + (LUNs / 2)
```

The following parameters are used in this formula:

- **SIZE_HIGH**The resulting value for High.
- **RECOMMENDATION**The value recommended by **kdumptool calibrate** for High.
- **LUNs**The maximum number of LUN kernel paths that you expect to ever create on the computer. Exclude multipath devices from this number, as these are ignored. To get the *current* number of LUNs available on your system, run the following command:

```
> cat /proc/scsi/scsi | grep Lun | wc -l
```

4. If the drivers for your device make many reservations in the DMA32 zone, the Low value also needs to be adjusted. However, there is no simple formula to calculate these. Finding the right size can therefore be a process of trial and error.

For the beginning, use the Low value recommended by **kdumptool calibrate**.

5. The values now need to be set in the correct location.

If you are changing the kernel command line directly

Append the following kernel option to your boot loader configuration:

```
crashkernel=SIZE_HIGH,high crashkernel=SIZE_LOW,low
```

Replace the placeholders *SIZE_HIGH* and *SIZE_LOW* with the appropriate value from the previous steps and append the letter M (for megabytes).

As an example, the following is valid:

```
crashkernel=36M,high crashkernel=72M,low
```

If you are using the YaST GUI:

Set *Kdump Low Memory* to the determined Low value.

Set *Kdump High Memory* to the determined High value.

If you are using the YaST command line interface:

Use the following command:

```
#yast kdump startup enable alloc_mem=LOW,HIGH
```

Replace *LOW* with the determined Low value. Replace *HIGH* with the determined HIGH value.

Procedure 20.2. Allocation size on POWER and IBM Z

1. To find out the basis value for the computer, run the following in a terminal:

```
#kdumptool calibrate
```

This command returns a list of values. All values are given in megabytes.

2. Write down the value of Low.

3. Adapt the Low value from the previous step for the number of LUN kernel paths (paths to storage devices) attached to the computer. A sensible value in megabytes can be calculated using this formula:

$$SIZE_LOW = RECOMMENDATION + (LUNs / 2)$$

The following parameters are used in this formula:

- **SIZE_LOW**The resulting value for Low.
 - **RECOMMENDATION**The value recommended by **kdumptool calibrate** for Low.
 - **LUNs**The maximum number of LUN kernel paths that you expect to ever create on the computer. Exclude multipath devices from this number, as these are ignored.
4. The values now need to be set in the correct location.

If you are working on the command line

Append the following kernel option to your boot loader configuration:

```
crashkernel=SIZE_LOW
```

Replace the placeholder `SIZE_LOW` with the appropriate value from the previous step and append the letter M (for megabytes).

As an example, the following is valid:

```
crashkernel=108M
```

If you are working in YaST

Set *Kdump Memory* to the determined Low value.



Excluding unused and inactive CCW devices on IBM Z

Depending on the number of available devices the calculated amount of memory specified by the `crashkernel` kernel parameter may not be sufficient. Instead of increasing the value, you may alternatively limit the amount of devices visible to the kernel. This lowers the required amount of memory for the `crashkernel` setting.

1. To ignore devices you can run the **`cio_ignore`** tool to generate an appropriate stanza to ignore all devices, except the ones currently active or in use.

```
>sudo cio_ignore -u -k  
cio_ignore=all,!da5d,!f500-f502
```

When you run **`cio_ignore -u -k`**, the blocklist becomes active and replaces any existing blocklist immediately. Unused devices are not being purged, so they still appear in the channel subsystem. But adding new channel devices (via CP ATTACH under z/VM or dynamic I/O configuration change in LPAR) treats them as blocked. To prevent this, preserve the original setting by running **`sudo cio_ignore -l`** first and reverting to that state after running **`cio_ignore -u -k`**. As an alternative, add the generated stanza to the regular kernel boot parameters.

2. Now add the `cio_ignore` kernel parameter with the stanza from above to `KDUMP_CMDLINE_APPEND` in `/etc/sysconfig/kdump`, for example:

```
KDUMP_COMMANDLINE_APPEND="cio_ignore=all,!da5d,!f500-f502"
```

3. Activate the setting by restarting `kdump`:

```
systemctl restart kdump.service
```

20.5. Basic Kexec usage

To use Kexec, ensure the respective service is enabled and running:

- Make sure the Kexec service is loaded at system start:

```
>sudo systemctl enable kexec-load.service
```

- Make sure the Kexec service is running:

```
>sudo systemctl start kexec-load.service
```

To verify if your Kexec environment works properly, try rebooting into a new Kernel with Kexec. Make sure no users are currently logged in and no important services are running on the system. Then run the following command:

```
systemctl kexec
```

The new kernel previously loaded to the address space of the older kernel rewrites it and takes control immediately. It displays the usual start-up messages. When the new kernel boots, it skips all hardware and firmware checks. Make sure no warning messages appear.



Using Kexec with the reboot command

To make **reboot** use Kexec rather than performing a regular reboot, run the following command:

```
ln -s /usr/lib/systemd/system/kexec.target /etc/systemd/system/reboot.target
```

You can revert this at any time by deleting `etc/systemd/system/reboot.target`.

20.6. How to configure Kexec for routine reboots

Kexec is often used for frequent reboots. For example, if it takes a long time to run through the hardware detection routines or if the start-up is not reliable.

The firmware and the boot loader are not used when the system reboots with Kexec. Any changes you make to the boot loader configuration are ignored until the computer performs a hard reboot.

20.7. Basic Kdump configuration

You can use Kdump to save kernel dumps. If the kernel crashes, it is useful to copy the memory image of the crashed environment to the file system. You can then debug the dump file to find the cause of the kernel crash. This is called “core dump”.

Kdump works similarly to Kexec (see *Chapter 20, Kexec and Kdump*). The capture kernel is executed after the running production kernel crashes. The difference is that Kexec replaces the production kernel with the capture kernel. With Kdump, you still have access to the memory space of the crashed production kernel. You can save the memory snapshot of the crashed kernel in the environment of the Kdump kernel.



Dumps over network

In environments with limited local storage, you need to set up kernel dumps over the network. Kdump supports configuring the specified network interface and bringing it up via `initrd`. Both LAN and VLAN interfaces are supported. Specify the network interface and the mode (DHCP or static) either with YaST, or using the `KDUMP_NETCONFIG` option in the `/etc/sysconfig/kdump` file.

Target file system for Kdump must be mounted during configuration



When configuring Kdump, you can specify a location to which the dumped images are saved (default: `/var/crash`). This location must be mounted when configuring Kdump, otherwise the configuration fails.

20.1. Manual Kdump configuration

Kdump reads its configuration from the `/etc/sysconfig/kdump` file. To make sure that Kdump works on your system, its default configuration is sufficient. To use Kdump with the default settings, follow these steps:

1. Determine the amount of memory needed for Kdump by following the instructions in *the section called “Calculating crashkernel allocation size”*. Make sure to set the kernel parameter `crashkernel`.
2. Reboot the computer.
3. Enable the Kdump service:

```
#systemctl enable kdump
```

4. You can edit the options in `/etc/sysconfig/kdump`. Reading the comments helps you understand the meaning of individual options.
5. Execute the init script once with **`sudo systemctl start kdump`**, or reboot the system.

After configuring Kdump with the default values, check if it works as expected. Make sure that no users are currently logged in and no important services are running on your system. Then follow these steps:

1. Switch to the rescue target with **`systemctl isolate rescue.target`**
2. Restart the Kdump service:

```
#systemctl start kdump
```

3. Unmount all the disk file systems except the root file system with:

```
#umount -a
```

4. Remount the root file system in read-only mode:

```
#mount -o remount,ro /
```

5. Invoke a “kernel panic” with the `procfs` interface to Magic SysRq keys:

```
#echo c > /proc/sysrq-trigger
```

Size of kernel dumps



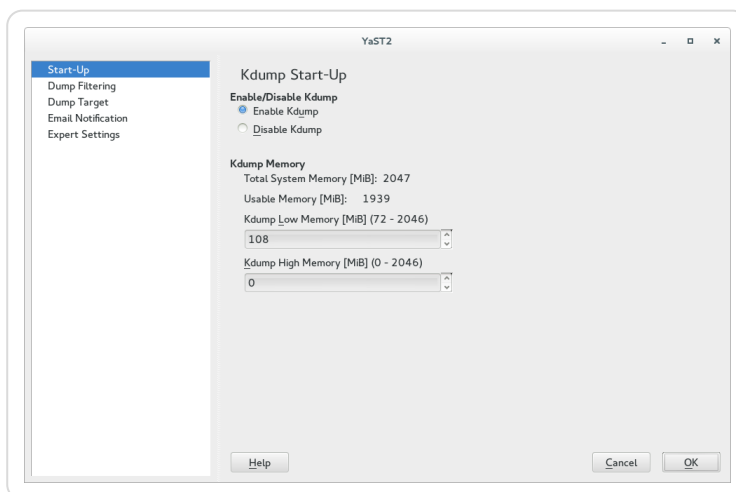
The `KDUMP_KEEP_OLD_DUMPS` option controls the number of preserved kernel dumps (default is 5). Without compression, the size of the dump can take up to the size of the physical memory or RAM. Make sure you have sufficient space on the `/var` partition.

The capture kernel boots and the crashed kernel memory snapshot is saved to the file system. The save path is given by the `KDUMP_SAVEDIR` option and it defaults to `/var/crash`. If `KDUMP_IMMEDIATE_REBOOT` is set to `yes`, the system automatically reboots the production kernel. Log in and check that the dump has been created under `/var/crash`.

20.2. YaST configuration

To configure Kdump with YaST, you need to install the `yast2-kdump` package. Then either start the *Kernel Kdump* module in the *System* category of *YaST Control Center*, or enter **yast2 kdump** in the command line as root.

Figure 20.1. YaST Kdump module: start-up page



In the *Start-Up* window, select *Enable Kdump*.

The values for *Kdump Memory* are automatically generated the first time you open the window. However, that does not mean that they are always sufficient. To set the right values, follow the instructions in the section called “*Calculating crashkernel allocation size*”.

After hardware changes, set *Kdump* memory values again



If you have set up Kdump on a computer and later decide to change the amount of RAM or hard disks available to it, YaST continues to display and use outdated memory values.

To work around this, determine the necessary memory again, as described in *the section called “Calculating crashkernel allocation size”*. Then set it manually in YaST.

Click *Dump Filtering* in the left pane, and check what pages to include in the dump. You do not need to include the following memory content to be able to debug kernel problems:

- Pages filled with zero
- Cache pages
- User data pages
- Free pages

In the *Dump Target* window, select the type of the dump target and the URL where you want to save the dump. If you selected a network protocol, such as FTP or SSH, you need to enter relevant access information as well.



Sharing the dump directory with other applications

It is possible to specify a path for saving Kdump dumps where other applications also save their dumps. When cleaning its old dump files, Kdump safely ignores other applications' dump files.

Fill the *Email Notification* window information if you want Kdump to inform you about its events via e-mail and confirm your changes with *OK* after fine tuning Kdump in the *Expert Settings* window. Kdump is now configured.

20.3. Kdump over SSH

Dump files usually contain sensitive data which should be protected from unauthorized disclosure. To allow transmission of such data over an insecure network, Kdump can save dump files to a remote machine using the SSH protocol.

1. The target host identity must be known to Kdump. This is needed to ensure that sensitive data is never sent to an imposter. When Kdump generates a new *initrd*, it runs **ssh-keygen -F TARGET_HOST** to query the target host's identity. This works only if

`TARGET_HOST` public key is already known. An easy way to achieve that is to make an SSH connection to `TARGET_HOST` as `root` on the Kdump host.

2. Kdump must be able to authenticate to the target machine. Only public key authentication is currently available. By default, Kdump uses `root`'s private key, but it is advisable to make a separate key for Kdump. This can be done with **ssh-keygen**:

1.

```
#ssh-keygen -f ~/.ssh/kdump_key
```

2. Press ☐ when prompted for passphrase (that is, do not use any passphrase).

3. Open `/etc/sysconfig/kdump` and set `KDUMP_SSH_IDENTITY` to `kdump_key`. You can use full path to the file if it is not placed under `~/.ssh`.

3. Set up the Kdump SSH key to authorize logins to the remote host.

```
#ssh-copy-id -i ~/.ssh/kdump_key TARGET_HOST
```

4. Set up `KDUMP_SAVEDIR`. There are two options:

Secure File Transfer Protocol (SFTP)

SFTP is the preferred method for transmitting files over SSH. The target host must enable the SFTP subsystem (SUSE Linux Enterprise Server default). Example:

```
KDUMP_SAVEDIR=sftp://TARGET_HOST/path/to/dumps
```

Secure Shell protocol (SSH)

Some other distributions use SSH to run certain commands on the target host. SUSE Linux Enterprise Server can also use this method. The Kdump user on the target host must have a login shell that can execute these commands: **mkdir**, **dd** and **mv**. Example:

```
KDUMP_SAVEDIR=ssh://TARGET_HOST/path/to/dumps
```

5. Restart the Kdump service to use the new configuration.

20.8. Analyzing the crash dump

After you obtain the dump, it is time to analyze it. There are several options.

The original tool to analyze the dumps is GDB. You can even use it in the latest environments, although it has several disadvantages and limitations:

- GDB was not specifically designed to debug kernel dumps.
- GDB does not support ELF64 binaries on 32-bit platforms.
- GDB does not understand other formats than ELF dumps (it cannot debug compressed dumps).

That is why the **crash** utility was implemented. It analyzes crash dumps and debugs the running system as well. It provides functionality specific to debugging the Linux kernel and is much more suitable for advanced debugging.

To debug the Linux kernel, install its debugging information package, too. Check if the package is installed on your system with:

```
>zypper se kernel | grep debug
```

Repository for packages with debugging information



If you subscribed your system for online updates, you can find “debuginfo” packages in the *-Debuginfo-Updates online installation repository relevant for SUSE Linux Enterprise Server15 SP7. Use YaST to enable the repository.

To open the captured dump in **crash** on the machine that produced the dump, use a command like this:

```
crash/boot/vmlinux-6.4.0-150700.38-default.gz \  
/var/crash/2024-04-23-11\ :17/vmcore
```

The first parameter represents the kernel image. The second parameter is the dump file captured by Kdump. You can find this file under /var/crash by default.



Getting basic information from a kernel crash dump

SUSE Linux Enterprise Server ships with the utility **kdumplib** (included in a package with the same name) for identifying unknown kernel dumps. It can be used to extract basic information such as architecture and kernel release. It supports lkcd, diskdump, Kdump files and ELF dumps. When called with the -v switch it tries to extract additional information such as machine type, kernel banner string and kernel configuration flavor.

20.1. Kernel binary formats

The Linux kernel comes in Executable and Linkable Format (ELF). This file is called `vmlinux` and is directly generated in the compilation process. Not all boot loaders support ELF binaries, especially on the AMD64/Intel 64 architecture. The following solutions exist on different architectures supported by SUSE® Linux Enterprise Server.

20.1. AMD64/Intel 64

Kernel packages for AMD64/Intel 64 from SUSE contain two kernel files: `vmlinux` and `vmlinux.gz`.

- **vmlinux** This is the file executed by the boot loader.

The Linux kernel consists of two parts: the kernel itself (`vmlinux`) and the setup code run by the boot loader. These two parts are linked together to create `vmlinux` (note the distinction: `z` compared to `x`).

In the kernel source tree, the file is called `bzImage`.

- **vmlinux.gz** This is a compressed ELF image that can be used by **crash** and GDB. The ELF image is never used by the boot loader itself on AMD64/Intel 64. Therefore, only a compressed version is shipped.

20.2. POWER

The yaboot boot loader on POWER also supports loading ELF images, but not compressed ones. In the POWER kernel package, there is an ELF Linux kernel file `vmlinux`. Considering **crash**, this is the easiest architecture.

If you decide to analyze the dump on another machine, you must check both the architecture of the computer and the files necessary for debugging.

You can analyze the dump on another computer only if it runs a Linux system of the same architecture. To check the compatibility, use the command **uname -i** on both computers and compare the outputs.

If you are going to analyze the dump on another computer, you also need the appropriate files from the `kernel` and `kernel debug` packages.

1. Put the kernel dump, the kernel image from `/boot`, and its associated debugging info file from `/usr/lib/debug/boot` into a single empty directory.
2. Additionally, copy the kernel modules from `/lib/modules/$(uname -r)/kernel/` and the associated debug info files from `/usr/lib/debug/lib/modules/$(uname -r)/kernel/` into a subdirectory named `modules`.
3. In the directory with the dump, the kernel image, its debug info file, and the `modules` subdirectory, start the **crash** utility:

```
>crashVMLINUX-VERSION vmcore
```

Regardless of the computer on which you analyze the dump, the `crash` utility produces output similar to this:

```

>crash/boot/vmlinux-6.4.0-150700.38-default.gz \
/var/crash/2024-04-23-11:17/vmcore
crash 7.2.1
Copyright (C) 2002-2017 Red Hat, Inc.
Copyright (C) 2004, 2005, 2006, 2010 IBM Corporation
Copyright (C) 1999-2006 Hewlett-Packard Co
Copyright (C) 2005, 2006, 2011, 2012 Fujitsu Limited
Copyright (C) 2006, 2007 VA Linux Systems Japan K.K.
Copyright (C) 2005, 2011 NEC Corporation
Copyright (C) 1999, 2002, 2007 Silicon Graphics, Inc.
Copyright (C) 1999, 2000, 2001, 2002 Mission Critical Linux, Inc.
This program is free software, covered by the GNU General Public License,
and you are welcome to change it and/or distribute copies of it under
certain conditions. Enter "help copying" to see the conditions.
This program has absolutely no warranty. Enter "help warranty" for details.

GNU gdb (GDB) 7.6
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law. Type "show copying"
and "show warranty" for details.
This GDB was configured as "x86_64-unknown-linux-gnu".

        KERNEL: /boot/vmlinux-6.4.0-150700.38-default.gz
        DEBUGINFO: /usr/lib/debug/boot/vmlinux-6.4.0-150700.38-default.debug
        DUMPFILE: /var/crash/2024-04-23-11:17/vmcore
        CPUS: 2
        DATE: Thu Apr 23 13:17:01 2024
        UPTIME: 00:10:41
LOAD AVERAGE: 0.01, 0.09, 0.09
        TASKS: 42
        NODENAME: eros
        RELEASE: 6.4.0-150700.38-default
        VERSION: #1 SMP 2024-03-31 14:50:44 +0200
        MACHINE: x86_64 (2999 Mhz)
        MEMORY: 16 GB
        PANIC: "SysRq : Trigger a crashdump"
        PID: 9446
        COMMAND: "bash"
        TASK: ffff88003a57c3c0 [THREAD_INFO: ffff880037168000]
        CPU: 1
        STATE: TASK_RUNNING (SYSRQ)
crash>

```

The command output prints first useful data: There were 42 tasks running at the moment of the kernel crash. The cause of the crash was a SysRq trigger invoked by the task with PID 9446. It was a Bash process because the **echo** that has been used is an internal command of the Bash shell.

The **crash** utility builds upon GDB and provides many additional commands. If you enter **bt** without any parameters, the backtrace of the task running at the moment of the crash is printed:

```

crash>bt
PID: 9446   TASK: ffff88003a57c3c0   CPU: 1   COMMAND: "bash"
#0 [ffff880037169db0] crash_kexec at ffffffff80268fd6
#1 [ffff880037169e80] __handle_sysrq at ffffffff803d50ed
#2 [ffff880037169ec0] write_sysrq_trigger at ffffffff802f6fc5
#3 [ffff880037169ed0] proc_reg_write at ffffffff802f068b
#4 [ffff880037169f10] vfs_write at ffffffff802b1aba
#5 [ffff880037169f40] sys_write at ffffffff802b1c1f
#6 [ffff880037169f80] system_call_fastpath at ffffffff8020bfbb
RIP: 00007fa958991f60   RSP: 00007fff61330390   RFLAGS: 00010246
RAX: 0000000000000001   RBX: ffffffff8020bfbb   RCX: 0000000000000001
RDX: 0000000000000002   RSI: 00007fa959284000   RDI: 0000000000000001
RBP: 0000000000000002   R8: 00007fa9592516f0   R9: 00007fa958c209c0
R10: 00007fa958c209c0   R11: 00000000000000246   R12: 00007fa958c1f780
R13: 00007fa959284000   R14: 0000000000000002   R15: 00000000595569d0
ORIG_RAX: 0000000000000001   CS: 0033   SS: 002b
crash>

```

Now it is clear what happened: The internal **echo** command of Bash shell sent a character to /proc/sysrq-trigger. After the corresponding handler recognized this character, it invoked the `crash_kexec()` function. This function called `panic()` and Kdump saved a dump.

In addition to the basic GDB commands and the extended version of **bt**, the crash utility defines other commands related to the structure of the Linux kernel. These commands understand the internal data structures of the Linux kernel and present their contents in a human readable format. For example, you can list the tasks running at the moment of the crash with **ps**. With **sym**, you can list all the kernel symbols with the corresponding addresses, or inquire an individual symbol for its value. With **files**, you can display all the open file descriptors of a process. With **kmem**, you can display details about the kernel memory usage. With **vm**, you can inspect the virtual memory of a process, even at the level of individual page mappings. The list of useful commands is long, and many of these accept a wide range of options.

The commands that we mentioned reflect the functionality of the common Linux commands, such as **ps** and **ls**. To find out the exact sequence of events with the debugger, you need to know how to use GDB and to have strong debugging skills. Both of these are out of the scope of this document. Additionally, you need to understand the Linux kernel. Several useful reference information sources are given at the end of this document.

20.9. Advanced Kdump configuration

The configuration for Kdump is stored in `/etc/sysconfig/kdump`. You can also use YaST to configure it. Kdump configuration options are available under *System > Kernel Kdump* in YaST Control Center. The following Kdump options may be useful for you.

You can change the directory for the kernel dumps with the `KDUMP_SAVEDIR` option. Keep in mind that the size of kernel dumps can be large. Kdump refuses to save the dump if the free disk space, subtracted by the estimated dump size, drops below the value specified by the `KDUMP_FREE_DISK_SIZE` option. `KDUMP_SAVEDIR` understands the URL format `PROTOCOL://SPECIFICATION`, where `PROTOCOL` is one of `file`, `ftp`, `sftp`, `nfs` or `cifs`, and

specification varies for each protocol. For example, to save kernel dump on an FTP server, use the following URL as a template: `ftp://username:password@ftp.example.com:123/var/crash`.

Kernel dumps are large and contain many pages that are not necessary for analysis. With `KDUMP_DUMPLEVEL` option, you can omit such pages. The option understands numeric value between 0 and 31. If you specify 0, the dump size is the largest. If you specify 31, it produces the smallest dump. For a complete table of possible values, see the manual page of **kdump** (**man 7 kdump**).

Sometimes it is useful to make the size of the kernel dump smaller. For example, you can do so to transfer the dump over the network or to save disk space in the dump directory. This can be done with `KDUMP_DUMPFORMAT` set to `compressed`. The **crash** utility supports dynamic decompression of the compressed dumps.

Changes to the Kdump configuration file



After making changes to the `/etc/sysconfig/kdump` file, you need to run **systemctl restart kdump.service**. Otherwise, the changes only take effect next time you reboot the system.

20.10. More information

There is no single comprehensive reference to Kexec and Kdump usage. However, there are helpful resources that deal with certain aspects:

- For the Kexec utility usage, see the manual page of **kexec** (**man 8 kexec**).
- IBM provides comprehensive documentation on how to use dump tools on the IBM Z architecture at <https://developer.ibm.com/technologies/linux/>.
- You can find general information about Kexec at <https://developer.ibm.com/technologies/linux/>.
- For more details on Kdump specific to SUSE Linux Enterprise Server, see <https://ftp.suse.com/pub/people/tiwai/kdump-training/kdump-training.pdf>.
- An in-depth description of Kdump internals can be found at <https://lse.sourceforge.net/kdump/documentation/ols2oo5-kdump-paper.pdf>.

For more details on **crash** dump analysis and debugging tools, use the following resources:

- In addition to the info page of GDB (**info gdb**), there are printable guides at <https://sourceware.org/gdb/documentation/>.
- The crash utility features a comprehensive online help. Use **helpCOMMAND** to display the online help for **command**.

- If you have the necessary Perl skills, you can use Alicia to make the debugging easier. This Perl-based front-end to the crash utility can be found at <https://alicia.sourceforge.net/> .
- If you prefer to use Python instead, you should install Pykdump. This package helps you control GDB through Python scripts.
- A comprehensive overview of the Linux kernel internals is given in *Understanding the Linux Kernel* by Daniel P. Bovet and Marco Cesati (ISBN 978-0-596-00565-8).

Chapter 21. Using `systemd-coredump` to debug application crashes

`systemd-coredump` collects and displays core dumps, for analyzing application crashes. The core dump contains an image of the process's memory at the time of termination. By default, when a process crashes (or all processes belonging to an application), it stores the core dump in the `/var/lib/systemd/coredump` file and logs the core dump to the `systemd` journal, including a backtrace if possible. You also have the option to examine the dump file with other tools such as **`gdb`** or **`crash`** (see the section called “Analyzing the crash dump”).

Core dumps stored in `/var/lib/systemd/coredump` are deleted after three days (see the `d /var/lib/systemd/coredump` line in `/usr/lib/tmpfiles.d/systemd.conf`).

There is an option to not store core dumps, but to log only to the journal, which may be useful to minimize the collection and storage of sensitive information.

21.1. Use and configuration

`systemd-coredump` is enabled and ready to run by default. The default configuration is in `/etc/systemd/coredump.conf`:

```
[Coredump]
#Storage=external
#Compress=yes
#ProcessSizeMax=2G
#ExternalSizeMax=2G
#JournalSizeMax=767M
#MaxUse=
#KeepFree=
```

Size units are B, K, M, G, T, P, and E. `ExternalSizeMax` also supports a value of infinity.

The following example shows how to use Vim for simple testing, by creating a segfault to generate journal entries and a core dump.

Procedure 21.1. Creating a core dump with Vim

1. Enable the `debuginfo-pool` and `debuginfo-update` repositories
2. Install `vim-debuginfo`
3. Launch **`vim testfile`** and type a few characters
4. Get the PID and generate a segfault:

```
>ps ax | grep vim
2345 pts/3    S+      0:00 vim testfile
#kill -s SIGSEGV 2345
```

Vim emits error messages:

```
Vim: Caught deadly signal SEGV
Vim: Finished.
Segmentation fault (core dumped)
```

5. List your core dumps, then examine them:

```
#coredumpctl
TIME                               PID  UID  GID SIG PRESENT EXE
Wed 2019-11-12 11:56:47 PST 2345 1000 100 11  *    /bin/vim

#coredumpctl info
PID: 2345 (vim)
UID: 0 (root)
GID: 0 (root)
Signal: 11 (SEGV)
Timestamp: Wed 2019-11-12 11:58:05 PST
Command Line: vim testfile
Executable: /bin/vim
Control Group: /user.slice/user-1000.slice/session-1.scope
Unit: session-1.scope
Slice: user-1000.slice
Session: 1
Owner UID: 1000 (tux)
Boot ID: b5c251b86ab34674a2222cef102c0c88
Machine ID: b43c44a64696799b985cafd95dc1b698
Hostname: linux-uoch
Coredump: /var/lib/systemd/coredump/core.vim.
0.b5c251b86ab34674a2222cef102
Message: Process 2345 (vim) of user 0 dumped core.

Stack trace of thread 2345:
#0  0x00007f21dd87e2a7 kill (libc.so.6)
#1  0x00000000000050cb35 may_core_dump (vim)
#2  0x00007f21ddbfc70 __restore_rt (libpthread.so.0)
#3  0x00007f21dd92ea33 __select (libc.so.6)
#4  0x00000000000050b4e3 RealWaitForChar (vim)
#5  0x00000000000050b86b mch_inchar (vim)
[...]
```

When you have multiple core dumps, **coredumpctl info** displays all of them. Filter them by PID, COMM (command), or EXE (full path to the executable). For example, all core dumps for Vim:

```
#coredumpctl info /bin/vim
```

See a single core dump by PID:

```
#coredumpctl info 2345
```

Output the selected core to **gdb**:

```
#coredumpctl gdb 2345
```

The asterisk in the **PRESENT** column indicates that a stored core dump is present. If the field is empty there is no stored core dump, and **coredumpctl** retrieves crash information from the journal. You can control this behavior in `/etc/systemd/coredump.conf` with the **Storage** option:

- **Storage=none**—core dumps are logged in the journal, but not stored. This is useful to minimize collecting and storing sensitive information, for example for General Data Protection Regulation (GDPR) compliance.

- `Storage=external`—cores are stored in `/var/lib/systemd/coredump`
- `Storage=journal`—cores are stored in the `systemd` journal

A new instance of `systemd-coredump` is invoked for every core dump, so configuration changes are applied with the next core dump, and there is no need to restart any services.

Core dumps are not preserved after a system restart. You may save them permanently with **`coredumpctl`**. The following example filters by the PID and stores the core in `vim.dump`:

```
#coredumpctl -o vim.dump dump 2345
```

See **`man systemd-coredump`**, **`man coredumpctl`**, **`man core`**, and **`man coredump.conf`** for complete command and option listings.

Part VII. Synchronized clocks with Precision Time Protocol

22 Precision Time Protocol 214

Chapter 22. Precision Time Protocol

For network environments, it is vital to keep the computer and other devices' clocks synchronized and accurate. There are several solutions to achieve synchronicity and accuracy, for example, the widely used Network Time Protocol (NTP) described in Chapter 39, Time synchronization with NTP in "[Administration Guide](#)".

The Precision Time Protocol (PTP) is a protocol capable of sub-microsecond accuracy, which is better than what NTP achieves. PTP support is divided between the kernel and user space. The kernel in SUSE Linux Enterprise Server includes support for PTP clocks, which are provided by network drivers.

22.1. Introduction to PTP

The clocks managed by PTP follow a master-slave hierarchy. The slaves are synchronized to their masters. The hierarchy is updated by the *best master clock* (BMC) algorithm, which runs on every clock. The clock with only one port can be either master or slave. Such a clock is called an *ordinary clock* (OC). A clock with multiple ports can be master on one port and slave on another. Such a clock is called a *boundary clock* (BC). The top-level master is called the *grandmaster clock*. The grandmaster clock can be synchronized with a Global Positioning System (GPS). This way disparate networks can be synchronized with a high degree of accuracy.

The hardware support is the main advantage of PTP. It is supported by several network switches and network interface controllers (NIC). While it is possible to use non-PTP enabled hardware within the network, having network components between all PTP clocks PTP hardware enabled achieves the best possible accuracy.

22.1. PTP Linux implementation

On SUSE Linux Enterprise Server, the implementation of PTP is provided by the `linuxptp` package. Install it with **`zypper install linuxptp`**. It includes the **`ptp4l`** and **`phc2sys`** programs for clock synchronization. **`ptp4l`** implements the PTP boundary clock and ordinary clock. When hardware time stamping is enabled, **`ptp4l`** synchronizes the PTP hardware clock to the master clock. With software time stamping, it synchronizes the system clock to the master clock. **`phc2sys`** is needed only with hardware time stamping to synchronize the system clock to the PTP hardware clock on the network interface card (NIC).

22.2. Using PTP

22.1. Network driver and hardware support

PTP requires that the used kernel network driver supports either software or hardware time stamping. Moreover, the NIC must support time stamping in the physical hardware. You can verify the driver and NIC time stamping capabilities with **`ethtool`**:

```
>sudo ethtool -T eth0
Time stamping parameters for eth0:
Capabilities:
hardware-transmit      (SOF_TIMESTAMPING_TX_HARDWARE)
software-transmit      (SOF_TIMESTAMPING_TX_SOFTWARE)
hardware-receive       (SOF_TIMESTAMPING_RX_HARDWARE)
software-receive       (SOF_TIMESTAMPING_RX_SOFTWARE)
software-system-clock   (SOF_TIMESTAMPING_SOFTWARE)
hardware-raw-clock     (SOF_TIMESTAMPING_RAW_HARDWARE)
PTP Hardware Clock: 0
Hardware Transmit Timestamp Modes:
off                    (HWTSTAMP_TX_OFF)
on                     (HWTSTAMP_TX_ON)
Hardware Receive Filter Modes:
none                   (HWTSTAMP_FILTER_NONE)
all                    (HWTSTAMP_FILTER_ALL)
```

Software time stamping requires the following parameters:

```
SOF_TIMESTAMPING_SOFTWARE
SOF_TIMESTAMPING_TX_SOFTWARE
SOF_TIMESTAMPING_RX_SOFTWARE
```

Hardware time stamping requires the following parameters:

```
SOF_TIMESTAMPING_RAW_HARDWARE
SOF_TIMESTAMPING_TX_HARDWARE
SOF_TIMESTAMPING_RX_HARDWARE
```

22.2. Using **ptp4l**

ptp4l uses hardware time stamping by default. As root, you need to specify the network interface capable of hardware time stamping with the **-i** option. The **-m** tells **ptp4l** to print its output to the standard output instead of the system's logging facility:

```
>sudo ptp4l -m -i eth0
selected eth0 as PTP clock
port 1: INITIALIZING to LISTENING on INITIALIZE
port 0: INITIALIZING to LISTENING on INITIALIZE
port 1: new foreign master 00a152.ffff.0b334d-1
selected best master clock 00a152.ffff.0b334d
port 1: LISTENING to UNCALIBRATED on RS SLAVE
master offset -25937 s0 freq +0 path delay      12340
master offset -27887 s0 freq +0 path delay      14232
master offset -38802 s0 freq +0 path delay      13847
master offset -36205 s1 freq +0 path delay      10623
master offset -6975 s2 freq -30575 path delay    10286
port 1: UNCALIBRATED to SLAVE on MASTER_CLOCK_SELECTED
master offset -4284 s2 freq -30135 path delay     9892
```

The `master_offset` value represents the measured offset from the master (in nanoseconds).

The `s0`, `s1` and `s2` indicators show the different states of the clock servo: `s0` is unlocked, `s1` is clock step, and `s2` is locked. If the servo is in the locked state (`s2`), the clock is not stepped (only slowly adjusted) if the `pi_offset_const` option is set to a negative value in the configuration file (see **man 8 ptp4l** for more information).

The `freq` value represents the frequency adjustment of the clock (in parts per billion, ppb).

The `path_delay` value represents the estimated delay of the synchronization messages sent from the master (in nanoseconds).

Port 0 is a Unix domain socket used for local PTP management. Port 1 is the `eth0` interface.

INITIALIZING, LISTENING, UNCALIBRATED and SLAVE are examples of port states which change on INITIALIZE, RS_SLAVE, and MASTER_CLOCK_SELECTED events. When the port state changes from UNCALIBRATED to SLAVE, the computer has successfully synchronized with a PTP master clock.

You can enable software time stamping with the `-S` option.

```
>sudo ptp4l -m -S -i eth3
```

You can also run **ptp4l** as a service:

```
>sudo systemctl start ptp4l
```

In this case, **ptp4l** reads its options from the `/etc/sysconfig/ptp4l` file. By default, this file tells **ptp4l** to read the configuration options from `/etc/ptp4l.conf`. For more information on **ptp4l** options and the configuration file settings, see **man 8 ptp4l**.

To enable the **ptp4l** service permanently, run the following:

```
>sudo systemctl enable ptp4l
```

To disable it, run

```
>sudo systemctl disable ptp4l
```

22.3. ptp4l configuration file

ptp4l can read its configuration from an optional configuration file. As no configuration file is used by default, you need to specify it with `-f`.

```
>sudo ptp4l -f /etc/ptp4l.conf
```

The configuration file is divided into sections. The global section (indicated as `[global]`) sets the program options, clock options and default port options. Other sections are port specific, and they override the default port options. The name of the section is the name of the configured port—for example, `[eth0]`. An empty port section can be used to replace the command line option.

```
[global]
verbose          1
time_stamping    software
[eth0]
```

The example configuration file is an equivalent of the following command's options:

```
>sudo ptp4l -i eth0 -m -S
```

For a complete list of **ptp4l** configuration options, see **man 8 ptp4l**.

22.4. Delay measurement

ptp4l measures time delay in two different ways: *peer-to-peer* (P2P) or *end-to-end* (E2E).

P2P

This method is specified with **-P**.

It reacts to changes in the network environment faster and is more accurate in measuring the delay. It is only used in networks where each port exchanges PTP messages with one other port. P2P needs to be supported by all hardware on the communication path.

E2E

This method is specified with **-E**. This is the default.

Automatic method selection

This method is specified with **-A**. The automatic option starts **ptp4l** in E2E mode, and changes to P2P mode if a peer delay request is received.

Common measurement method



All clocks on a single PTP communication path must use the same method to measure the time delay. A warning is printed if either a peer delay request is received on a port using the E2E mechanism, or an E2E delay request is received on a port using the P2P mechanism.

22.5. PTP management client: **pmc**

You can use the **pmc** client to obtain more detailed information about **ptp4l**. It reads from the standard input—or from the command line—actions specified by name and management ID. Then it sends the actions over the selected transport, and prints any received replies. There are three actions supported: **GET** retrieves the specified information, **SET** updates the specified information, and **CMD** (or **COMMAND**) initiates the specified event.

By default, the management commands are addressed to all ports. The **TARGET** command can be used to select a particular clock and port for the subsequent messages. For a complete list of management IDs, run **pmc help**.

```
>sudo pmc -u -b 0 'GET TIME_STATUS_NP'
sending: GET TIME_STATUS_NP
90f2ca.ffff.20d7e9-0 seq 0 RESPONSE MANAGEMENT TIME_STATUS_NP
      master_offset          283
      ingress_time          1361569379345936841
      cumulativeScaledRateOffset +1.000000000
      scaledLastGmPhaseChange 0
      gmTimeBaseIndicator     0
      lastGmPhaseChange       0x0000'0000000000000000.0000
      gmPresent               true
      gmIdentity              00b058.feef.0b448a
```

The `-b` option specifies the boundary hops value in sent messages. Setting it to zero limits the boundary to the local **ptp4l** instance. Increasing the value retrieves the messages also from PTP nodes that are further from the local instance. The returned information may include:

stepsRemoved

The number of communication nodes to the grandmaster clock.

offsetFromMaster, master_offset

The last measured offset of the clock from the master clock (nanoseconds).

meanPathDelay

The estimated delay of the synchronization messages sent from the master clock (nanoseconds).

gmPresent

If `true`, the PTP clock is synchronized to the master clock; the local clock is not the grandmaster clock.

gmIdentity

This is the grandmaster's identity.

For a complete list of **pmc** command line options, see **man 8 pmc**.

22.3. Synchronizing the clocks with phc2sys

Use **phc2sys** to synchronize the system clock to the PTP hardware clock (PHC) on the network card. The system clock is considered a *slave*, while the network card a *master*. PHC itself is synchronized with **ptp4l** (see *the section called "Using PTP"*). Use `-s` to specify the master clock by device or network interface. Use `-w` to wait until **ptp4l** is in a synchronized state.

```
>sudo phc2sys -s eth0 -w
```

PTP operates in *International Atomic Time* (TAI), while the system clock uses *Coordinated Universal Time* (UTC). If you do not specify `-w` to wait for **ptp4l** synchronization, you can specify the offset in seconds between TAI and UTC with `-0`:

```
>sudo phc2sys -s eth0 -0 -35
```

You can run **phc2sys** as a service as well:

```
>sudo systemctl start phc2sys
```

In this case, **phc2sys** reads its options from the `/etc/sysconfig/phc2sys` file. For more information on **phc2sys** options, see **man 8 phc2sys**.

To enable the **phc2sys** service permanently, run the following:

```
>sudo systemctl enable phc2sys
```

To disable it, run

```
>sudo systemctl disable phc2sys
```

22.1. Verifying time synchronization

When PTP time synchronization is working properly and hardware time stamping is used, **ptp4l** and **phc2sys** output messages with time offsets and frequency adjustments periodically to the system log.

An example of the **ptp4l** output:

```
ptp4l[351.358]: selected /dev/ptp0 as PTP clock
ptp4l[352.361]: port 1: INITIALIZING to LISTENING on INITIALIZE
ptp4l[352.361]: port 0: INITIALIZING to LISTENING on INITIALIZE
ptp4l[353.210]: port 1: new foreign master 00a069.eefe.0b442d-1
ptp4l[357.214]: selected best master clock 00a069.eefe.0b662d
ptp4l[357.214]: port 1: LISTENING to UNCALIBRATED on RS SLAVE
ptp4l[359.224]: master offset      3304 s0 freq      +0 path delay      9202
ptp4l[360.224]: master offset      3708 s1 freq    -28492 path delay      9202
ptp4l[361.224]: master offset     -3145 s2 freq    -32637 path delay      9202
ptp4l[361.224]: port 1: UNCALIBRATED to SLAVE on MASTER_CLOCK_SELECTED
ptp4l[362.223]: master offset      -145 s2 freq    -30580 path delay      9202
ptp4l[363.223]: master offset      1043 s2 freq    -28436 path delay      8972
[...]
ptp4l[371.235]: master offset        285 s2 freq    -28511 path delay      9199
ptp4l[372.235]: master offset        -78 s2 freq    -28788 path delay      9204
```

An example of the **phc2sys** output:

```
phc2sys[616.617]: Waiting for ptp4l...
phc2sys[628.628]: phc offset      66341 s0 freq      +0 delay      2729
phc2sys[629.628]: phc offset      64668 s1 freq    -37690 delay      2726
[...]
phc2sys[646.630]: phc offset      -333 s2 freq    -37426 delay      2747
phc2sys[646.630]: phc offset       194 s2 freq    -36999 delay      2749
```

ptp4l normally writes messages with high frequency. You can reduce the frequency with the `summary_interval` directive. Its value is an exponent of the 2^N expression. For example, to

reduce the output to every 1024 (which is equal to 2^{10}) seconds, add the following line to the `/etc/ptp4l.conf` file:

```
summary_interval 10
```

You can also reduce the frequency of the **phc2sys** command's updates with the `-u SUMMARY-UPDATES` option.

22.4. Examples of configurations

This section includes several examples of **ptp4l** configuration. The examples are not full configuration files but rather a minimal list of changes to be made to the specific files. The string *ethX* stands for the actual network interface name in your setup.

Example 22.1. Slave clock using software time stamping

`/etc/sysconfig/ptp4l:`

```
OPTIONS="-f /etc/ptp4l.conf -i ethX"
```

No changes made to the distribution `/etc/ptp4l.conf`.

Example 22.2. Slave clock using hardware time stamping

`/etc/sysconfig/ptp4l:`

```
OPTIONS="-f /etc/ptp4l.conf -i ethX"
```

`/etc/sysconfig/phc2sys:`

```
OPTIONS="-s ethX -w"
```

No changes made to the distribution `/etc/ptp4l.conf`.

Example 22.3. Master clock using hardware time stamping

`/etc/sysconfig/ptp4l:`

```
OPTIONS="-f /etc/ptp4l.conf -i ethX"
```

`/etc/sysconfig/phc2sys:`

```
OPTIONS="-s CLOCK_REALTIME -c ethX -w"
```

`/etc/ptp4l.conf:`

```
priority1 127
```

Example 22.4. Master clock using software time stamping (not generally recommended)

`/etc/sysconfig/ptp4l:`

```
OPTIONS="-f /etc/ptp4l.conf -i ethX"
```

/etc/ptp4l.conf:

```
priority1 127
```

22.5. PTP and NTP

NTP and PTP time synchronization tools can coexist, synchronizing time from one to another in both directions.

22.1. NTP to PTP synchronization

When `chronyd` is used to synchronize the local system clock, you can configure the **ptp4l** to be the grandmaster clock distributing the time from the local system clock via PTP. Include the `priority1` option in `/etc/ptp4l.conf`:

```
[global]
priority1 127
[eth0]
```

Then run **ptp4l**:

```
>sudo ptp4l -f /etc/ptp4l.conf
```

When hardware time stamping is used, you need to synchronize the PTP hardware clock to the system clock with **phc2sys**:

```
>sudo phc2sys -c eth0 -s CLOCK_REALTIME -w
```

22.2. Configuring PTP-NTP bridge

If a highly accurate PTP grandmaster is available in a network without switches or routers with PTP support, a computer may operate as a PTP slave and a stratum-1 NTP server. Such a computer needs to have two or more network interfaces, and be close to the grandmaster or have a direct connection to it. This ensures highly accurate synchronization in the network.

Configure the **ptp4l** and **phc2sys** programs to use one network interface to synchronize the system clock using PTP. Then configure `chronyd` to provide the system time using the other interface:

```
bindaddress 192.0.131.47
hwtimestamp eth1
local stratum 1
```



NTP and DHCP

When the DHCP client command **dhclient** receives a list of NTP servers, it adds them to NTP configuration by default. To prevent this behavior, set

```
NETCONFIG_NTP_POLICY=""
```

in the `/etc/sysconfig/network/config` file.

Appendix A. GNU licenses

This appendix contains the GNU Free Documentation License version 1.2.

A.1. GNU Free Documentation License

Copyright (C) 2000, 2001, 2002 Free Software Foundation, Inc. 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA. Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

0. PREAMBLE

The purpose of this License is to make a manual, textbook, or other functional and useful document "free" in the sense of freedom: to assure everyone the effective freedom to copy and redistribute it, with or without modifying it, either commercially or non-commercially. Secondly, this License preserves for the author and publisher a way to get credit for their work, while not being considered responsible for modifications made by others.

This License is a kind of "copyleft", which means that derivative works of the document must themselves be free in the same sense. It complements the GNU General Public License, which is a copyleft license designed for free software.

We have designed this License to use it for manuals for free software, because free software needs free documentation: a free program should come with manuals providing the same freedoms that the software does. But this License is not limited to software manuals; it can be used for any textual work, regardless of subject matter or whether it is published as a printed book. We recommend this License principally for works whose purpose is instruction or reference.

1. APPLICABILITY AND DEFINITIONS

This License applies to any manual or other work, in any medium, that contains a notice placed by the copyright holder saying it can be distributed under the terms of this License. Such a notice grants a world-wide, royalty-free license, unlimited in duration, to use that work under the conditions stated herein. The "Document", below, refers to any such manual or work. Any member of the public is a licensee, and is addressed as "you". You accept the license if you copy, modify or distribute the work in a way requiring permission under copyright law.

A "Modified Version" of the Document means any work containing the Document or a portion of it, either copied verbatim, or with modifications and/or translated into another language.

A "Secondary Section" is a named appendix or a front-matter section of the Document that deals exclusively with the relationship of the publishers or authors of the Document to the Document's overall subject (or to related matters) and contains nothing that could fall directly within that overall

subject. (Thus, if the Document is in part a textbook of mathematics, a Secondary Section may not explain any mathematics.) The relationship could be a matter of historical connection with the subject or with related matters, or of legal, commercial, philosophical, ethical or political position regarding them.

The "Invariant Sections" are certain Secondary Sections whose titles are designated, as being those of Invariant Sections, in the notice that says that the Document is released under this License. If a section does not fit the above definition of Secondary then it is not allowed to be designated as Invariant. The Document may contain zero Invariant Sections. If the Document does not identify any Invariant Sections then there are none.

The "Cover Texts" are certain short passages of text that are listed, as Front-Cover Texts or Back-Cover Texts, in the notice that says that the Document is released under this License. A Front-Cover Text may be at most 5 words, and a Back-Cover Text may be at most 25 words.

A "Transparent" copy of the Document means a machine-readable copy, represented in a format whose specification is available to the general public, that is suitable for revising the document straightforwardly with generic text editors or (for images composed of pixels) generic paint programs or (for drawings) some widely available drawing editor, and that is suitable for input to text formatters or for automatic translation to a variety of formats suitable for input to text formatters. A copy made in an otherwise Transparent file format whose markup, or absence of markup, has been arranged to thwart or discourage subsequent modification by readers is not Transparent. An image format is not Transparent if used for any substantial amount of text. A copy that is not "Transparent" is called "Opaque".

Examples of suitable formats for Transparent copies include plain ASCII without markup, Texinfo input format, LaTeX input format, SGML or XML using a publicly available DTD, and standard-conforming simple HTML, PostScript or PDF designed for human modification. Examples of transparent image formats include PNG, XCF and JPG. Opaque formats include proprietary formats that can be read and edited only by proprietary word processors, SGML or XML for which the DTD and/or processing tools are not generally available, and the machine-generated HTML, PostScript or PDF produced by some word processors for output purposes only.

The "Title Page" means, for a printed book, the title page itself, plus such following pages as are needed to hold, legibly, the material this License requires to appear in the title page. For works in formats which do not have any title page as such, "Title Page" means the text near the most prominent appearance of the work's title, preceding the beginning of the body of the text.

A section "Entitled XYZ" means a named subunit of the Document whose title either is precisely XYZ or contains XYZ in parentheses following text that translates XYZ in another language. (Here XYZ stands for a specific section name mentioned below, such as "Acknowledgements", "Dedications", "Endorsements", or "History".) To "Preserve the Title" of such a section when you modify the Document means that it remains a section "Entitled XYZ" according to this definition.

The Document may include Warranty Disclaimers next to the notice which states that this License applies to the Document. These Warranty Disclaimers are considered to be included by reference in this License, but only as regards disclaiming warranties: any other implication that these Warranty Disclaimers may have is void and has no effect on the meaning of this License.

2. VERBATIM COPYING

You may copy and distribute the Document in any medium, either commercially or non-commercially, provided that this License, the copyright notices, and the license notice saying this License applies to the Document are reproduced in all copies, and that you add no other conditions whatsoever to those of this License. You may not use technical measures to obstruct or control the reading or further copying of the copies you make or distribute. However, you may accept compensation in exchange for copies. If you distribute a large enough number of copies you must also follow the conditions in section 3.

You may also lend copies, under the same conditions stated above, and you may publicly display copies.

3. COPYING IN QUANTITY

If you publish printed copies (or copies in media that commonly have printed covers) of the Document, numbering more than 100, and the Document's license notice requires Cover Texts, you must enclose the copies in covers that carry, clearly and legibly, all these Cover Texts: Front-Cover Texts on the front cover, and Back-Cover Texts on the back cover. Both covers must also clearly and legibly identify you as the publisher of these copies. The front cover must present the full title with all words of the title equally prominent and visible. You may add other material on the covers in addition. Copying with changes limited to the covers, as long as they preserve the title of the Document and satisfy these conditions, can be treated as verbatim copying in other respects.

If the required texts for either cover are too voluminous to fit legibly, you should put the first ones listed (as many as fit reasonably) on the actual cover, and continue the rest onto adjacent pages.

If you publish or distribute Opaque copies of the Document numbering more than 100, you must either include a machine-readable Transparent copy along with each Opaque copy, or state in or with each Opaque copy a computer-network location from which the general network-using public has access to download using public-standard network protocols a complete Transparent copy of the Document, free of added material. If you use the latter option, you must take reasonably prudent steps, when you begin distribution of Opaque copies in quantity, to ensure that this Transparent copy will remain thus accessible at the stated location until at least one year after the last time you distribute an Opaque copy (directly or through your agents or retailers) of that edition to the public.

It is requested, but not required, that you contact the authors of the Document well before redistributing any large number of copies, to give them a chance to provide you with an updated version of the Document.

4. MODIFICATIONS

You may copy and distribute a Modified Version of the Document under the conditions of sections 2 and 3 above, provided that you release the Modified Version under precisely this License, with the Modified Version filling the role of the Document, thus licensing distribution and modification of the Modified Version to whoever possesses a copy of it. In addition, you must do these things in the Modified Version:

1. Use in the Title Page (and on the covers, if any) a title distinct from that of the Document, and from those of previous versions (which should, if there were any, be listed in the History section of the Document). You may use the same title as a previous version if the original publisher of that version gives permission.
2. List on the Title Page, as authors, one or more persons or entities responsible for authorship of the modifications in the Modified Version, together with at least five of the principal authors of the Document (all of its principal authors, if it has fewer than five), unless they release you from this requirement.
3. State on the Title page the name of the publisher of the Modified Version, as the publisher.
4. Preserve all the copyright notices of the Document.
5. Add an appropriate copyright notice for your modifications adjacent to the other copyright notices.
6. Include, immediately after the copyright notices, a license notice giving the public permission to use the Modified Version under the terms of this License, in the form shown in the Addendum below.
7. Preserve in that license notice the full lists of Invariant Sections and required Cover Texts given in the Document's license notice.
8. Include an unaltered copy of this License.
9. Preserve the section Entitled "History", Preserve its Title, and add to it an item stating at least the title, year, new authors, and publisher of the Modified Version as given on the Title Page. If there is no section Entitled "History" in the Document, create one stating the title, year, authors, and publisher of the Document as given on its Title Page, then add an item describing the Modified Version as stated in the previous sentence.
10. Preserve the network location, if any, given in the Document for public access to a Transparent copy of the Document, and likewise the network locations given in the Document for previous versions it was based on. These may be placed in the "History" section. You may omit a network location for a work that was published at least four years

before the Document itself, or if the original publisher of the version it refers to gives permission.

11. For any section Entitled "Acknowledgements" or "Dedications", Preserve the Title of the section, and preserve in the section all the substance and tone of each of the contributor acknowledgements and/or dedications given therein.
12. Preserve all the Invariant Sections of the Document, unaltered in their text and in their titles. Section numbers or the equivalent are not considered part of the section titles.
13. Delete any section Entitled "Endorsements". Such a section may not be included in the Modified Version.
14. Do not retitle any existing section to be Entitled "Endorsements" or to conflict in title with any Invariant Section.
15. Preserve any Warranty Disclaimers.

If the Modified Version includes new front-matter sections or appendices that qualify as Secondary Sections and contain no material copied from the Document, you may at your option designate some or all of these sections as invariant. To do this, add their titles to the list of Invariant Sections in the Modified Version's license notice. These titles must be distinct from any other section titles.

You may add a section Entitled "Endorsements", provided it contains nothing but endorsements of your Modified Version by various parties--for example, statements of peer review or that the text has been approved by an organization as the authoritative definition of a standard.

You may add a passage of up to five words as a Front-Cover Text, and a passage of up to 25 words as a Back-Cover Text, to the end of the list of Cover Texts in the Modified Version. Only one passage of Front-Cover Text and one of Back-Cover Text may be added by (or through arrangements made by) any one entity. If the Document already includes a cover text for the same cover, previously added by you or by arrangement made by the same entity you are acting on behalf of, you may not add another; but you may replace the old one, on explicit permission from the previous publisher that added the old one.

The author(s) and publisher(s) of the Document do not by this License give permission to use their names for publicity for or to assert or imply endorsement of any Modified Version.

5. COMBINING DOCUMENTS

You may combine the Document with other documents released under this License, under the terms defined in section 4 above for modified versions, provided that you include in the combination all of the Invariant Sections of all of the original documents, unmodified, and list them all as Invariant Sections of your combined work in its license notice, and that you preserve all their Warranty Disclaimers.

The combined work need only contain one copy of this License, and multiple identical Invariant Sections may be replaced with a single copy. If there are multiple Invariant Sections with the same name but different contents, make the title of each such section unique by adding at the end of it, in parentheses, the name of the original author or publisher of that section if known, or else a unique number. Make the same adjustment to the section titles in the list of Invariant Sections in the license notice of the combined work.

In the combination, you must combine any sections Entitled "History" in the various original documents, forming one section Entitled "History"; likewise combine any sections Entitled "Acknowledgements", and any sections Entitled "Dedications". You must delete all sections Entitled "Endorsements".

6. COLLECTIONS OF DOCUMENTS

You may make a collection consisting of the Document and other documents released under this License, and replace the individual copies of this License in the various documents with a single copy that is included in the collection, provided that you follow the rules of this License for verbatim copying of each of the documents in all other respects.

You may extract a single document from such a collection, and distribute it individually under this License, provided you insert a copy of this License into the extracted document, and follow this License in all other respects regarding verbatim copying of that document.

7. AGGREGATION WITH INDEPENDENT WORKS

A compilation of the Document or its derivatives with other separate and independent documents or works, in or on a volume of a storage or distribution medium, is called an "aggregate" if the copyright resulting from the compilation is not used to limit the legal rights of the compilation's users beyond what the individual works permit. When the Document is included in an aggregate, this License does not apply to the other works in the aggregate which are not themselves derivative works of the Document.

If the Cover Text requirement of section 3 is applicable to these copies of the Document, then if the Document is less than one half of the entire aggregate, the Document's Cover Texts may be placed on covers that bracket the Document within the aggregate, or the electronic equivalent of covers if the Document is in electronic form. Otherwise they must appear on printed covers that bracket the whole aggregate.

8. TRANSLATION

Translation is considered a kind of modification, so you may distribute translations of the Document under the terms of section 4. Replacing Invariant Sections with translations requires special

permission from their copyright holders, but you may include translations of some or all Invariant Sections in addition to the original versions of these Invariant Sections. You may include a translation of this License, and all the license notices in the Document, and any Warranty Disclaimers, provided that you also include the original English version of this License and the original versions of those notices and disclaimers. In case of a disagreement between the translation and the original version of this License or a notice or disclaimer, the original version will prevail.

If a section in the Document is Entitled "Acknowledgements", "Dedications", or "History", the requirement (section 4) to Preserve its Title (section 1) will typically require changing the actual title.

9. TERMINATION

You may not copy, modify, sublicense, or distribute the Document except as expressly provided for under this License. Any other attempt to copy, modify, sublicense or distribute the Document is void, and will automatically terminate your rights under this License. However, parties who have received copies, or rights, from you under this License will not have their licenses terminated so long as such parties remain in full compliance.

10. FUTURE REVISIONS OF THIS LICENSE

The Free Software Foundation may publish new, revised versions of the GNU Free Documentation License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns. See <https://www.gnu.org/copyleft/>.

Each version of the License is given a distinguishing version number. If the Document specifies that a particular numbered version of this License "or any later version" applies to it, you have the option of following the terms and conditions either of that specified version or of any later version that has been published (not as a draft) by the Free Software Foundation. If the Document does not specify a version number of this License, you may choose any version ever published (not as a draft) by the Free Software Foundation.

ADDENDUM: How to use this License for your documents

```
Copyright (c) YEAR YOUR NAME.
Permission is granted to copy, distribute and/or modify this document
under the terms of the GNU Free Documentation License, Version 1.2
or any later version published by the Free Software Foundation;
with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts.
A copy of the license is included in the section entitled "GNU
Free Documentation License".
```

If you have Invariant Sections, Front-Cover Texts and Back-Cover Texts, replace the "with...Texts." line with this:

with the Invariant Sections being LIST THEIR TITLES, with the Front-Cover Texts being LIST, and with the Back-Cover Texts being LIST.

If you have Invariant Sections without Cover Texts, or some other combination of the three, merge those two alternatives to suit the situation.

If your document contains nontrivial examples of program code, we recommend releasing these examples in parallel under your choice of free software license, such as the GNU General Public License, to permit their use in free software.